The block below carries no module guard, so docstrip copies it into EVERY generated file. That is what we want for the TeX files and wrong for the latexmkrc, which is Perl and does not understand lines opening with two percent signs – hence the negated guard around it.

The guard lines themselves are hidden behind a false conditional. docstrip reads a guard line either way, but ltxdoc, which typesets this file, would otherwise set the literal text of the guard as a stray paragraph in the body of the manual.

Nothing in this area may name a TeX conditional, not even in prose: outside the meta-comment at the top of the file, every documentation line here IS LaTeX source and is executed, and a conditional named in passing opens or closes a real one. That is not hypothetical – an earlier draft of this very comment swallowed the rest of the manual.

Three conditional primitives are deliberately absent from the lists below: the ones that close or branch a conditional. `\DoNotIndex` scans its argument with TEX's own conditional machinery live, so a bare one of them closes the wrong conditional. The manual ended its run with “Too many }'s” and exited with status 1, which failed the docs step of the harness even though the PDF had been written. Leaving them out of the lists means they get indexed, which is harmless. Note that they cannot be named here either, not even inside `...` shorthand split across a line.

The coppe document class
Version 4.0

Geraldo Xexéo Vicente H. F. Batista
George O. Ainsworth Jr. Eduardo Mangeli

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Abstract

In this work, it is described the coppe document class as well as other files distributed by the CoppeTEX project. This class is suitable for writing academic dissertations, thesis and qualifying exams according to the formatting rules of the Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering. The minimalist set of macro commands allows its users to concentrate most of their efforts on text composition rather than on the document layout.

1 Introduction

Writing documents in LaTeX may be a laborious task when the authors have to prepare their manuscripts rigorously respecting formatting rules imposed by publishers. Regardless of difficulty, a lot of thesis presented to the Coordination of Graduate Studies and Research in Engineering of the Federal University of Rio de Janeiro (COPPE/UFRJ) is typeseted in LaTeX. This demand motivated the creation of the CoppeTEX project, which tries to facilitate and encourage the use of LaTeX within the COPPE/UFRJ scope.

The coppe document class is the main product of CoppeTEX. It was designed to be clear and succinct. It enables the creation of dissertations, qualifying exams and thesis in a simple and automatic way. The main goal of the coppe class is to maintain authors strictly focused on text composition without worrying about margins sizes, line spacing, paper size, vertical and horizontal alignment, etc. The CoppeTEX project comprehends also BibTEX and MakeIndex style files for creating lists of references, symbols and abbreviations. Although there aren't official guidelines to write qualifying exams, we provide this option just for convenience, as this exam is a requisite to obtain the DSc degree and, for some of the programs, the MSc degree.

In which follows, it is described the user interface of the coppe class. Some details about using the style files cited above are also given. We use the term *thesis* to generally refer to dissertation, qualifying exam, and thesis itself.

2 License

Each file belonging to this package contains a copyright notice. Its use is protected by the GNU General Public License (GPL) version 3, so that users are free for copying, distributing or modifying the source code, among other acts covered by this license.

To see the full text of the GNU GPL license, go to the `COPYING` file attached to this package.

3 Support

We maintain a WhatsApp group where users can send questions, comments, and bugs to. More details can be found here.

However, as the project maintenance was transferred to a new repository, bug reports, as well as new feature requests, should be directed to `https://github.com/COPPE-UFRJ/CoppeTeX`.

4 User interface

`\frontmatter` `\mainmatter` `\backmatter` A thesis to be approved by the Academic Registry at COPPE/UFRJ must contain three-parts: *front*, *main* and *back* matters (COPPE/UFRJ Academic Registry guidelines). Each one of these parts is started by calling its corresponding macro `\frontmatter`, `\mainmatter` or `\backmatter`. The front matter of a thesis consists of front cover and face, cataloging page, dedication, acknowledgments, abstracts, table of contents, and lists of tables, algorithms, symbols and abbreviations. The main matter is just composed by chapters, while the back matter usually consists of bibliographic references, appendices and index.

You must invoke the `\frontmatter` macro immediately after the `\maketitle` one. The `\mainmatter` command comes right before the first chapter, and `\backmatter` must be typed before the list of references.

Cover (capa)

The ABNT cover suggested in the manual's Annex A. The `\maketitle` command builds it automatically, ahead of the title page: the university, the full COPPE name, and the graduate program (taken from `\department`), followed by the author, the title, and the city and year, all centred and in uppercase. The UFRJ and COPPE logos head the page. The capa is not counted.

Front face

The front face is unnumbered. There, it is not allowed to use hyphenation (COPPE/UFRJ Academic Registry guidelines). It is constructed by calling `\maketitle`. Next, it is described the commands used to enter the information required to create it.

`\author` The `\author` command was redefined. Here, it takes two arguments: the author's first names and surname, e.g., `\author{First Names}{Surname}`. The words should be typed with only first letters in uppercase.

`\title` `\foreigntitle` `\titlein` The macros `\title` and `\foreigntitle` are used to enter the titles of your monograph in Brazilian Portuguese and English respectively (the two built-in languages of CoppeTeX 4.x). For documents whose main language is neither brazilian nor english – e.g. spanish, french, italian, or a user-supplied language –, also call \titlein{<lang>}{<text>} to register the title in that language; the cover and folha de rosto print whichever title matches the main language declared in the class option. See the section “Multilingual support” below for the complete picture. The babel package is loaded automatically by coppe.cls, so you do not need to load it yourself.

\advisor \examiner Every COPPE student is coordinated by at least one advisor. M.Sc. and D.Sc. students can have at most 2 and 3 advisors, respectively. Their names must be provided by issuing the command \advisor as below:

```
\advisor{Title}{Advisor's Name}{Surname}{Degree}
\advisor{Title}{Second Advisor's Name}{Surname}{Degree}
\advisor{Title}{Third Advisor's Name}{Surname}{Degree}
```

The advisors are not necessarily members of the thesis examination board. Thus, it is required to enter the names of all examiners using the \examiner macro. The examiners' names are entered differently:

```
\examiner{Title}{First Examiner's Name Surname}{Degree}
\examiner{Title}{Second Examiner's Name Surname}{Degree}
...
\examiner{Title}{N-th Examiner's Name Surname}{Degree}
```

Remember that all names must be given before calling \maketitle.

\department The Alberto Luiz Coimbra institute is divided into 12 academic units: Biomedical Engineering (PEB), Civil Engineering (PEC), Electrical Engineering (PEE), Mechanical Engineering (PEM), Metallurgical and Materials Science Engineering (PEMM), Nuclear Engineering (PEN), Ocean Engineering (PENO), Energy Planning (PPE), Production Engineering (PEP), Chemical Engineering (PEQ), Systems Engineering and Computer Science (PESC), and Transportation Engineering (PET). You must specify your department using one of the above abbreviations, e.g., \department{PEC}.

\date This macro is used to set the month and year of defense. This information is required to create the front face, cataloging details page and abstracts. For example, October 2007 should be entered as \date{10}{2007}.

\keyword The keywords should describe the concentration areas of your work. You must provide them as follows:

```
\keyword{First Keyword}
\keyword{Second Keyword}
...
\keyword{N-th Keyword}
```

Usually, six words are enough.

Folha adicional: Coleta CAPES

Since August 2026 every thesis and dissertation must carry a *folha adicional* immediately after the folha de rosto, holding the fields the CAPES/MEC collection requires plus the ficha catalografica (SiBI manual, 9th rev. ed. 2026, 3.1.2.1.2

and Annex H). The sheet is filled in by the graduate programme, and is neither counted nor numbered.

The macros below feed that sheet. They are institutional data, so the sheet is always typeset in Portuguese regardless of the main language of the work, as the COPPE norm requires of institutional identity.

`\areaconcentracao` The area of concentration, e.g. `\areaconcentracao{Engenharia de Sistemas e Computacao}`. It appears on the folha de rosto, on the folha de aprovacao and on the folha adicional.

`\linhapesquisa` The research line, required by the CAPES model of the folha de rosto.

`\tipoproducao` The kind of intellectual output, one of `bibliografica`, `artistica`, `tecnologica` or `tecnica`. Anything else raises a warning.

`\projetovinculado` `\nomeprojeto` Whether the work belongs to a research project (`sim` or `nao`) and, if so, its name.

`\agenciafomento` A funding agency, given as full name and acronym; repeat once per agency:

```
\agenciafomento{Conselho Nacional de Desenvolvimento
                Cientifico e Tecnologico}{CNPq}
\agenciafomento{Fundacao Carlos Chagas Filho de Amparo a
                Pesquisa do Estado do Rio de Janeiro}{FAPERJ}
```

Cataloging details

This page contains cataloging information useful for librarians. Fortunately, it is automatically generated from the data you entered at the time you call `\maketitle`. It is not needed in qualifying exams, though.

Dedication (optional)

`\dedication` This macro was added for convenience. The input text is placed at the right bottom of a blank page. It is emphasized and in normal size.

Abstracts

`abstract` (*env.*) `foreignabstract` (*env.*) As stated by the COPPE/UFRJ Academic Registry, abstracts must be in one page each, with at most 250 words. We recommended that they should be only one paragraph long. They must be defined inside the environments `abstract` and `foreignabstract`.

Lists of symbols and abbreviations (optional)

`\abbrev` `\symbl` The lists of symbols and abbreviations are optional, although highly recommended. It is a good practice to define a symbol/abbreviation in its first occurrence in the text. To define a symbol use `\symbl[alphabetic symbol]{Symbol}{Symbol Definition}`, and for abbreviations `\abbrev[alphabetic symbol]{Abbreviation}{Abbreviation Definition}`. These commands are called *dummy*, since they don't output anything at the place they are executed, just an entry in the correspondent list.

`\makeloabbreviations` `\makelosymbols` `\printloabbreviations` `\printlosymbols` These lists are lexicographically sorted by using the MakeIndex program, which is part of any LaTeX implementation. For `\symbl`, if the optional parameter is provided, it will used as sort key. This was later, in 2024, implemented also for

\abbrev, otherwise Symbol, or Abreviation will be used as sort key, what can result in an undesirable order if it contains LaTeXcommands, mathematical symbols, or mix of uppercase and lowercase. MakeIndex needs two commands to create a final sorted list: one which generates a list of entries and the other that indicates the position where the list will be printed out. To generate the lists of symbols and abbreviations, the coppe class provides the commands \makeloabbreviations and \makelosymbols, respectively. They must be called in the document preamble. The commands \printlosymbols and \printloabbreviations have to be invoked at the point where you want these lists appear, e.g., following the list of tables as showed in the example. Once you call latex, it will be created two files with extensions abx and syx, which contain MakeIndex input data. They must be processed with makeindex in order to get the lists correctly produced, redirecting the output to files with extension lab and los respectively:

```
makeindex -s coppe.ist -o example.lab example.abx
makeindex -s coppe.ist -o example.los example.syx
```

Note the -s option for specifying the style coppe.ist. Now, rerun latex twice to get the references solved and you are done.

References

CoppeTeX produces citations and the reference list with biblatex and the biber backend, through its own bibliography style implementing the post-2020 ABNT NBR 6023 rules. The class loads this style automatically: there is no \bibliographystyle to set and no BibTeX .bst file to choose (the former coppe-plain and coppe-unsrt BibTeX styles have been retired).

By default, citations follow the author–date style, e.g., "Chartier (2002)". Passing the numbers class option switches to the numeric style, in which citations are bracketed numbers and the reference list is numbered in order of citation (unsorted).

Declare your bibliography database in the preamble with \addbibresource and print the list where it should appear with \printbibliography:

```
\addbibresource{example.bib}   % in the preamble
...
\printbibliography             % where the list goes
```

Entry types and fields may be written in English (@book, title, author) or with their unaccented Portuguese synonyms (@livro, titulo, autor).

Run in sequence LaTeX, biber, and twice again LaTeX to resolve the citations and references. The style is natbib compatible (via the biblatex natbib=true option), so you may freely issue \citet and \citep, as well as the other natbib commands.

Reference entry types

Every ABNT document category maps to a biblatex entry type, each with an unaccented Portuguese synonym (a worked example of every type is in the sample manuscript's reference list):

Academic @book (@livro); @article (@artigo); @artigodejornal (newspaper); @incollection (@partedelivro); @inbook (@emlivro); @inreference (@verbete); @proceedings (@anais); @inproceedings (@trabalhodeevento); @thesis (@tese/@dissertacao, with the presets @dissertacaomestrado, @tesedoutorado, @tesephd, @tcc); @report (@relatorio); @periodical (@periodico); @manual; @misc (@diverso); @online.

Special (NBR 6023 §4.2.5–4.2.14) @patent (@patente); @legislation (@legislacao/@atonormat @jurisdiction (@jurisprudencia, with a relator field); @legal (@documentojuridico); @standard (@norma); @music (@partitura); @map (@mapa); @image (@iconografico); @artwork (@obradearte); @video (@audiovisual); @movie (@filme); @letter (@correspondencia); @dataset (@conjuntodedados).

Games and TV @game (@jogo); @boardgame (@jogodetabuleiro); @videogame (@jogoeletronico); @tvshow (@programadetv).

When an entry has no author or editor, it is entered by title, with the first significant word (and a leading article) in capitals.

Appendix and Annex (Optional)

\appendix \annex Appendices and annexes are optional chapters that are part of the back matter. The \appendix command is a standard LaTeX command used before all appendices. CoppeTeX introduces the \annex command, which should be used only in the back matter (i.e., after the \backmatter command) and only after all existing \appendix chapters. This restriction is due to implementation constraints.

Therefore, the order for backmatter is:

```
\backmatter
\printbibliography
\appendix
\chapter{An Appendix}
\chapter{Another Appendix}
\annex
\chapter{First Annex}
\chapter{Second Annex}
```

Annex was introduced in v3.5

Final colophon page (\coppetexfinalpage)

\coppetexfinalpage Optional closing page modelled on the printer's *colophon*: it records, in Portuguese, the tooling and edition used to typeset the thesis. Call it once, after every other piece of content, immediately before \end{document}. The macro

- forces a verso (even) page, so in a two-sided print the colophon sits opposite the rear cover;
- types the body at the foot of the page (\vfill);
- renders the text in Portuguese unconditionally (Brazilian babel), even when the thesis main language is English, Spanish, French or Italian — the colophon is institutional metadata, not body content.

Three slot macros are user-overridable. Redefine them with `\renewcommand` before `\coppetexfinalpage` if the document deviates from the defaults:

`\coppefinalengine` the TEX engine identifier. Default: auto-detected via `iftex` (`\hologo{pdfLaTeX}`, `\hologo{LuaLaTeX}` or `\hologo{XeLaTeX}`).

`\coppefinalfont` the text body font. Default: *Computer Modern (padrão LATEX)*. Redefine if you loaded `lmodern`, `mathptmx`, an OpenType math font, etc.

`\coppefinalmanual` the reference manual title (typeset in italics). Default: the current COPPE/UFRJ norm. Redefine if your work follows an earlier edition.

The macro also prints the COPPETEX class version (captured into `\coppe@version` at class-load time, so it survives any later package that may redefine the global `\fileversion`) and the GitHub repository URL, https://github.com/COPPE-UFRJ/CoppeTeX.

5 Class options

There are some options users can specify in order to customize the appearance of the output produced by the `coppe` class. These options can be passed to `coppe` as follows: `\documentclass[option1, option2]{coppe}`. In which follows, we give a brief description of all supported options.

`dsc, msc, dscexam, mscexam` The `coppe` class is able to produce thesis, dissertations, and qualifying exams, which are enabled by the `dsc`, `msc`, `mscexam`, and `dscexam` options, respectively.

`doublespacing` The default line spacing is one-and-a-half. For enabling double spacing between lines, use the `doublespacing` option.

`numbers` The default citation style is the author–date scheme. For numbered citations, specify the `numbers` option to the `coppe` class; the reference list is then numbered in order of citation (unsorted). The matching `biblatex` style is selected automatically in either case.

`brazilian, english, spanish, french, italian` Pick the document's *main* language. The class loads Babel with that language as the main one (the last in the Babel option list); inside the document `\selectlanguage{...}` and `\begin{otherlanguage}{...}` keep working as usual. The default, when no language option is given, is `brazilian`.

- `brazilian` and `english` are *built-in*: their strings ship inside `coppe.cls` and need no extra file.
- `spanish`, `french`, `italian` are shipped as *language packs*: two files per language (`coppe-lang-<lang>.def` and `<lang>-coppe.lbx`) that must sit next to `coppe.cls` (or in your project directory). They are auto-loaded by the class. Picking one of them as the main language without the matching pack files installed raises a clear class error.

- Any other Babel-known language can be plugged in by writing the same two files and either passing the language name as a class option or calling \usecoppelanguage{<lang>} in the preamble.

See the section “Multilingual support” below for the three-slot architecture, the \titlein / \brazilianabstract API, and how to create a new language pack.

5.1 Changing document identification

\freeconfig

The user could *optionally* use the command freeconfig to modify the parameters that print the document identification. The command freeconfig needs all those paramerters, which are degree initials, degree name, title, foreign title, local doctype, and foreign doctype as in the following example:

```
\freeconfig{Dr.}{Philosophiae Doctor}{PhD}{Doutor}{Dissertation}{Tese}
```

5.2 Multilingual support

CoppeTeX 4.x reserves three fixed language slots:

main the language of the body, captions, TOC labels and the abstract environment. Picked by the class option (brazilian by default; one of english, spanish, french, italian also accepted out of the box; any other Babel language possible with a user-supplied pack).

foreign the language of the foreignabstract environment. english by default for every main language; switches automatically to brazilian when main = english.

third optional. Loaded on demand with \usecoppelanguage{<lang>} in the preamble, useful when the body quotes or cites in a third language (e.g. a Spanish-main thesis citing a French source). Does not have a class option of its own.

\titlein
brazilianabstract (*env.*)
\usecoppelanguage

Two pieces of user-facing API come with the multilingual model:

- \titlein{<lang>}{<text>} records the title in any language. \title{...} still writes the Brazilian-Portuguese title and \foreigntitle{...} still writes the English one (back-compat); for a main language outside the brazilian/english pair, also call \titlein{<lang>}{...}. The cover and folha de rosto print whichever title matches \coppe@mainlang.
- The brazilianabstract environment is the optional third abstract used when the main language is neither brazilian nor english – typically a Spanish/French/Italian thesis that also wants a Portuguese resumo for UFRJ examiners. It wraps its body in \begin{otherlanguage}{brazilian} and uses the Portuguese template strings. A document with main = brazilian can simply keep using abstract; brazilianabstract is redundant then.
- \usecoppelanguage{<lang>} loads an extra language pack in the preamble without making it the main or foreign language.

Institutional names stay Portuguese

The cover, folha de rosto and ficha catalográfica are a Brazilian institutional template (NBR 14724 + COPPE manual). The strings universityname, cityname, statename, countryname, the departments declared with \department and the surrounding "apresentada/submetida ao…" wording are deliberately kept in Portuguese for every main language. Only the title (printed via \coppe@selecttitle) follows the main language.

Writing a new language pack

A language pack is just two files dropped next to coppe.cls (or anywhere on the TEXINPUTS path):

coppe-lang-<lang>.def class-level strings. Populate the dispatcher table with \copperdefstring{<lang>}{<key>}{<value>} for every key already populated for brazilian/english in coppe.cls (appendix, annex, frame, listframe, source, program, listprogram, references, in:, art, platform, directedby, producedby, listabbreviation, listsymbol, glossary, advisor, advisors, dept, approvedmale, approvedfemale, universityname, cityname, statename, countryname, monthname1…monthname12, algorithm2eopt, babelname). Finish with

```
\AtBeginDocument{\DeclareLanguageMapping{<lang>}{<lang>-coppe}}
```

so the biblatex mapping is registered after the bibliography engine is loaded by coppe.cls (which happens *after* the language pack is read).

<lang>-coppe.lbx biblatex localization strings. \InheritBibliographyExtras{<lang>} then \DeclareBibliographyStrings for the keys availablefrom, urlseen, in, editor, editors, depositor, coppescale, mscdiss, dscthesis, phdthesis, tcc. The shipped spanish-coppe.lbx, french-coppe.lbx and italian-coppe.lbx are the reference templates to copy from.

Tip: the algorithm2eopt key holds whatever algorithm2e calls the language. Mind the spelling: algorithm2e ships portuguese, english, spanish, french and italiano – not italian.

Example: a Spanish-main thesis

```
\documentclass[spanish,dsc]{coppe}
\addbibresource{example.bib}
\begin{document}
  \title{T\'\i tulo em portugu\^es}     % Brazilian abstract slot
  \foreigntitle{English title}           % English abstract slot
  \titlein{spanish}{T\'\i tulo en espa\~nol}  % cover/folha
  \author{Juan}{P\'erez}
  \advisor{Prof.}{Mar\'\i a}{Garc\'\i a}{D.Sc.}
  \department{PESC}\date{01}{2026}
  \keyword{multilenguaje}
  \maketitle
  \frontmatter
```

```
  \begin{abstract}        Resumen principal en espa\~nol.   \end{abstract}
  \begin{foreignabstract} English foreign abstract.         \end{foreignabstract}
  \begin{brazilianabstract} Resumo em portugu\^es para a banca. \end{brazilianabstract}
  \tableofcontents
  \mainmatter
  \chapter{Introducci\'on} Cuerpo en espa\~nol\dots
\end{document}
```

6 Files in your project folder

In a normal install where CoppeTeX is part of your `TEXINPUTS` path (e.g. inside the local texmf tree of MiKTeX or TeX Live), you do not need to think about this list at all — the files travel with the class. This section is for two situations where you do:

- shipping the class together with your manuscript (a self-contained submission package); or
- installing CoppeTeX by hand, dropping the files next to `coppe.cls` in the project directory.

Always required

These files must be findable at every compilation, regardless of the main language or any class option:

`coppe.cls` the document class itself.

`coppe.dbx` custom `biblatex` datamodel (declares the ABNT-specific entry types `@game`, `@videogame`, `@boardgame`, `@tvshow` and `@map`, plus a few custom fields).

`coppe.bbx` and `coppe.cbx` `biblatex` bibliography and citation styles for the author–date scheme (the default).

`brazilian-coppe.lbx` and `english-coppe.lbx` `biblatex` localisation tables for Portuguese and English. Both are required *even when neither language is the main one*, because `coppe.bbx` registers

```
\DeclareLanguageMapping{brazilian}{brazilian-coppe}
\DeclareLanguageMapping{english}{english-coppe}
```

unconditionally, so `biblatex` tries to read both LBX files at every compilation.

Conditional on class options

`coppe-numeric.bbx` and `coppe-numeric.cbx` only when the `numbers` class option is given (`\documentclass[numbers]{coppe}`). Without it, `biblatex` never tries to read them.

coppe.ist only when the glossary, list-of-symbols, or list-of-abbreviations features are used. It is the makeindex style file invoked by the makeindex -s coppe.ist ... step described in subsection *Lists of symbols and abbreviations* above.

Conditional on the main language

The main language is chosen by the matching class option (brazilian, english, spanish, french, italian). The two built-in languages ship their strings inside coppe.cls and so need no extra file. The three language packs add two files each — one class-level (coppe-lang-<lang>.def) and one biblatex-level (<lang>-coppe.lbx):

Main language	Extra files needed
brazilian (default)	— (built into coppe.cls)
english	— (built into coppe.cls)
spanish	coppe-lang-spanish.def, spanish-coppe.lbx
french	coppe-lang-french.def, french-coppe.lbx
italian	coppe-lang-italian.def, italian-coppe.lbx

Loading a third language at preamble time via \usecoppelanguage{<lang>} brings in the same pair of files for that language. A user-supplied language pack follows the same scheme: drop coppe-lang-<lang>.def and <lang>-coppe.lbx next to coppe.cls.

Cover assets

The institutional title page produced by \maketitle needs two PDF logo files at compile time:

ufrj-logo.pdf and coppe-logo.pdf the UFRJ shield and the COPPE/UFRJ wordmark. Both ship with the CoppeTeX distribution; they must be findable from the project directory (or on TEXINPUTS).

7 Required LaTeX packages

The class is built on the standard book class (12pt, a4paper, oneside) and brings in the stock CTAN packages listed below with \RequirePackage. On a full TeX Live or MiKTeX install they are all present already, so this list matters only for a minimal install — or when you simply want to know which commands are already in scope. You do *not* need to \usepackage any of these yourself; every command and environment they provide is available in your manuscript. Options in brackets are the ones the class passes.

Engine and encoding

iftex detects the running engine, so the class can branch its setup between pdfTeX and the Unicode engines (LuaTeX/XeTeX).

inputenc (utf8) UTF-8 input; loaded *only under pdfTeX* (the Unicode engines read UTF-8 natively).

fontenc (T1) T1 font encoding, for correctly composed accented glyphs and hyphenation.

Mathematics and symbols

`amsmath` the AMS mathematics environments.

`amssymb` the AMS symbol fonts; loaded *only under pdfTEX*. Under a Unicode engine load `unicode-math` (or `amssymb`) yourself — the class warns at end of document if neither is present.

Page layout and spacing

`geometry` A4 one-sided margins (3/3/2/2 cm) and header height.

`fancyhdr` running headers and footers.

`titlesec` chapter and section title formatting.

`setspace` one-and-a-half line spacing (double with the `doublespacing` option).

Lists, tables and rules

`enumitem` customisable list spacing and labels.

`tabularx` tables with automatic-width columns.

`booktabs` professional horizontal rules (`\toprule`, `\midrule`, `\bottomrule`).

`longtable` tables that break across pages.

Floats, graphics and captions

`graphicx` image inclusion.

`float` float placement control (e.g. the `[H]` specifier).

`caption` caption typography (the ABNT caption-on-top layout).

`subcaption` subfigures and subtables.

Code, verbatim and algorithms

`listings` source-code listings.

`fancyvrb` extended verbatim.

`algorithm2e` (`linesnumbered, lined, algochapter, ruled`) typeset algorithms; the natural-language option tracks the main language.

Bibliography, quotations and language

`babel` multilingual typesetting; the language options are computed from the main and foreign language slots.

`csquotes` context-sensitive quotation marks (the recommended companion of `biblatex`).

`biblatex` (`backend=biber`) the bibliography engine and the ABNT `coppe` styles (`coppe` or, with the `numbers` option, `coppe-numeric`). The backend is **biber**, not bibTEX.

Hyperlinks

`hyperref` clickable cross-references, PDF bookmarks and document metadata.

Low-level utilities

`etoolbox` e-TeX programming toolkit.

`ltxcmds` low-level command helpers (`\ltx@ifpackageloaded`, ...).

`ifthen` the `\ifthenelse` conditional.

`hyphenat` hyphenation control (`\nohyphens`, used on the cover).

`lastpage` `\pageref{LastPage}` for the cataloging page count.

`hologo` TeX-family logos (`\hologo{LaTeX}`, ...).

`xcolor` colour support (used by `listings` and `hyperref`).

8 Quick, useful tips

Pictures. The default picture format of LaTeX is the Encapsulated PostScript (EPS). If you use pdfLaTeX, the default format becomes the PDF, but you can equally load PNG files. For such, you must enter the name of your image file without extension, e.g., `\includegraphics{filename}`, and `pdflatex` will firstly look for a file called `filename.pdf` and after for file `filename.png`. For producing high quality pictures with embedded fonts we recommend the Ipe drawing software available here.

Fonts. The default font in LaTeX is the Computer Modern. If you would like to try its enhanced version, consider using the `lmodern` package. To use Times, it is recommended to load the package `mathptmx`, rather than the deprecated `times`. There is also an enhanced Times version available with the `tgtermes` package. You can still use the Arial font face with the package `uarial`.

Hyperref. When working with PDF's, there is the possibility to add extra information to the file as the author's name, document title, subject, keywords, etc. This is easily done with the `hyperref` package, which is loaded by the `coppe` class itself (since v4.0): the user must *not* load it again in the manuscript preamble. Author, title, subject and keyword metadata are filled in automatically by `\maketitle`.

Math (`amssymb` vs. `unicode-math`). The class always loads `amsmath`. The companion AMS symbol package `amssymb`, however, is loaded *only* under `pdfLaTeX`: it is a legacy 8-bit package built around the `msam`/`msbm` Type 1 fonts and clashes with `unicode-math`, the modern OpenType-math package used on Unicode engines (LuaLaTeX, XeLaTeX). On those engines the class skips `amssymb` and schedules an *end-of-document* check: if by then the author has loaded neither `unicode-math` nor `amssymb` (which they could have loaded by hand, though it is discouraged), a `ClassWarning` reminds them to add

```
\usepackage{unicode-math}
\setmathfont{Latin Modern Math}  % or STIX Two Math, ...
```

to the manuscript preamble. The class deliberately does *not* auto-load `unicode-math`, because picking an OpenType math font is a typographic choice that belongs to the author. **Never load `amssymb` together with** `unicode-math`: it would re-establish the 8-bit AMS fonts on top of the OpenType math font and split symbols across two incompatible encodings.

Printing. To get your work correctly printed, you must ensure that any page scaling option (e.g., fit or shrink to printable area) isn't enabled. This kind of option often comes in print dialogs of document visualization softwares.

`longquote` **Quotation** To quote text larger than three lines, according to ABNT, you must increase the left margin to 4 cm, do not use quotation marks, and use a smaller font. The `coppe` class provides the `longquote` environment to easily make these adjustments.

9 A simple example

```
⟨*example⟩
\documentclass[dsc]{coppe}

\usepackage{rotating}       % rodando coisas, como tabelas
\usepackage[most]{tcolorbox} % caixas de texto
\usepackage{tikz}           % gráficos vetoriais (o tcolorbox já o traz)
% Sob LuaLaTeX/XeLaTeX, descomente as duas linhas a seguir para obter os
% símbolos AMS (\mathbb, \hbar, \varnothing, ...) via unicode-math. Sob
% pdfLaTeX, a classe já carrega amssymb automaticamente; deixe comentado.
%\usepackage{unicode-math}
%\setmathfont{Latin Modern Math}    % ou STIX Two Math, TeX Gyre Termes Math...
\DeclareMathOperator{\RMSE}{RMSE}   % operadores próprios (cap. de matemática)
\DeclareMathOperator{\diag}{diag}
% A classe coppe (v4.0) já carrega: amsmath, booktabs, longtable, subcaption,
% hyperref, biblatex, listings, fancyvrb, algorithm2e -- não recarregue esses
% pacotes aqui (em particular, hyperref deve ficar a cargo da classe para que
% a ordem com setspace e biblatex permaneça correta).

\addbibresource{example.bib}% base de referências (biblatex/biber)

\makelosymbols
\makeloabbreviations
% Algumas coisas opcioanais que não cabem no coppetex.sty
%\renewcommand{\familydefault}{\sfdefault} % para usar fontes SEM serifa no default
\renewcommand{\familydefault}{\rmdefault} % para usar fontes COM serifa no default


\begin{document}
```

```
\title{Título da Tese}
\foreigntitle{Thesis Title}
\author{Nome do Autor}{Sobrenome}
\advisor[UFRJ]{Prof.}{Nome do Primeiro Orientador}{Sobrenome}{D.Sc.}
\advisor[UFRJ]{Prof.}{Nome do Segundo Orientador}{Sobrenome}{Ph.D.}
% A classe aceita de um a tres \advisor; se houver apenas um, o rotulo na
% capa e na folha de rosto sai como "Orientador" (singular), com dois ou
% mais sai como "Orientadores" (plural). Para o caso 1/3 orientadores,
% veja tests/test_brazilian_one_advisor.tex e tests/test_brazilian_three_advisors.tex.

\examiner[UFRJ]{Prof.}{Nome do Primeiro Examinador Sobrenome}{D.Sc.}
\examiner[UFRJ]{Prof.}{Nome do Segundo Examinador Sobrenome}{Ph.D.}
\examiner[UFF]{Prof.}{Nome do Terceiro Examinador Sobrenome}{D.Sc.}
\examiner[UNICAMP]{Prof.}{Nome do Quarto Examinador Sobrenome}{Ph.D.}
\examiner[USP]{Prof.}{Nome do Quinto Examinador Sobrenome}{Ph.D.}
\department{PESC}
\date{06}{2026}

\keyword{Primeira palavra-chave}
\keyword{Segunda palavra-chave}
\keyword{Terceira palavra-chave}

% Dados da folha adicional (Coleta CAPES), obrigatoria desde agosto de 2026.
% A folha e preenchida pelo Programa; estes campos apenas a compoem.
\areaconcentracao{Engenharia de Sistemas e Computa\c c\~ao}
\linhapesquisa{Engenharia de Dados e Conhecimento}
\tipoproducao{bibliografica}   % bibliografica | artistica | tecnologica | tecnica
\projetovinculado{sim}         % sim | nao
\nomeprojeto{Nome do Projeto de Pesquisa ao qual o trabalho se vincula}
\agenciafomento{Conselho Nacional de Desenvolvimento Cientifico e
                Tecnologico}{CNPq}
\agenciafomento{Fundacao Carlos Chagas Filho de Amparo a Pesquisa do
                Estado do Rio de Janeiro}{FAPERJ}
% A ficha oficial vem de fichacatalografica.sibi.ufrj.br; tendo o arquivo,
% descomente a linha abaixo e a moldura vazia da folha adicional some.
% \fichacatalografica{ficha.pdf}

\maketitle

\frontmatter
\dedication{A alguém cujo valor é digno desta dedicatória.}

\chapter*{Agradecimentos}

Gostaria de agradecer a todos os que contribu\'iram para este trabalho:
orientadores, colegas, familiares e \`as ag\^encias de fomento. Lorem ipsum
dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
ut labore et dolore magna aliqua.

\begin{abstract}

Apresenta-se, nesta tese, um exemplo de resumo. Lorem ipsum dolor sit amet,
consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
```

```
ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
aspernatur aut odit aut fugit.

\end{abstract}

\begin{foreignabstract}

In this work, we present an example abstract. Lorem ipsum dolor sit amet,
consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
aspernatur aut odit aut fugit.

\end{foreignabstract}

% Ordem pre-textual (NBR 14724/6027, R55): listas (ilustracoes, tabelas,
% abreviaturas e siglas, simbolos) ANTES do sumario; sumario por ultimo.
\listoffigures
\listoftables
\listofquadros
\listofprogramas
\listofalgorithms
\printloabbreviations
\printlosymbols
\tableofcontents

\mainmatter
\chapter{Introdução}

Este é um documento exemplo para o uso da classe CoppeTeX, destinado a ajudar os alunos da

A classe \verb|coppe| foi criada originalmente por Vicente Helano e George Ainsworth, poré
Em 2026, Geraldo Xexéo, com forte ajuda das ferramentas Claude 4.7 Opus e Max e 4.8 Opus M

 Provavelmente Geraldo Xexéo, professor do Programa de Engenharia de Sistemas e Computação

A versão mais atual dessa classe é mantida no GitHub, no repositório \url{https://github.c

Esse documento segue a norma de formatação de teses e dissertações da COPPE. Ele também po

Há um grupo de WhatsApp para perguntas e discussões do estilo: \href{https://github.com/CO
```

```
 Bugs devem ser reportados na \href{https://github.com/COPPE-UFRJ/CoppeTeX/issues}{página d

 Esse documento é usado como exemplo de coisas que podem ser feitas. Ele está configurado p

 É importante de notar que essa classe não foi construída sobre a classe \LaTeX \  para a A

 Essa classe implementa citações ABNT modernas e vários tipos de entrada descritos na norma

 Este documento não substitui, mas complementa, o documento que descreve a classe.



\chapter{Configurações Iniciais}

  A primeira coisa a fazer é escolher o tipo de documento. Isso é feito como uma opção no c

  Como pode ser visto nesse documento, muita coisa pode ser configurada, o que gerará o tra

  Recomendo ler o documento ``The \verb*|coppe| document class'' para entender melhor todas

\section{Linguagem principal do texto}

 Essa classe considera que o texto principal está em português e algumas partes específicas
 \begin{itemize}
 \item A opção \verb|english| deve ser usada no comando \verb|\documentclass|.
 \item A bibliografia segue automaticamente o idioma do Babel (português ou inglês); não é p
 \end{itemize}

 A variação de linguagem, em inglês ou português apenas, já é suportada pela classe \CoppeT


\section{Por que usar o \LaTeX}

 Há uma grande discussão entre usuário de Word e \LaTeX, principalmente, quanto ao uso dess

 Nós escolhemos o \LaTeX por alguns motivos: grande facilidade de seguir um estilo sem se p

 As principais desvantagens são: idiossincrasias que podem gastar tempo, pouco controle sob

\section{Como e onde usar o \LaTeX}

 Existem muitos tutoriais de \LaTeX, mas basicamente, em 2024, ele é usado em dois ambiente
 \begin{enumerate}
 \item Na sua máquina, instalando uma versão completa como o \hologo{MiKTeX}, típico do Win
 \item Usar um ambiente na rede, como o Overleaf.
 \end{enumerate}

 Em todo caso, recomendo fortemente que, ao mesmo tempo, mantenha versões no Git e faça o b


\chapter{Algumas Regras da COPPE}

 Todas abreviaturas e símbolos devem ser definida antes de utilizada. Isso é facilmente fei

```

```
 É imprescindível definir os símbolos, tal como o
 conjunto dos números reais $\mathbb{R}$ e o conjunto vazio $\emptyset$.
 \symbl{$\mathbb{R}$}{Conjunto dos números reais}
 \symbl{$\emptyset$}{Conjunto vazio}. Usamos esse exemplo aqui justamente para mostrar como

 Para as listas de abreviaturas e símbolos funcionarem no Overleaf é necessário rodar o \ve

 Como as listas de símbolos e de abreviaturas usamo o mesmo comando usado para criar índice

\section{Citações}
\label{sec:citacoes}

Citações curtas podem ser feitas \enquote{o comando quote} ou direto com ``duas crases e do

 \begin{longquote}
 Um exemplo de citação longa nas regras da ABNT (4cm de recuo e fonte menor)
 feita com o ambiente  \verb*|longquote| The primary objective of this
 investigation was to determine the feasibility of detecting corrosion in
 aluminum Naval aircraft components with neutron radiographic interrogation
 and the use of standard corrosion penetrameters. Secondary objectives
 included the determination of the effect of object thickness on image quality,
 the defining of minimum levels of detectability and a preliminary investigation
 of a means whereby the degree of corrosion could be quantified with neutron
 radiographic data. \citep{article-example}
 \end{longquote}

 Citações devem apontar as referências. Para isso, está disponível o ótimo pacote \verb*|na

 Em todo caso, \textbf{deve se tomar enorme atenção com as citações, para evitar ocorrer em

 Quando se cita uma obra por meio de outra (\textit{apud}), pela ABNT entra na lista de ref

\chapter{Floats}

Grande parte dos problemas de iniciantes, e veteranos, em \LaTeX é da localização dos \texti

A regra geral de posicionamento é que uma figura ou quadro só pode aparecer a partir da mesm

\textbf{Pela norma atual (manual2024), a legenda} \verb|\caption| \textbf{de toda ilustração

Quadros são ilustrações cujos dados são delimitados por linhas em \emph{todas} as bordas (ao


\section{Tabelas e Figuras Padrão}

Vamos ver uma tabela padrão, como a \autoref{tab:exemplo_numeros}.

\begin{table}[ht]
\centering % Centraliza a tabela
\caption{Exemplo de Tabela de Números}
\label{tab:exemplo_numeros}
\begin{tabular}{ccc} % Define a quantidade de colunas
\hline % Linha superior
\textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ % Cabeçalhos
```

```
\hline % Linha média
1 & 2 & 3 \\ % Primeira linha de dados
\hline
4 & 5 & 6 \\ % Segunda linha de dados
\hline
7 & 8 & 9 \\ % Terceira linha de dados
\hline
10 & 11 & 12 \\ % Quarta linha de dados
\hline % Linha inferior
\end{tabular}
\fonte{Elabora\c{c}\~ao pr\'opria.} % fonte obrigatoria, abaixo do float
\end{table}

Já a \autoref{fig:exemplo_figura} é uma figura padrão, com controle da largura.

\begin{figure}[ht]
\centering % Centraliza a figura
\caption{Exemplo de Figura com Legenda Acima} % Legenda em cima (norma atual)
\label{fig:exemplo_figura} % Etiqueta para referência cruzada
\includegraphics[width=0.5\textwidth]{coppe-logo.pdf} % Inclui a imagem com metade da largur
\source{COPPE/UFRJ.} % fonte obrigatoria, abaixo da figura
\end{figure}

Quadros têm a mesma regra (legenda acima, fonte abaixo), mas com linhas em todas as bordas.

\begin{quadro}[ht]
\centering
\caption{Exemplo de Quadro}
\label{qua:exemplo}
\begin{tabular}{|l|l|}
\hline
\textbf{Ilustração} & \textbf{Delimitação} \\
\hline
Tabela & Linhas apenas no topo e na base. \\
\hline
Quadro & Linhas em todas as bordas. \\
\hline
\end{tabular}
\fonte{Elabora\c{c}\~ao pr\'opria.}
\end{quadro}

\section{Tabelas mais elegantes}

Atualmente a tendência é usar tabelas mais leves, como \autoref{tab:exemplo_numerosbom}. Iss

\begin{table}[ht]
\centering % Centraliza a tabela
\caption{Exemplo de Tabela de Números mais elegantes}
```

```
\label{tab:exemplo_numerosbom}
\begin{tabular}{ccc} % Define a quantidade de colunas
\toprule % Linha superior
\textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ % Cabeçalhos
\midrule % Linha média
1 & 2 & 3 \\ % Primeira linha de dados
4 & 5 & 6 \\ % Segunda linha de dados
7 & 8 & 9 \\ % Terceira linha de dados
10 & 11 & 12 \\ % Quarta linha de dados
\bottomrule % Linha inferior
\end{tabular}
\fonte{Elabora\c{c}\~ao pr\'opria.}
\end{table}

\section{Tabelas Longas ou Largas}

Se sua tabela é muito longa ou larga, existem várias opções.
\begin{itemize}
    \item alterar o tamanho da letra
    \item Usar o longtable
    \item rodar a tabela, fazendo ela em \textit{landscape}
    \item fazer a tabela dentro de um minibox
\end{itemize}


\subsection{Tabelas largas demais}

É comum em teses que as tabelas sejam largas demais. Há várias formas de resolver isso.

A \autoref{tab:tabela_largafns} é larga demais, e nela isso é resolvido diminuindo a fonte p

\begin{table}[ht]
\centering % Centraliza a tabela
\caption{Exemplo de Tabela Larga com Fonte Menor}
\label{tab:tabela_largafns}
\footnotesize % Aplica uma fonte menor para a tabela
\begin{tabular}{cccccccc} % Aumente o número de colunas conforme necessário
\toprule
\textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} & \textbf{Col. 4} & \textbf{Col. 5} & \t
\midrule
Dado 1.1 & Dado 1.2 & Dado 1.3 & Dado 1.4 & Dado 1.5 & Dado 1.6 & Dado 1.7 & Dado 1.8 \\
Dado 2.1 & Dado 2.2 & Dado 2.3 & Dado 2.4 & Dado 2.5 & Dado 2.6 & Dado 2.7 & Dado 2.8 \\
Dado 3.1 & Dado 3.2 & Dado 3.3 & Dado 3.4 & Dado 3.5 & Dado 3.6 & Dado 3.7 & Dado 3.8 \\
\bottomrule
\end{tabular}
\fonte{Elabora\c{c}\~ao pr\'opria.}
\end{table}

O comando \verb|\resizebox{width}{height}{content}| permite ajustar o tamanho de qualquer co

\begin{table}[ht]
\centering
\caption{Exemplo de Tabela Redimensionada}
\label{tab:examplerb}
```

```
\resizebox{\textwidth}{!}{%
\begin{tabular}{llll}
\toprule
Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 \\
\midrule
Dados 1 & Dados 2 & Dados 3 & Dados 4 \\
Dados 5 & Dados 6 & Dados 7 & Dados 8 \\
\bottomrule
\end{tabular}%
}
\fonte{Elabora\c{c}\~ao pr\'opria.}
\end{table}


Para rodar uma tabela muito larga em 90 graus no LaTeX, você pode usar o pacote \verb*|rotat

Aqui está um exemplo de como usar o ambiente \verb*|sidewaystable| para girar uma tabela. Pr

\begin{sidewaystable}
\centering
\caption{Sua Legenda Aqui}
\label{tab:sua_tabela}
\begin{tabular}{lll}
\toprule
Coluna 1 & Coluna 2 & Coluna 3 \\
\midrule
Item 1 & Item 2 & Item 3 \\
Item 4 & Item 5 & Item 6 \\
\bottomrule
\end{tabular}
\fonte{Elabora\c{c}\~ao pr\'opria.}
\end{sidewaystable}

Se a tabela for muito longa, o ambiente \verb|longtable| é o ideal. Ele fornece comandos par

% Exemplo de tabela longa que se estende por várias páginas
\begin{longtable}{|c|c|c|}
% primeiro cabeçalho (é o caption)
\caption{Exemplo de Tabela Longa}\label{tab:longa} \\
\hline \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ \hline
\endfirsthead
% cabeçalho normal
\multicolumn{3}{c}%
{{\tablename\ \thetable{} -- continuação da página anterior}} \\
\hline \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ \hline
\endhead
% pé normal
\hline \multicolumn{3}{|r|}{{Continua na próxima página}} \\ \hline
\endfoot
\hline
% último pé
\multicolumn{3}{|r|}{{Continua na próxima página}}\\
\hline \hline
\endlastfoot
```

```

% Conteúdo da tabela
1 & 2 & 3 \\
4 & 5 & 6 \\
1 & 2 & 3 \\
4 & 5 & 6 \\
1 & 2 & 3 \\
4 & 5 & 6 \\
1 & 2 & 3 \\
4 & 5 & 6 \\
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```
4 & 5 & 6 \\
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4 & 5 & 6 \\
1 & 2 & 3 \\
4 & 5 & 6 \\
1 & 2 & 3 \\
```

```
4 & 5 & 6 \\

% Repetir linhas semelhantes conforme necessário para estender a tabela por 3 páginas
\end{longtable}
\fonte{Elabora\c{c}\~ao pr\'opria.}% a longtable não é float: a fonte vem após o ambiente

\section{Subfiguras (subcaption)}

Para colocar várias imagens (ou tabelas) lado a lado, cada uma com sua própria legenda, use

\begin{figure}[ht]
  \centering
  \begin{subfigure}[b]{0.42\textwidth}
    \centering\includegraphics[width=\linewidth]{example-image-a}
    \caption{Primeira vista}\label{fig:sub-a}
  \end{subfigure}
  \hfill
  \begin{subfigure}[b]{0.42\textwidth}
    \centering\includegraphics[width=\linewidth]{example-image-b}
    \caption{Segunda vista}\label{fig:sub-b}
  \end{subfigure}
  \caption{Figura com duas subfiguras}
  \label{fig:subs}
  \fonte{Elabora\c{c}\~ao pr\'opria.}
\end{figure}

Referencie a figura inteira com \verb|\autoref{fig:subs}|. Cada parte tem
rótulo próprio: \verb|\autoref{fig:sub-a}| leva à \autoref{fig:sub-a}.

\section{Imagens (graphicx)}

A classe já carrega o \verb|graphicx|. Insira imagens com \verb|\includegraphics|; as opções
\begin{itemize}
  \item \verb|[width=0.8\textwidth]| ou \verb|[scale=0.5]| --- tamanho;
  \item \verb|[angle=90]| --- rotação;
  \item \verb|[trim=l b r t, clip]| --- recorte (corta as bordas \emph{esquerda, baixo, dire
\end{itemize}
Formatos aceitos com \hologo{pdfLaTeX}/\hologo{LuaLaTeX}: \texttt{pdf}, \texttt{png}, \textt

\section{Introdução ao tikz}

O \verb|tikz| (disponível também via \verb|tcolorbox|) desenha gráficos vetoriais no própric

\begin{figure}[ht]
  \centering
  \begin{tikzpicture}
    \node[draw, rounded corners] (a) at (0,0) {Início};
    \node[draw, rounded corners] (b) at (4,0) {Fim};
    \draw[->, thick] (a) -- (b) node[midway, above] {processo};
  \end{tikzpicture}
  \caption{Diagrama simples em tikz}
  \label{fig:tikz}
  \fonte{Elabora\c{c}\~ao pr\'opria.}
\end{figure}
```

```
Para gráficos de dados (curvas, barras), veja o \verb|pgfplots|, construído sobre o \verb|ti

\section{Float ou direto no texto?}

Colocar a tabela ou imagem dentro de \verb|figure|/\verb|table| (um \emph{float}) deixa o \L

 \chapter{Using BibLaTeX}
 \label{cap:biblatex}

 Esta classe gera as referências com BibLaTeX e o \emph{backend} \texttt{biber}, no padrão

 \textbf{É obrigatório compilar com \texttt{biber}} (e não com \hologo{BibTeX}). A sequênci

 Os comandos do \texttt{natbib} (\verb|\citet|, \verb|\citep|) continuam disponíveis porque

 \begin{table}[h]
 \caption{Exemplos de cita{\c c}\~oes utilizando o comando padr\~ao
   \texttt{\textbackslash cite} do \LaTeX\ e
   o comando \texttt{\textbackslash citep},
   disponível pela opção \texttt{natbib} do BibLaTeX.}
 \label{tab:citation}
 \centering
 {\footnotesize
 \begin{tabular}{|c|c|c|}
   \hline
   Tipo da Publicação & \verb|\cite| & \verb|\citep|\\
   \hline
   Livro & \cite{book-example} & \citep{book-example}\\
   Artigo & \cite{article-example} & \citep{article-example}\\
   Relatório & \cite{techreport-example} & \citep{techreport-example}\\
   Relatório & \cite{techreport-exampleIn} & \citep{techreport-exampleIn}\\
   Anais de Congresso & \cite{inproceedings-example} &
     \citep{inproceedings-example}\\
   Séries & \cite{incollection-example} & \citep{incollection-example}\\
   Em Livro & \cite{inbook-example} & \citep{inbook-example}\\
   Dissertação de mestrado & \cite{mastersthesis-example} &
     \citep{mastersthesis-example}\\
   Tese de doutorado & \cite{phdthesis-example} & \citep{phdthesis-example}\\
   \hline
 \end{tabular}}
 \fonte{Elabora\c{c}\~ao pr\'opria.}
 \end{table}

\begin{table}[h]
 \caption{Exemplos de cita{\c c}\~oes utilizando o comando padr\~ao
   \texttt{\textbackslash cite} do \LaTeX\ e
   o comando \texttt{\textbackslash citet},
   disponível pela opção \texttt{natbib} do BibLaTeX. Além disso, usando o booktabs.}
 \label{tab:citation1}
 \centering
 {\footnotesize
 \begin{tabular}{ccc}
   \toprule
```

```
  Tipo da Publicação & \verb|\cite| & \verb|\citet|\\
  \midrule
  Livro & \cite{book-example} & \citet{book-example}\\
  Artigo & \cite{article-example} & \citet{article-example}\\
  Relatório & \cite{techreport-example} & \citet{techreport-example}\\
  Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\\
  Anais de Congresso & \cite{inproceedings-example} &
    \citet{inproceedings-example}\\
  Séries & \cite{incollection-example} & \citet{incollection-example}\\
  Em Livro & \cite{inbook-example} & \citet{inbook-example}\\
  Dissertação de mestrado & \cite{mastersthesis-example} &
    \citet{mastersthesis-example}\\
  Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\\
  \bottomrule
\end{tabular}}
\fonte{Elabora\c{c}\~ao pr\'opria.}
\end{table}

\section{Comandos de citação}

Os principais comandos são: \verb|\cite| (referência simples), \verb|\citet| (autor por ex

\subsection{Parâmetro opcional: página e ‘‘tradução nossa’’}

Todos os comandos aceitam um parâmetro opcional para a localização (página, seção) e obser

\subsection{Citação de citação (\emph{apud})}

Para citar uma obra através de outra, use \verb|\citetapud| e \verb|\citepapud| (demonstra

\subsection{Múltiplas bibliografias}

Para separar as referências (por tipo, capítulo ou tema), use o ambiente \verb|refsection|

\section{Tipos de entrada bibliográfica}

Cada tipo de documento da ABNT corresponde a um tipo de entrada do BibLaTeX, com um sinôni

\subsection{Tipos acadêmicos principais}
\begin{itemize}
  \item Livro / monografia --- \verb|@book| (\verb|@livro|)
  \item Artigo de periódico --- \verb|@article| (\verb|@artigo|); matéria de jornal --- \v
  \item Capítulo de livro --- \verb|@incollection| (\verb|@partedelivro|), \verb|@inbook|
  \item Evento no todo (anais) --- \verb|@proceedings| (\verb|@anais|); trabalho em evento
  \item Tese/dissertação --- \verb|@thesis| (\verb|@tese|, \verb|@dissertacao|); com tipo
  \item Relatório --- \verb|@report| (\verb|@relatorio|, \verb|@relatoriotecnico|); periód
  \item Manual --- \verb|@manual|; diversos --- \verb|@misc| (\verb|@diverso|); página/sit
\end{itemize}

\subsection{Tipos especiais (ABNT §4.2.5--4.2.14)}
\begin{itemize}
  \item Patente --- \verb|@patent| (\verb|@patente|)
  \item Legislação / ato normativo --- \verb|@legislation| (\verb|@legislacao|, \verb|@ato
  \item Norma técnica --- \verb|@standard| (\verb|@norma|); partitura --- \verb|@music| (\
```

```
  \item Documento cartográfico (mapa) --- \verb|@map| (\verb|@mapa|, \verb|@cartografico|)
  \item Audiovisual --- \verb|@video| (\verb|@audiovisual|); filme --- \verb|@movie| (\ver
 \end{itemize}

 \subsection{Jogos e TV (tipos próprios da CoppeTeX)}
 \begin{itemize}
  \item Jogo --- \verb|@game| (\verb|@jogo|); de tabuleiro --- \verb|@boardgame| (\verb|@j
  \item Programa de TV / série --- \verb|@tvshow| (\verb|@programadetv|, \verb|@serietv|)
 \end{itemize}

 Os campos também aceitam sinônimos em português (sem acentos): \verb|autor|, \verb|titulo|

\chapter{Alguns outros exemplo úteis}

\begin{tcolorbox}[title=Meu Textbox]
Este é o conteúdo do meu textbox. Você pode adicionar qualquer texto aqui, bem como incluir
\end{tcolorbox}

\begin{tcolorbox}
Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
\end{tcolorbox}

\begin{figure}[ht]
  \centering
  \begin{tikzpicture}
    \node[anchor=south west,inner sep=0] (image) at (0,0) {\includegraphics[width=0.5\te
    \begin{scope}[x={(image.south east)},y={(image.north west)}]
      % Definindo o textbox dentro da figura
      \node[anchor=north west, text width=0.3\textwidth, fill=white, opacity=0.7, text
        \begin{tcolorbox}[colback=red!5!white,colframe=red!75!black,title=Textbox de
          Este textbox fala sobre como inserir um textbox dentro de uma figura usa
        \end{tcolorbox}
      };
    \end{scope}
  \end{tikzpicture}
  \caption{Figura com Textbox}
  \label{fig:figura_com_textbox1}
  \fonte{Elabora\c{c}\~ao pr\'opria.}
\end{figure}


\begin{figure}[ht]
  \centering
 \begin{tcolorbox}
Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
\end{tcolorbox}
  \caption{Figura com Textbox simples}
  \label{fig:figura_com_textbox}
  \fonte{Elabora\c{c}\~ao pr\'opria.}
\end{figure}

\section{Listagens de código e algoritmos}

O CoppeTeX traz embutido o suporte a listagens de programas (pacote \verb|listings|) e a alg
```

```

\begin{python}[caption={Exemplo de programa em Python},label=prog:exemplo]
def saudacao(nome):
    # imprime uma saudação acentuada
    print(f"Olá, {nome}!")

saudacao("COPPE")
\end{python}
\source{Elaboração própria.}

Para inserir código no meio do texto use \verb|\pythoninline|, como em \pythoninline{saudaca

\begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
Olá, COPPE!
\end{Verbatim}

Há estilos prontos também para \verb|java|, \verb|xml|, \verb|html| e \verb|prolog|.
Há também \verb|\pythoninline| e \verb|\javainline| para código no meio do
texto, e o ambiente \verb|Verbatim| (com as opções acima) para a saída de programas. Para al

\begin{algorithm}[H]
\caption{Soma dos elementos de um vetor}
\label{alg:soma}
\Dados{vetor $v$ com $n$ elementos}
\Resultado{soma $s$ dos elementos de $v$}
$s \leftarrow 0$\;
\Para{$i \leftarrow 1$ \Ate{} $n$}{
  $s \leftarrow s + v[i]$\;
}
\Retorna $s$\;
\end{algorithm}

  \chapter{Escrevendo matemática}

  A classe carrega \texttt{amsmath} em qualquer motor. Sob \hologo{pdfLaTeX} ela também carr

  \section{Alinhamento (align)}
  O ambiente \verb|align| alinha equações pelo \verb|&| e numera cada linha (\verb|\\| quebr
  \begin{align}
    f(x) &= (x+1)^2 \\
         &= x^2 + 2x + 1 .
  \end{align}

  \section{Matrizes (matrix)}
  O \texttt{amsmath} traz \verb|pmatrix|, \verb|bmatrix|, \verb|vmatrix|, entre outros:
  \begin{equation}
    A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}
    \qquad
    \det A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}
  \end{equation}

  \section{Subequações (subequations)}
  Para numerar um grupo de equações sob o mesmo número, com letras (a), (b):
  \begin{subequations}
```

```
\begin{align}
  a &= b + c , \\
  d &= e - f .
\end{align}
\end{subequations}

\section{Estruturas alinhadas (array)}
O \verb|array| monta estruturas dentro do modo matemático, como definições por partes:
\begin{equation}
  |x| = \left\{ \begin{array}{ll}
    x,  & \text{se } x \ge 0 , \\
    -x, & \text{caso contrário.}
  \end{array} \right.
\end{equation}

\section{Operadores (\textbackslash DeclareMathOperator)}
Para operadores com a tipografia correta (romano, espaçamento adequado), declare-os no pre
\begin{verbatim}
\usepackage{amsmath}
\DeclareMathOperator{\RMSE}{RMSE}
\DeclareMathOperator{\diag}{diag}
\end{verbatim}
Assim:
\begin{itemize}
  \item \verb|$\RMSE(y,\hat{y})$| produz $\RMSE(y,\hat{y})$;
  \item \verb|$\diag(\lambda_1,\lambda_2)$| produz
        $\diag(\lambda_1,\lambda_2)$.
\end{itemize}

\section{Unicode e LuaLaTeX (unicode-math)}
\label{sec:unicode-math}
Ao migrar para \hologo{LuaLaTeX} (ou \hologo{XeLaTeX}), \textbf{não carregue} \texttt{amss
\begin{verbatim}
\usepackage{amsmath}      % a classe já carrega
\usepackage{unicode-math} % NÃO use \usepackage{amssymb} junto
\setmathfont{Latin Modern Math}  % ou STIX Two Math
\end{verbatim}
O código clássico continua valendo ($\mathbb{R}$, $\int_a^b f(x)\,dx$, $\alpha$) e você ai

\chapter{Referências cruzadas, notas e índices}
\label{cap:refs}

Este capítulo reúne os recursos para \emph{apontar} para coisas: rótulos, notas de rodapé,

\section{Rótulos e referências cruzadas}

Marque um elemento com \verb|\label{tipo:nome}| e aponte para ele com \verb|\ref| (só o nú

\section{Notas de rodapé}

Use \verb|\footnote{texto}|\footnote{Como esta nota, que a classe imprime em espaçamento s

\section{hyperref e marcadores}

```

```
A classe carrega o \verb|hyperref|: o sumário, as citações e os \verb|\ref|/\verb|\autoref

\section{URLs}

Para endereços, use \verb|\url{https://exemplo.org}|: o \verb|hyperref| cria o link e queb
use \verb|\href{https://exemplo.org}{texto}|. Evite digitar URLs como texto comum --- elas n

\section{Índice remissivo}

Marque os termos com \verb|\index{termo}| e imprima o índice com \verb|\printindex| (carre

\section{Traduções (NBR 10520)}

Ao citar um trecho que você mesmo traduziu, coloque a tradução no corpo do texto e escreva

\chapter{Truques de \LaTeX}

\section{Comentários mágicos (\texttt{\% !TeX})}
Linhas \verb|% !TeX| no topo do arquivo dizem ao editor (\hologo{TeX}studio, Overleaf, \ho
\begin{verbatim}
% !TeX program = lualatex
% !TeX encoding = UTF-8
% !TeX spellcheck = pt_BR
% !BIB program = biber
\end{verbatim}
Com isso o documento compila igual em qualquer ambiente. O \texttt{latexmkrc} cumpre papel

\section{Espaço engolido}
Um comando sem argumento engole o espaço seguinte: \verb|\LaTeX é bom| sai como ‘‘\LaTeX é

\section{Escopo (grupos)}
Chaves \verb|{...}| (ou \verb|\begingroup|\dots\verb|\endgroup|) criam um escopo: mudanças

\section{Descarregar figuras: \texttt{\textbackslash clearpage} × \texttt{\textbackslash n
\verb|\newpage| apenas quebra a página e leva os floats pendentes adiante, acumulando-os.

\section{\texttt{latexmkrc}, perl e Overleaf}
O \verb|latexmk| automatiza a compilação (roda \LaTeX, \verb|biber| e \verb|makeindex| qua

\chapter{Método Proposto}
\chapter{Resultados e Discuss\~oes}

\section{Algumas Demonstra{\c c}\~oes}

A  Lista de Símbolos precisa usar comandos específicos. Aqui vamos usar os símbolos $\alph
\symbl[beta]{Beta}{A palavra Beta more e corrigida}
\symbl[zzbeta]{$\beta$}{A letra $\beta$ corrigida}
\symbl{beta}{A palavra beta}
\symbl{alpha}{A palavra alpha}
\symbl[alpha]{Alpha}{A palavra Alpha}
\symbl[zzalpha]{$\alpha$}{A letra $\alpha$ corrigida}
\symbl[marco]{Marco}{A palavra Marco corrigida}

A Lista de Abreviações segue, a partir de 2024, a mesma regra, e aqui seguem alguns exempl
```

```
  \abbrev{GoT}{Game of Thrones}
  \abbrev[GOT]{GoT}{Game of Thrones ordenado como GOT}
  \abbrev[iot]{IoT}{IoT ordenado como iot}
  \abbrev[IoT]{IoT}{IoT ordenado como IoT}
  \abbrev[IOT]{IoT}{IoT ordenado como IOT}
  \abbrev{IoT}{IoT com ordenação default}
  \abbrev[ITU]{ITU}{ITU mesmo}

  \newsigla{ufrj}{UFRJ}{Universidade Federal do Rio de Janeiro}
  \newsigla{cpgp}{CPGP}{Comissão de Pós-Graduação e Pesquisa de Engenharia}

  As siglas (manual2024, seção 2.8) são expandidas na primeira ocorrência e registradas auto




  \chapter{Conclusões}

  \backmatter

  % Tipos exóticos da ABNT (NBR 6023): patente, legislação, norma, mapa,
  % partitura --- declarados via sinônimos em português em example.bib.
  \nocite{patente-example,legislacao-example,norma-example,mapa-example,partitura-example}
  \nocite{jurisprudencia-example}
  \nocite{videogame-example,filme-example,semautor-example,semautorart-example}
  \nocite{jornalcaderno-example,jornal-example}
  \printbibliography



\appendix

\chapter{Um apêndice}

Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilizaçãc

Apêndice: O apêndice é um texto ou documento elaborado pelo autor do trabalho com o objetivo


\chapter{Outro apêndice}

\annex


\chapter{Um Anexo}
Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilizaçãc



Anexo: O anexo, por sua vez, consiste em um texto ou documento não elaborado pelo autor, que

No modelo \CoppeTeX os anexos devem obrigtoriamente vir depois dos apêndices e usam o comand
```

```
\chapter{Outro Anexo}


\coppetexfinalpage% colofão institucional (página par, face do contracapa)

\end{document}

⟨/example⟩
```

10 Implementation

10.1 The ‘coppe.cls’ file

```
⟨*class⟩
\def\filename{coppe.dtx}
\def\fileversion{v4.0}
\def\filedate{2026/05/28}
\NeedsTeXFormat{LaTeX2e}[1995/12/01]
\ProvidesClass{coppe}[\filedate\ \fileversion\ COPPE Dissertations and Thesis]
% Save the class version and date into private macros immediately, before any
% required package can redefine the very common \fileversion / \filedate
% global names (LaTeX has no convention reserving them for the host class).
% \coppe@version / \coppe@date are what user-facing macros (e.g.
% \coppetexfinalpage) should reference.
\let\coppe@version\fileversion
\let\coppe@date\filedate
% Document date stored in private slots so \date can set them without ever
% touching TeX's \month/\year primitives (which would break \today). They
% default to the current system month/year, so a document without \date
% still renders sensibly on the cover and folha de rosto.
\edef\coppe@docmonth{\the\month}
\edef\coppe@docyear{\the\year}
\LoadClass[12pt,a4paper,oneside]{book}
% setspace, hyperref and the bibliography engine (biblatex+biber) are loaded at
% the END of the class (after every class hook is defined), in that order:
% setspace -> hyperref -> biblatex. hyperref must come late so all the macros
% it patches (TOC, refs, lists) are already defined, and biblatex must come
% after hyperref so its citation links and back-references work.
\RequirePackage{hyphenat}
\RequirePackage{lastpage}
\RequirePackage{ifthen}
\RequirePackage{graphicx}
\RequirePackage{tabularx}
\RequirePackage{booktabs}      % regras de tabela profissionais (toprule/midrule/bottomrule)
\RequirePackage{longtable}     % tabelas que quebram entre páginas
\RequirePackage{etoolbox}
\RequirePackage{ltxcmds}
\RequirePackage{hologo}        % TeX-family logos: \hologo{TeX}, \hologo{LaTeX}, ...
% Render \TeX, \LaTeX and friends via hologo (\CoppeTeX is a custom logo, kept).
\def\TeX{\hologo{TeX}}
\def\LaTeX{\hologo{LaTeX}}
\def\LaTeXe{\hologo{LaTeXe}}
\def\BibTeX{\hologo{BibTeX}}
\def\pdfLaTeX{\hologo{pdfLaTeX}}
```

```
\def\pdfTeX{\hologo{pdfTeX}}
\def\LuaLaTeX{\hologo{LuaLaTeX}}
\def\XeLaTeX{\hologo{XeLaTeX}}
\def\MiKTeX{\hologo{MiKTeX}}
% Engine-aware UTF-8 input: pdfLaTeX needs inputenc(utf8); LuaLaTeX (and XeTeX)
% are UTF-8 native, so inputenc must NOT be loaded there (it would clash). Source
% files may use accented Latin characters (áéíóú) directly. Under pdfLaTeX we also
% suggest LuaLaTeX, which handles Unicode/fonts better (at the cost of speed).
\RequirePackage{iftex}
\ifPDFTeX
  \RequirePackage[utf8]{inputenc}
  \AtEndOfClass{\ClassWarningNoLine{coppe}{%
    Although slower, you should try compiling with LuaLaTeX}}
\fi
% lmodern MUST come with the T1 encoding. Without it, T1 falls back to the EC
% fonts, which exist only as METAFONT bitmaps in most installations, and the
% whole document is typeset in Type3 bitmap fonts. Measured before this change:
% example.pdf carried 87 Type3 fonts against 6 Type1, and coppe.pdf 133 against
% 2. The consequences are not cosmetic:
%
%   * the text renders blurred at any zoom and prints poorly;
%   * copying and searching text in the PDF is unreliable;
%   * PDF/A -- required for deposit by 2.2(d) of the 2026 manual -- is not
%     attainable with bitmap text.
%
% Latin Modern is the vector successor of Computer Modern, metrically and
% visually near-identical, and ships with every current distribution, so the
% page layout does not move.
\RequirePackage{lmodern}
\RequirePackage[T1]{fontenc}
% Math support. amsmath is universal and works on every engine, so the class
% brings it in unconditionally (it must precede hyperref, which is loaded at
% the end of the class). amssymb is the legacy AMS symbol package, built
% around the 8-bit Type 1 fonts msam/msbm; on a Unicode engine (LuaLaTeX or
% XeLaTeX) it clashes with unicode-math, which already provides all of its
% symbols through Unicode math codes. So amssymb is loaded ONLY under
% pdfLaTeX; Unicode-engine users get a ClassWarning pointing them at
% unicode-math (the class does not auto-load unicode-math because choosing
% an OpenType math font is a typographic decision that belongs to the author).
\RequirePackage{amsmath}
\ifPDFTeX
  \RequirePackage{amssymb}
\else
  % Unicode engine: defer the check to end of document. Only warn if the
  % user did NOT add either unicode-math (the modern OpenType-math route)
  % or amssymb (the legacy route, which they would have loaded manually).
  % This avoids nagging authors who already followed the documented
  % \usepackage{unicode-math} workflow.
  \AtEndDocument{%
    \@ifpackageloaded{unicode-math}{}{%
      \@ifpackageloaded{amssymb}{}{%
        \ClassWarningNoLine{coppe}{%
          You are compiling with a Unicode engine (LuaLaTeX or XeLaTeX) and\MessageBreak
          neither unicode-math nor amssymb is loaded. AMS-style symbols\MessageBreak
```

```
          (\string\mathbb, \string\hbar, \string\varnothing, ...) will be undefined.\Message
          Add to your preamble:\MessageBreak
          \space\space\string\usepackage{unicode-math}\MessageBreak
          \space\space\string\setmathfont{Latin Modern Math}\MessageBreak
          Do NOT load amssymb together with unicode-math -- it would clash\MessageBreak
          with the OpenType math font setup}}}}
\fi
% Margins per the SiBI manual, 9th rev. ed. (2026), 2.3: left 3cm, top 3cm,
% right 2cm, bottom 2cm -- ONE-SIDED. The 2025 edition also specified verso
% margins (right 3cm / left 2cm) and recommended printing the body on both
% sides; the 2026 revision DELETED the verso margins entirely, following CEPG
% Resolution 246/2023, which ended the printed deposit. A digital-only document
% has no gutter, so the mirrored layout (previously inner=outer=2cm plus a 1cm
% bindingoffset) no longer has any normative basis and is now opt-in via the
% \texttt{twoside} class option, handled after \ProcessOptions below.
%
% headheight=15pt is given here (not patched later) so geometry's own page
% arithmetic stays self-consistent: fancyhdr needs >= 14.49998pt of header
% height to host the running head, and the standard 12pt default triggers
% its "headheight is too small" warning. Letting geometry know up front
% keeps \topmargin a derived value -- we never touch it by hand.
% (geometry has no \texttt{oneside} key -- only \texttt{twoside}; with the key
% absent it follows the class, which \LoadClass set to one-sided above.)
% headsep is given explicitly to place the folio. 2.7 measures it the way it
% measures the right margin -- to the digits themselves: "a 2cm da borda
% superior, ficando o ultimo algarismo a 2cm da borda direita". The right edge
% already landed at 2.02cm; the top of the digits sat at 1.87cm, too high.
% geometry derives \topmargin from top - 1in - headheight - headsep, so
% shortening headsep lowers the header and leaves the body top untouched:
% topmargin + headheight + headsep is 13.088pt either way, and 3cm is where the
% body still starts. The value is CALIBRATED, not computed -- the folio sits a
% little lower than the arithmetic alone predicts, because fancyhdr's box does
% not put the digits flush with the top of \headheight. Two measured points,
% taken from the rendered page rather than from the PDF's font metrics:
%   headsep 18.07pt -> digits at 1.8669cm      headsep 14.29pt -> 2.0616cm
% which put 2.0000cm at 15.49pt. Measured back on the rendered page, the top of
% the digits and their right edge now come out at the same 2.019cm -- and the
% right margin is 2cm by construction, so that shared 0.019cm is the ruler's
% own bias, not a residual error. Change the body font size or headheight and
% this number has to be measured again, not scaled.
\RequirePackage[a4paper,headheight=15pt,headsep=15.49pt,%
top=3cm,bottom=2cm,left=3cm,right=2cm]{geometry}
% The first paragraph of every section must be indented like any other. LaTeX's
% default is the Anglo-Saxon convention -- no indent after a heading -- which is
% not what Brazilian typography (nor the SiBI manual's own examples) uses.
% indentfirst was mentioned only as a commented suggestion in the example
% preamble, so every document built with this class came out with the first
% paragraph of each section flush left. Loading it in the class fixes it for
% everybody instead of asking each author to remember.
%
% Loaded BEFORE titlesec so titlesec's \@afterindent handling sees the patched
% \@afterindentfalse.
% Long unbreakable tokens -- \verb|...| in running prose, command names, URLs --
% cannot be hyphenated, so TeX overfills the line rather than stretching the
```

```
% spaces. example.tex alone produced 17 overfull boxes this way, the worst 86pt,
% which is the "text does not always respect the margins" a recent review
% reported. \emergencystretch gives TeX a final pass in which it may stretch
% interword spaces instead of running into the margin. It cannot save a single
% token longer than the measure, but it clears the ordinary cases.
\setlength\emergencystretch{3em}
\RequirePackage{indentfirst}
\RequirePackage{titlesec}
% --- Section heading formats (manual2024 2.6 / proposta P22) -----------------
% chapter (numbered): "N  TITLE", ALL CAPS, bold, flush left (no "Capitulo N").
% APENDICE/ANEXO chapters print "APENDICE A -- Title" (R75/R76); normal chapters
% keep "N Title". \appendix and \annex raise the flag (see below).
\newif\if@coppeappendix \@coppeappendixfalse
\titleformat{\chapter}[hang]
  {\normalfont\Large\bfseries}
  {\if@coppeappendix\MakeUppercase{\appendixname}\ \thechapter\ \textendash\else\thechapter\
  {1ex}{\MakeUppercase}
\titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}
\let\coppe@orig@appendix\appendix
\renewcommand\appendix{\coppe@orig@appendix\@coppeappendixtrue}
% Sumario (TOC) entries for APENDICE/ANEXO chapters read "APENDICE A -- Title"
% / "ANEXO A -- Title" (R75/R76), not the bare "A". The label word is chosen by
% which macro is written (so it is correct at .toc-render time even though the
% appendixname has changed back); the chapter letter is baked in at write time.
\newif\if@coppeannex \@coppeannexfalse
\newcommand\coppe@tocapp[1]{%
  \MakeUppercase{\coppestring{appendix}}%
  \nobreakspace#1\nobreakspace\textendash\nobreakspace}
\newcommand\coppe@tocanx[1]{%
  \MakeUppercase{\coppestring{annex}}%
  \nobreakspace#1\nobreakspace\textendash\nobreakspace}
\def\@chapter[#1]#2{%
  \ifnum \c@secnumdepth >\m@ne
    \if@mainmatter
      \refstepcounter{chapter}%
      \typeout{\@chapapp\space\thechapter.}%
      \if@coppeappendix
        \if@coppeannex
          \addcontentsline{toc}{chapter}{\protect\coppe@tocanx{\thechapter}#1}%
        \else
          \addcontentsline{toc}{chapter}{\protect\coppe@tocapp{\thechapter}#1}%
        \fi
      \else
        \addcontentsline{toc}{chapter}{\protect\numberline{\thechapter}#1}%
      \fi
    \else
      \addcontentsline{toc}{chapter}{#1}%
    \fi
  \else
    \addcontentsline{toc}{chapter}{#1}%
  \fi
  \chaptermark{#1}%
  \addtocontents{lof}{\protect\addvspace{10\p@}}%
  \addtocontents{lot}{\protect\addvspace{10\p@}}%
```

```
  \if@twocolumn
    \@topnewpage[\@makechapterhead{#2}]%
  \else
    \@makechapterhead{#2}%
    \@afterheading
  \fi}
% chapter (unnumbered): centred ALL CAPS bold (errata, lists, resumo, sumario,
% referencias, glossario, ... -- unnumbered headings are centred, P29-P30).
\titleformat{name=\chapter,numberless}[hang]
  {\normalfont\Large\bfseries\filcenter}{}{0pt}{\MakeUppercase}
\titlespacing*{name=\chapter,numberless}{0pt}{10pt}{1.5\baselineskip}
% section: ALL CAPS, normal weight -- the \bfseries here contradicted this very
% comment until v4.1, and made the secondary level as heavy as the primary one,
% defeating the "destaque gradativo" of 2.6. The gradation is now
% chapter = ALL CAPS bold, section = ALL CAPS regular, subsection = Title Case
% bold, subsubsection = Title Case bold italic, as read from the manual.
\titleformat{\section}
  {\normalfont\large}{\thesection}{1ex}{\MakeUppercase}
% subsection: bold (author writes Title Case).
\titleformat{\subsection}
  {\normalfont\bfseries}{\thesubsection}{1ex}{}
% subsubsection: bold italic.
\titleformat{\subsubsection}
  {\normalfont\bfseries\itshape}{\thesubsubsection}{1ex}{}
% paragraph = quinary section: italic, the author capitalising only the first
% word. 2.6 recommends going as deep as the quinary, and the annotation on the
% 2025 manual (specs/ANOTACOES.md) describes exactly this shape.
%
% book makes \paragraph a RUN-IN heading, so the text continued on the same
% line -- 2.6 requires "o texto deve ser iniciado em outra linha". The positive
% "after" spacing in \titlespacing turns it into a displayed heading.
\titleformat{\paragraph}[hang]
  {\normalfont\itshape}{\theparagraph}{1ex}{}
\titlespacing*{\paragraph}{0pt}{2.5ex plus 1ex minus .2ex}{1.5ex plus .2ex}
% book defaults secnumdepth to 2, so \subsubsection and \paragraph came out
% UNNUMBERED however carefully \titleformat named \thesubsubsection. 2.6 has
% the numbering following the whole chain down to the quinary, and the manual's
% own sumario is four levels deep, so tocdepth follows.
\setcounter{secnumdepth}{4}
\setcounter{tocdepth}{4}
% 2.6: "no sumario, as secoes devem ser grafadas conforme apresentadas no corpo
% do trabalho". The body sets chapter and section in ALL CAPS (via the
% \MakeUppercase in their \titleformat), but that case change belongs to the
% printed heading alone -- the .toc keeps the title as typed, so the sumario
% came out in ordinary case and contradicted the body.
%
% \l@chapter and \l@section are the two TOC line formatters; wrapping their
% entry argument is enough, and leaves \l@subsection and below untouched, which
% is what we want since those levels are NOT uppercased in the body either.
% The entry carries \numberline{N} plus the title; \MakeUppercase passes the
% digits through unchanged and leaves robust commands (including hyperref's
% link wrappers) intact.
\let\coppe@orig@lchapter\l@chapter
\renewcommand\l@chapter[2]{\coppe@orig@lchapter{\MakeUppercase{#1}}{#2}}
```

```
\let\coppe@orig@lsection\l@section
\renewcommand\l@section[2]{\coppe@orig@lsection{\MakeUppercase{#1}}{#2}}
% --- Captions (manual2024 2.10-2.11 / proposta P6): smaller font, ABNT
%     "Figura 1 -- titulo" travessao separator (sources handled by \source). ---
\RequirePackage{caption}
% justification=centering makes MULTI-LINE captions centre like the one-line
% ones. Without it, caption's singlelinecheck centres a caption that fits on one
% line and justifies one that wraps, so in a document with mixed caption lengths
% a couple of tables looked left-aligned while the rest looked centred -- which
% is what a recent review noticed about tables 5.1 and 5.2. 2.10 does not
% prescribe the alignment; it does not tolerate two alignments either.
\captionsetup{font=footnotesize,labelsep=endash,justification=centering}
% subcaption (NÃO subfig/subfigure) for subfigures / subtables; load right after
% caption so it picks up the ABNT captionsetup above.
\RequirePackage{subcaption}
% --- Floats: caption on TOP and a mandatory "Fonte:" line at the bottom.
%     manual2024 2.10-2.11 (R46/R51: identification on top; R48/R52: the source
%     at the bottom is mandatory). The float package's plaintop style forces the
%     caption above the body for figure, table and the new quadro float. -------
\RequirePackage{float}
\floatstyle{plaintop}
\restylefloat{figure}
\restylefloat{table}
% New "quadro" float -- "Quadro" (pt) / "Frame" (en) (R53) -- with its own list
% file (.loq). A quadro's data is bordered on all sides (drawn by the author),
% unlike a tabela (rules on top and bottom only).
\providecommand{\quadroname}{\coppestring{frame}}
\providecommand{\listquadroname}{\coppestring{listframe}}
\newfloat{quadro}{htbp}{loq}[chapter]
\floatname{quadro}{\quadroname}
\providecommand{\quadroautorefname}{\quadroname}
% Format the Lista de Quadros entries exactly like the Lista de Figuras ones
% (float's own \l@quadro lays the number/title out incorrectly).
\let\l@quadro\l@figure
\newcommand\listofquadros{%
  \chapter*{\listquadroname}%
  \addcontentsline{toc}{chapter}{\listquadroname}%
  \@mkboth{\MakeUppercase\listquadroname}{\MakeUppercase\listquadroname}%
  \@starttoc{loq}}
% \source / \fonte: the illustration source (ABNT-mandatory). Typeset below the
% float as a smaller, single-spaced, centred line; it does NOT advance any
% caption counter. \cpsource/\cpfonte are clash-safe aliases (the rename rule);
% the label is "Fonte" (pt) / "Source" (en).
\newcommand{\cpsourcename}{\coppestring{source}}
\newcommand{\cpsource}[1]{%
  \par\addvspace{\smallskipamount}%
  {\footnotesize\singlespacing\centering\cpsourcename: #1\par}}
\let\cpfonte\cpsource
\@ifundefined{source}{\let\source\cpsource}{}%
\@ifundefined{fonte}{\let\fonte\cpsource}{}%
% \newcoppefloat{env}{caption name}{list title}: declare an extra caption-on-top
% float (e.g. "picture") that supports \source and gains a \listof<env>s command.
\newcommand{\newcoppefloat}[3]{%
  {\floatstyle{plaintop}\newfloat{#1}{htbp}{lo#1}}%
```

```
  \floatname{#1}{#2}%
  \expandafter\newcommand\csname listof#1s\endcsname{%
    \chapter*{#3}\addcontentsline{toc}{chapter}{#3}%
    \@mkboth{\MakeUppercase{#3}}{\MakeUppercase{#3}}%
    \@starttoc{lo#1}}}
% --- Footnotes (manual2024 / R16, R8, R21): single spacing and a 5 cm rule
%     (filete) from the left margin; footnotesize is the book default. --------
\let\coppe@orig@footnotetext\@footnotetext
\long\def\@footnotetext#1{\coppe@orig@footnotetext{\singlespacing#1}}
\renewcommand\footnoterule{\kern-3\p@\hrule\@width 5cm\kern 2.6\p@}
% --- alineas: list keyed by lowercase letters a) b) c); subitems with "--"
%     (manual2024 2.6 / proposta P24). ---------------------------------------
\RequirePackage{enumitem}
\newlist{alineas}{enumerate}{2}
\setlist[alineas,1]{label=\alph*),leftmargin=1.8em,itemsep=0pt}
\setlist[alineas,2]{label=--,leftmargin=1.8em,itemsep=0pt}
% --- Page numbering: arabic in the top OUTER corner on textual pages
%     (manual2024 2.7 / proposta P25-P26). Pre-textual numbering is set by
%     \frontmatter (unnumbered by default; roman with the numeraisromanos
%     class option). ---------------------------------------------------------
\RequirePackage{fancyhdr}
% \headheight is set to 15pt by geometry up above (see the geometry call),
% which is what fancyhdr needs (>= 14.49998pt) for the running head.
\fancypagestyle{coppe}{\fancyhf{}\fancyhead[RO,LE]{\thepage}%
  \renewcommand{\headrulewidth}{0pt}}
% Blank pages inserted by \cleardoublepage (two-sided) must be empty: no page
% number may leak onto them (e.g. the verso right after the cover).
\renewcommand{\cleardoublepage}{%
  \clearpage
  \if@twoside
    \ifodd\c@page\else
      \hbox{}\thispagestyle{empty}\newpage
      \if@twocolumn\hbox{}\newpage\fi
    \fi
  \fi}
\def\CoppeTeX{{\rm C\kern-.05em{\sc o\kern-.025em p\kern-.025em
p\kern-.025em e}}\kern-.08em
T\kern-.1667em\lower.5ex\hbox{E}\kern-.125emX\spacefactor1000}

\newboolean{maledoc}
\setboolean{maledoc}{false}
%
% Class options.
% Multilingual model (CoppeTeX 4.x, branch nlinguas). The class holds three
% fixed language slots:
%   * main     -- the language of the body, captions, and the main abstract.
%                 Set by a class option: brazilian (default), english,
%                 spanish, french, italian, or any user-supplied language
%                 that ships a coppe-lang-<name>.def file.
%   * foreign  -- the language used by the foreignabstract environment and
%                 by \foreign@... registers. Defaults to english (or to
%                 brazilian when main=english) for back-compat.
%   * third    -- optional, loaded on demand with \usecoppelanguage{<name>}
%                 in the preamble (for citations or quotations in a third
%                 language). Has no class option of its own.
```

```
% Note: the legacy \if@english boolean (defined and toggled here in
% v3.x for backward compatibility) was removed in v4.0 -- nothing
% inside the class or in the biblatex style files (.bbx/.cbx/.dbx/.lbx)
% tests it. Code that needs to branch on the main language should test
% \ifx\coppe@mainlang\coppe@@englishstr instead.
\def\coppe@mainlang{brazilian}
\def\coppe@foreignlang{english}
\DeclareOption{brazilian}{%
  \def\coppe@mainlang{brazilian}\def\coppe@foreignlang{english}}
\DeclareOption{english}{%
  \def\coppe@mainlang{english}\def\coppe@foreignlang{brazilian}}
\DeclareOption{spanish}{%
  \def\coppe@mainlang{spanish}\def\coppe@foreignlang{english}}
\DeclareOption{french}{%
  \def\coppe@mainlang{french}\def\coppe@foreignlang{english}}
\DeclareOption{italian}{%
  \def\coppe@mainlang{italian}\def\coppe@foreignlang{english}}
\DeclareOption{msc}{%
  \newcommand{\@degree}{M.Sc.}
  \newcommand{\@degreename}{Mestrado}
  \newcommand{\local@degname}{Mestre}
  \newcommand{\foreign@degname}{Master}
  \newcommand\local@doctype{Disserta{\c c}{\~ a}o}
  \newcommand\foreign@doctype{Dissertation}
}
\DeclareOption{dscexam}{%
  \newcommand{\@degree}{D.Sc.}
  \newcommand{\@degreename}{Doutorado}
  \newcommand{\local@degname}{Doutor}
  \newcommand{\foreign@degname}{Doctor}
  \setboolean{maledoc}{true}
  \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
  \newcommand\foreign@doctype{Qualifying Exam}
}
\DeclareOption{mscexam}{%
 \newcommand{\@degree}{M.Sc.}
  \newcommand{\@degreename}{Mestrado}
  \newcommand{\local@degname}{Mestre}
  \newcommand{\foreign@degname}{Master}
  \setboolean{maledoc}{true}
  \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
  \newcommand\foreign@doctype{Qualifying Exam}
}
\DeclareOption{dsc}{%
  \newcommand{\@degree}{D.Sc.}
  \newcommand{\@degreename}{Doutorado}
  \newcommand{\local@degname}{Doutor}
  \newcommand{\foreign@degname}{Doctor}
  \newcommand\local@doctype{Tese}
  \newcommand\foreign@doctype{Thesis}
}
\newif\if@coppenumeric\@coppenumericfalse
\DeclareOption{numbers}{\@coppenumerictrue}
% numeraisromanos: show roman numerals on the pre-textual pages. Default: the
```

```
% pre-textual pages are counted but not numbered (manual2024 2.7).
\newif\if@copperoman\@copperomanfalse
\DeclareOption{numeraisromanos}{\@copperomantrue}
```

Default line spacing is one-and-a-half; the `doublespacing` option switches to double spacing. Since `setspace` is loaded at the end of the class (after `hyperref`), the actual `\onehalfspacing`/`\doublespacing` call is deferred: the option only flips a flag, which the late-load block reads.

```
\newif\if@coppedoublespace\@coppedoublespacefalse
\DeclareOption{doublespacing}{\@coppedoublespacetrue}
% Two-sided layout is opt-in since v4.1: the 2026 manual dropped the verso
% margins and the deposit is digital (CEPG 246/2023). Given \texttt{twoside},
% the mirrored 3cm inner / 2cm outer layout is restored after \ProcessOptions.
% With no \fichacatalografica file given, the folha adicional leaves a framed
% blank where the ficha goes. The rascunhoficha option fills that blank with the
% class's own LaTeX-composed ficha instead -- useful while writing, never valid
% for deposit: the official ficha comes from the SiBI generator.
% Annex D of the manual draws one signature rule per board member, with the
% identification underneath. The COPPE folha de aprovacao has listed the names
% compactly, without rules, for years. The compact list stays the default and
% the rules are opt-in, so existing theses do not change shape overnight.
% PDF/A output. Opt-in for now: 2.2(d) of the 2026 manual requires the deposited
% file to be PDF/A, but the constraints (embedded fonts, colour profile, no
% encryption, restricted transparency) can trip up a document that pulls in
% arbitrary images or packages, and breaking existing theses to enforce it would
% be worse than letting the author switch it on when depositing.
\newif\if@coppepdfa \@coppepdfafalse
\DeclareOption{pdfa}{\@coppepdfatrue}
\newif\if@coppeassin \@coppeassinfalse
\DeclareOption{assinaturas}{\@coppeassintrue}
\newif\if@copperascunho \@copperascunhofalse
\DeclareOption{rascunhoficha}{\@copperascunhotrue}
\newif\if@coppetwoside \@coppetwosidefalse
\DeclareOption{twoside}{\@coppetwosidetrue}
\DeclareOption{oneside}{\@coppetwosidefalse}
```

```
\ProcessOptions\relax
% \LoadClass and geometry ran before \ProcessOptions (see above), so the
% two-sided layout can only be applied here. geometry reads left/right as
% inner/outer once twoside is on, so the 3cm/2cm pair set above becomes the
% mirrored gutter with no further arithmetic. \@twosidetrue is what book's own
% twoside option sets, and is what \cleardoublepage and fancyhdr test.
% In one-sided mode fancyhdr treats every page as odd, so the \fancyhead[RO,LE]
% declarations elsewhere already put the folio in the top right corner of every
% folha, as 2.7 requires.
\if@coppetwoside
  \@twosidetrue
  \geometry{twoside}%
\fi
% Built-in language names (used by the loader and the dispatcher below).
\def\coppe@@brazilianstr{brazilian}
\def\coppe@@englishstr{english}
% Load Babel with the foreign language first and the main language last
% (Babel treats the LAST option in the list as the main language). With
% pt-main this produces [english,brazilian]{babel} and with en-main
```

```
% [brazilian,english] -- identical to the historical \if@english calls.
\edef\coppe@babelopts{\coppe@foreignlang,\coppe@mainlang}
\expandafter\RequirePackage\expandafter[\coppe@babelopts]{babel}
% --- Multilingual string dispatcher ---------------------------------------
% \copperdefstring{<lang>}{<key>}{<value>} adds one entry to the table for
% language <lang>. The three retrievers below look strings up either at the
% current babel language (\coppestring -- follows \selectlanguage), or pinned
% to the main slot (\coppemainstring) or the foreign slot (\coppeforeignstring).
\providecommand{\copperdefstring}[3]{%
  \expandafter\gdef\csname coppe@str@#1@#2\endcsname{#3}}
\providecommand{\coppestring}[1]{\@nameuse{coppe@str@\languagename @#1}}
\providecommand{\coppemainstring}[1]{\@nameuse{coppe@str@\coppe@mainlang @#1}}
\providecommand{\coppeforeignstring}[1]{\@nameuse{coppe@str@\coppe@foreignlang @#1}}
% Load an extra language pack at any point in the preamble (third-language
% slot for citations or quotations beyond the main+foreign pair).
\providecommand{\usecoppelanguage}[1]{%
  \def\coppe@@cur{#1}%
  \ifx\coppe@@cur\coppe@@brazilianstr\else
  \ifx\coppe@@cur\coppe@@englishstr\else
    \InputIfFileExists{coppe-lang-#1.def}{}
      {\ClassWarning{coppe}{Language pack coppe-lang-#1.def not found}}%
  \fi\fi}
% \titlein{<lang>}{<text>}: record the title in any language. Will be
% consulted by the cover/folha renderers once they are migrated in step 2.
% In step 1 it is just a no-op store -- the historical \title / \foreigntitle
% still drive every visible site, so behavior for pt/en is unchanged.
\providecommand{\titlein}[2]{%
  \expandafter\gdef\csname coppe@title@#1\endcsname{#2}}
% --- Built-in string tables: brazilian + english. The values mirror the
% literals currently scattered through the class; step-1 only populates the
% tables -- no site reads them yet, so the rendered output is unchanged. ----
\copperdefstring{brazilian}{appendix}{Ap\^endice}
\copperdefstring{english}  {appendix}{Appendix}
\copperdefstring{brazilian}{annex}   {Anexo}
\copperdefstring{english}  {annex}   {Annex}
\copperdefstring{brazilian}{frame}        {Quadro}
\copperdefstring{english}  {frame}        {Frame}
\copperdefstring{brazilian}{listframe}    {Lista de Quadros}
\copperdefstring{english}  {listframe}    {List of Frames}
\copperdefstring{brazilian}{source}       {Fonte}
\copperdefstring{english}  {source}       {Source}
\copperdefstring{brazilian}{program}      {Programa}
\copperdefstring{english}  {program}      {Program}
\copperdefstring{brazilian}{listprogram}  {Lista de Programas}
\copperdefstring{english}  {listprogram}  {List of Programs}
\copperdefstring{brazilian}{references}   {Refer\^encias}
\copperdefstring{english}  {references}   {References}
\copperdefstring{brazilian}{in:}          {em}
\copperdefstring{english}  {in:}          {in}
\copperdefstring{brazilian}{art}          {Ilustra\c{c}\~ao}
\copperdefstring{english}  {art}          {Art}
\copperdefstring{brazilian}{platform}     {Plataforma}
\copperdefstring{english}  {platform}     {Platform}
\copperdefstring{brazilian}{directedby}   {Dire\c{c}\~ao}
```

```
\copperdefstring{english}  {directedby}   {Directed by}
\copperdefstring{brazilian}{producedby}   {Produ\c{c}\~ao}
\copperdefstring{english}  {producedby}   {Produced by}
\copperdefstring{brazilian}{listabbreviation}{Lista de Abreviaturas e Siglas}
\copperdefstring{english}  {listabbreviation}{List of Abbreviations and Acronyms}
\copperdefstring{brazilian}{listsymbol}      {Lista de S{\' i}mbolos}
\copperdefstring{english}  {listsymbol}      {List of Symbols}
\copperdefstring{brazilian}{glossary}        {Gloss{\' a}rio}
\copperdefstring{english}  {glossary}        {Glossary}
\copperdefstring{brazilian}{advisor}      {Orientador}
\copperdefstring{english}  {advisor}      {Advisor}
\copperdefstring{brazilian}{advisors}     {Orientadores}
\copperdefstring{english}  {advisors}     {Advisors}
\copperdefstring{brazilian}{dept}         {Programa}
\copperdefstring{english}  {dept}         {Department}
\copperdefstring{brazilian}{approvedmale}  {Aprovado por}
\copperdefstring{brazilian}{approvedfemale}{Aprovada por}
\copperdefstring{brazilian}{approvedonmale}  {Aprovado em}
\copperdefstring{brazilian}{approvedonfemale}{Aprovada em}
\copperdefstring{brazilian}{keywordsname} {Palavras-chave}
\copperdefstring{brazilian}{volumename} {Volume}
\copperdefstring{brazilian}{coadvisor}  {Coorientador}
\copperdefstring{brazilian}{coadvisors} {Coorientadores}
\copperdefstring{brazilian}{ofname}     {de}
\copperdefstring{english}  {approvedmale}  {Approved by}
\copperdefstring{english}  {approvedfemale}{Approved by}
\copperdefstring{english}  {approvedonmale}  {Approved on}
\copperdefstring{english}  {approvedonfemale}{Approved on}
\copperdefstring{english}  {keywordsname} {Keywords}
\copperdefstring{english}  {volumename} {Volume}
\copperdefstring{english}  {coadvisor}  {Co-advisor}
\copperdefstring{english}  {coadvisors} {Co-advisors}
\copperdefstring{english}  {ofname}     {of}
% Institutional names: always Portuguese on the cover/folha (UFRJ template).
\copperdefstring{brazilian}{universityname}{Universidade Federal do Rio de Janeiro}
\copperdefstring{english}  {universityname}{Universidade Federal do Rio de Janeiro}
\copperdefstring{brazilian}{cityname}     {Rio de Janeiro}
\copperdefstring{english}  {cityname}     {Rio de Janeiro}
\copperdefstring{brazilian}{statename}    {RJ}
\copperdefstring{english}  {statename}    {RJ}
\copperdefstring{brazilian}{countryname}  {Brasil}
\copperdefstring{english}  {countryname}  {Brasil}
% Month names.
\copperdefstring{brazilian}{monthname1} {Janeiro}
\copperdefstring{brazilian}{monthname2} {Fevereiro}
\copperdefstring{brazilian}{monthname3} {Mar{\c c}o}
\copperdefstring{brazilian}{monthname4} {Abril}
\copperdefstring{brazilian}{monthname5} {Maio}
\copperdefstring{brazilian}{monthname6} {Junho}
\copperdefstring{brazilian}{monthname7} {Julho}
\copperdefstring{brazilian}{monthname8} {Agosto}
\copperdefstring{brazilian}{monthname9} {Setembro}
\copperdefstring{brazilian}{monthname10}{Outubro}
\copperdefstring{brazilian}{monthname11}{Novembro}
```

```
\copperdefstring{brazilian}{monthname12}{Dezembro}
\copperdefstring{english}{monthname1} {January}
\copperdefstring{english}{monthname2} {February}
\copperdefstring{english}{monthname3} {March}
\copperdefstring{english}{monthname4} {April}
\copperdefstring{english}{monthname5} {May}
\copperdefstring{english}{monthname6} {June}
\copperdefstring{english}{monthname7} {July}
\copperdefstring{english}{monthname8} {August}
\copperdefstring{english}{monthname9} {September}
\copperdefstring{english}{monthname10}{October}
\copperdefstring{english}{monthname11}{November}
\copperdefstring{english}{monthname12}{December}
% algorithm2e package option for this language (empty -> english default).
\copperdefstring{brazilian}{algorithm2eopt}{portuguese}
\copperdefstring{english}  {algorithm2eopt}{}
% Babel package name for the foreignabstract \begin{otherlanguage}{...}.
\copperdefstring{brazilian}{babelname}{brazilian}
\copperdefstring{english}  {babelname}{english}
% Auto-load language packs for any non-built-in language listed in babelopts.
% Built-ins (brazilian, english) ship their string tables inside coppe.cls
% (defined just above) and need no extra file; others (spanish/french/italian
% /...) ship as coppe-lang-<name>.def alongside the class. The dispatcher
% must already be defined when these run, so the loader sits AFTER the tables.
\@for\coppe@@lang:=\coppe@babelopts\do{%
  \edef\coppe@@cur{\coppe@@lang}%
  \ifx\coppe@@cur\coppe@@brazilianstr\else
  \ifx\coppe@@cur\coppe@@englishstr\else
    \InputIfFileExists{coppe-lang-\coppe@@cur.def}
      {\ClassInfo{coppe}{Loaded language pack coppe-lang-\coppe@@cur.def}}
      {\ClassError{coppe}
        {Missing language pack 'coppe-lang-\coppe@@cur.def'.}
        {Install it next to coppe.cls, or pick a built-in language.}}%
  \fi\fi}
% Quotation marks: csquotes gives the language-aware \enquote; \citacao is its
% Portuguese alias for short, inline quotations (proposta P3).
\RequirePackage{csquotes}
\newcommand{\citacao}[1]{\enquote{#1}}
% --- Code listings & algorithms (TODO3 "tratar listagens"): the listings
%     package gives the "Programa" listing (caption on top, ABNT R46) with
%     \listofprogramas; algorithm2e gives algorithms. Names are babel-aware and
%     the language styles below are based on the user's listagens.tex. ---------
\RequirePackage{xcolor}
\RequirePackage{listings}
\RequirePackage{fancyvrb}
\DeclareFixedFont{\ttb}{T1}{txtt}{bx}{n}{10}
\DeclareFixedFont{\ttm}{T1}{txtt}{m}{n}{10}
\lstset{captionpos=t,extendedchars=true,basicstyle=\ttfamily,inputencoding=utf8,%
  literate=%
  {ã}{{\~{a}}}1 {á}{{\'{a}}}1 {â}{{\^{a}}}1 {à}{{\`{a}}}1 {é}{{\'{e}}}1%
  {è}{{\`{e}}}1 {ê}{{\^{e}}}1 {î}{{\^{i}}}1 {í}{{\'{i}}}1 {ô}{{\^{o}}}1%
  {ó}{{\'{o}}}1 {õ}{{\~{o}}}1 {û}{{\^{u}}}1 {ú}{{\'{u}}}1 {ç}{{\c{c}}}1%
  {Ç}{{\c{C}}}1 {À}{{\`{A}}}1 {Â}{{\^{A}}}1 {Á}{{\'{A}}}1 {É}{{\'{E}}}1%
  {Ê}{{\^{E}}}1 {Í}{{\'{I}}}1 {Ó}{{\'{O}}}1 {Ô}{{\^{O}}}1 {Õ}{{\~{O}}}1%
```

```
  {Ú}{{\'{U}}}1%
}
% Babel-aware listing names, set after listings' own babel setup.
\AtBeginDocument{%
  \renewcommand\lstlistingname{\coppestring{program}}%
  \renewcommand\lstlistlistingname{\coppestring{listprogram}}}
% \listofprogramas in the coppe list style (heading + ToC + running head).
\newcommand\listofprogramas{%
  \chapter*{\lstlistlistingname}%
  \addcontentsline{toc}{chapter}{\lstlistlistingname}%
  \@mkboth{\MakeUppercase\lstlistlistingname}{\MakeUppercase\lstlistlistingname}%
  \@starttoc{lol}}
% Per-language styles + environments (python, java, xml, html, prolog).
% The red hook that marks a wrapped line is built ONCE, into a box, and the
% styles below insert the box. Writing the colour inline -- postbreak=\mbox{%
% \textcolor{red}{$\hookrightarrow$}\space} -- worked for years and broke the
% moment the pdfa option was switched on: pdfx puts xcolor into a converting
% mode (\selectcolormodel plus \convertcolorsU/D), and a colour command run
% from inside listings' line-breaking machinery then fails with "Missing
% number, treated as zero" at the first wrapped line. Every listing in the
% document dies, so the class's own example could not be compiled as PDF/A at
% all. A box is immune: it is built here, in ordinary horizontal mode where
% xcolor works normally, and listings only ever copies it.
%
% The \sbox waits for \begin{document} because xcolor's colours are not
% defined while the class is still loading.
\newsavebox\coppe@lstarrow
\AtBeginDocument{\sbox\coppe@lstarrow{\textcolor{red}{$\hookrightarrow$}\space}}
\newcommand\pythonstyle{\lstset{language=Python,basicstyle=\ttm,numbers=left,%
  numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
  morekeywords={self,as},keywordstyle=\ttb\color[rgb]{0,0,0.5},%
  emph={MyClass,__init__},emphstyle=\ttb\color[rgb]{0.6,0,0},%
  stringstyle=\color[rgb]{0,0.5,0},frame=tb,showstringspaces=false,%
  extendedchars=true,firstnumber=auto,inputencoding=utf8,breaklines=true,%
  escapechar=`,postbreak=\usebox\coppe@lstarrow}}
\lstnewenvironment{python}[1][]{\pythonstyle\lstset{#1}%
  \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
\newcommand\pythoninline[1]{{\pythonstyle\lstinline!#1!}}
\newcommand\pythonexternal[2][]{{\pythonstyle\lstinputlisting[#1]{#2}}}
\newcommand\xmlstyle{\lstset{language=XML,basicstyle=\ttm,numbers=left,%
  numberstyle=\small\ttfamily,numbersep=0.5em,%
  morekeywords={xs:schema,xs:element,xs:sequence,xs:complexType},%
  keywordstyle=\ttb\color[rgb]{0,0,0.5},%
  emph={name,type,attributeFormDefault,elementFormDefault,xmlns:xs},%
  emphstyle=\ttb\color[rgb]{0.6,0,0},stringstyle=\color[rgb]{0,0.5,0},frame=tb,%
  showstringspaces=false,extendedchars=true,inputencoding=utf8,breaklines=true,%
  postbreak=\usebox\coppe@lstarrow}}
\lstnewenvironment{xml}[1][]{\xmlstyle\lstset{#1}}{}
\newcommand\htmlstyle{\lstset{language=HTML,basicstyle=\ttm,numbers=left,%
  numberstyle=\small\ttfamily,numbersep=0.5em,%
  keywordstyle=\ttb\color[rgb]{0,0,0.5},emphstyle=\ttb\color[rgb]{0.6,0,0},%
  stringstyle=\color[rgb]{0,0.5,0},frame=tb,showstringspaces=false,%
  extendedchars=true,inputencoding=utf8,breaklines=true,%
  postbreak=\usebox\coppe@lstarrow}}
```

```
\lstnewenvironment{html}[1][]{\htmlstyle\lstset{#1}}{}
\newcommand\prologstyle{\lstset{language=Prolog,basicstyle=\ttm,numbers=left,%
  numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
  keywordstyle=\ttb\color[rgb]{0,0,0.5},emph={MyClass,__init__},%
  emphstyle=\ttb\color[rgb]{0.6,0,0},stringstyle=\color[rgb]{0,0.5,0},frame=tb,%
  showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
  breaklines=true,escapechar=`,%
  postbreak=\usebox\coppe@lstarrow}}
\lstnewenvironment{prolog}[1][]{\prologstyle\lstset{#1}%
  \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
\newcommand\javastyle{\lstset{language=Java,basicstyle=\ttm,numbers=left,%
  numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
  morekeywords={self,as},keywordstyle=\ttb\color[rgb]{0,0,0.5},%
  emphstyle=\ttb\color[rgb]{0.6,0,0},stringstyle=\color[rgb]{0,0.5,0},frame=tb,%
  showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
  breaklines=true,escapechar=`,%
  postbreak=\usebox\coppe@lstarrow}}
\lstnewenvironment{java}[1][]{\javastyle\lstset{#1}%
  \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
\newcommand\javainline[1]{{\javastyle\lstinline!#1!}}
\newcommand\javaexternal[2][]{{\javastyle\lstinputlisting[#1]{#2}}}
% fancyvrb's Verbatim environment is available for program output; the
% \DefineVerbatimEnvironment{gxoutput}/{gxoutputs} aliases that used to live
% here were author-personal (gx prefix) and were removed in v4.0. Use, for
% example,
%   \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
%   ...
%   \end{Verbatim}
% (or \DefineVerbatimEnvironment{...}{Verbatim}{...} in the manuscript preamble).
\renewcommand{\theFancyVerbLine}{{\small\ttfamily\arabic{FancyVerbLine}}}
% Algorithms: algorithm2e, language driven by the main-slot \coppemainstring
% lookup (key "algorithm2eopt"). Empty means use algorithm2e's default
% language (English) -- needed because algorithm2e errors on an empty option.
\edef\coppe@@algolang{\coppemainstring{algorithm2eopt}}
\ifx\coppe@@algolang\@empty
  \RequirePackage[linesnumbered,lined,algochapter,ruled]{algorithm2e}
\else
  \expandafter\RequirePackage\expandafter[\coppe@@algolang,linesnumbered,lined,algochapter,r
\fi
% algorithm2e composes its own caption and therefore ignores the class-wide
% \captionsetup{font=footnotesize,labelsep=endash}. Left alone it printed
% "Algoritmo 6.1:Titulo" in bold at body size, while every other illustration
% printed "Figura 6.1 -- Titulo" in footnotesize -- 2.10 asks for the
% designating word, the number, a TRAVESSAO and the title, uniformly.
%
% The separator and the caption fonts are therefore set to match. The `ruled'
% layout (rules above and below) is kept: 2.10 prescribes where the caption and
% the source go, not whether the float is delimited, and the rules read here
% like the top-and-bottom delimitation a table carries.
\SetAlgoCaptionSeparator{\ \textendash}
\SetAlCapFnt{\footnotesize}
\SetAlCapNameFnt{\footnotesize}
\SetAlCapSty{textnormal}
\SetAlCapNameSty{textnormal}
```

```
\SetKwFor{Para}{para}{faça}{fim do para}
\SetKwFor{Enquanto}{enquanto}{faça}{fim do enquanto}
% setspace + hyperref + biblatex are loaded at the very end of the class
% (see the block just before %</class>), so their hooks see every macro the
% class defines, and the load order setspace -> hyperref -> biblatex is
% respected.
```

`\department` This macro is used to set the author's affiliation. There are twelve options which correspond to all academic units at COPPE/UFRJ. It defines the current and the foreign names of these units.

```
\newcommand\coppe@deptcode{}
\newcommand\meta@deptname{}
\newcommand\department[1]{%
% The programme CODE is kept alongside the resolved name, and each name is
% declared TWICE. \local@deptname keeps the TeX accent escapes with braces
% (Computa{\c c}{\~ a}o) that the cover and the folha de rosto typeset;
% \meta@deptname is the same name in literal UTF-8, and is the one that goes
% into the XMP. pdfx copies whatever it is given straight through, so feeding it
% the escaped form produced "Computa{c}o" in dc:description -- which is why the
% subject line used to fall back to the bare programme code. Both forms are
% written out here rather than derived from one another because there is no
% reliable way to strip TeX accent syntax at the point where the .xmpdata is
% written, and a table of thirteen entries is cheaper than that machinery.
  \gdef\coppe@deptcode{#1}%
  \ifthenelse{\equal{#1}{PEB}}
    {\global\def\local@deptname{Engenharia Biom{\' e}dica}
     \global\def\meta@deptname{Engenharia Biomédica}
     \global\def\foreign@deptname{Biomedical Engineering}}{}
  \ifthenelse{\equal{#1}{PEC}}
    {\global\def\local@deptname{Engenharia Civil}
     \global\def\meta@deptname{Engenharia Civil}
     \global\def\foreign@deptname{Civil Engineering}}{}
  \ifthenelse{\equal{#1}{PEE}}
    {\global\def\local@deptname{Engenharia El{\' e}trica}
     \global\def\meta@deptname{Engenharia Elétrica}
     \global\def\foreign@deptname{Electrical Engineering}}{}
  \ifthenelse{\equal{#1}{PEM}}
    {\global\def\local@deptname{Engenharia Mec{\^ a}nica}
     \global\def\meta@deptname{Engenharia Mecânica}
     \global\def\foreign@deptname{Mechanical Engineering}}{}
  \ifthenelse{\equal{#1}{PEMM}}
    {\global\def\local@deptname{Engenharia Metal{\' u}rgica e de Materiais}
     \global\def\meta@deptname{Engenharia Metalúrgica e de Materiais}
\global\def\foreign@deptname{Metallurgical and Materials Engineering}}{}
  \ifthenelse{\equal{#1}{PEN}}
    {\global\def\local@deptname{Engenharia Nuclear}
     \global\def\meta@deptname{Engenharia Nuclear}
     \global\def\foreign@deptname{Nuclear Engineering}}{}
  \ifthenelse{\equal{#1}{PENO}}
    {\global\def\local@deptname{Engenharia Oce{\^ a}nica}
     \global\def\meta@deptname{Engenharia Oceânica}
     \global\def\foreign@deptname{Ocean Engineering}}{}
  \ifthenelse{\equal{#1}{PPE}}
    {\global\def\local@deptname{Planejamento Energ{\' e}tico}
```

```
     \global\def\meta@deptname{Planejamento Energético}
     \global\def\foreign@deptname{Energy Planning}}{}
  \ifthenelse{\equal{#1}{PEP}}
    {\global\def\local@deptname{Engenharia de Produ{\c c}{\~ a}o}
     \global\def\meta@deptname{Engenharia de Produção}
     \global\def\foreign@deptname{Production Engineering}}{}
  \ifthenelse{\equal{#1}{PEQ}}
    {\global\def\local@deptname{Engenharia Qu{\' i}mica}
     \global\def\meta@deptname{Engenharia Química}
     \global\def\foreign@deptname{Chemical Engineering}}{}
  \ifthenelse{\equal{#1}{PESC}}
    {\global\def\local@deptname{Engenharia de Sistemas e Computa{\c c}{\~ a}o}
     \global\def\meta@deptname{Engenharia de Sistemas e Computação}
     \global\def\foreign@deptname{Systems Engineering and Computer Science}}{}
  \ifthenelse{\equal{#1}{PET}}
    {\global\def\local@deptname{Engenharia de Transportes}
     \global\def\meta@deptname{Engenharia de Transportes}
     \global\def\foreign@deptname{Transportation Engineering}}{}
  \ifthenelse{\equal{#1}{PENT}}
    {\global\def\local@deptname{Engenharia de Nanotecnologia}
     \global\def\meta@deptname{Engenharia de Nanotecnologia}
      \global\def\foreign@deptname{Nanotechnology Engineering}}{}
}
```

`\title` Used to enter the title in Brazilian Portuguese. Mirrors into `\coppe@title@brazilian` so that the per-language title slot used by the cover/folha renderers (`\coppe@selecttitle`) picks the right text for a pt-main thesis.

```
\renewcommand\title[1]{%
  \global\def\local@title{#1}%
  \expandafter\gdef\csname coppe@title@brazilian\endcsname{#1}%
}
```

`\foreigntitle` Used to enter the foreign title. Also mirrors into `\coppe@title@english` so that the cover renders the right text for an english-main thesis.

```
\newcommand\foreigntitle[1]{%
  \global\def\foreign@title{#1}%
  \expandafter\gdef\csname coppe@title@english\endcsname{#1}%
}
```

`\coppe@selecttitle` Expands to the title stored for the current main language. For pt-main this is `\local@title`; for en-main `\foreign@title`; for any other main language the user must call `\titlein{<lang>}{...}` to populate it.

```
\newcommand\coppe@selecttitle{\@nameuse{coppe@title@\coppe@mainlang}}
% Subtitle, mirroring the title machinery slot for slot: \subtitle feeds the
% brazilian slot, \foreignsubtitle the english one, \subtitlein any other.
% 3.1.1 and 3.1.2.1.1(c) require the subtitle on the capa and on the folha de
% rosto, subordinated to the title by a colon.
\newcommand\subtitle[1]{%
  \expandafter\gdef\csname coppe@subtitle@brazilian\endcsname{#1}}
\newcommand\foreignsubtitle[1]{%
  \expandafter\gdef\csname coppe@subtitle@english\endcsname{#1}}
\newcommand\subtitlein[2]{%
  \expandafter\gdef\csname coppe@subtitle@#1\endcsname{#2}}
```

```
\newcommand\coppe@selectsubtitle{\@nameuse{coppe@subtitle@\coppe@mainlang}}
% Title as the capa and the folha de rosto print it. Annexes A and B show the
% title in caps and the subtitle in ordinary case after a colon, so the case
% change is applied to the title alone.
\newcommand\coppe@titleblock{%
  \MakeUppercase{\coppe@selecttitle}%
  \@ifundefined{coppe@subtitle@\coppe@mainlang}{}{: \coppe@selectsubtitle}}
% Number of volumes (3.1.1 and 3.1.2.1.1(d)): printed only when there is more
% than one, since a single-volume work says nothing about volumes.
% Coorientador (3.1.2.1.1(f)). Same shape as \advisor, including the optional
% institution; the label switches to the plural on the second call, as the
% advisor label already did.
\newcount\@coadvisor\@coadvisor0
\newcommand\coadvisor[5][]{%
  \global\@namedef{CoppeCoadvisorInst:\expandafter\the\@coadvisor}{#1}
  \global\@namedef{CoppeCoadvisorTitle:\expandafter\the\@coadvisor}{#2}
  \global\@namedef{CoppeCoadvisorName:\expandafter\the\@coadvisor}{#3}
  \global\@namedef{CoppeCoadvisorSurname:\expandafter\the\@coadvisor}{#4}
  \global\@namedef{CoppeCoadvisorDegree:\expandafter\the\@coadvisor}{#5}
  \global\advance\@coadvisor by 1
}
\newcommand\local@coadvisorstring{%
  \ifnum\@coadvisor>1
    \coppemainstring{coadvisors}%
  \else
    \coppemainstring{coadvisor}%
  \fi}
\newcommand\coppe@nvolumes{1}
\newcommand\coppe@thisvolume{1}
\newcommand\volumes[1]{\gdef\coppe@nvolumes{#1}}
\newcommand\volume[1]{\gdef\coppe@thisvolume{#1}}
\newcommand\coppe@volumeline{%
  \ifnum\coppe@nvolumes>1
    \par\vspace*{4mm}%
    \coppemainstring{volumename}\space\coppe@thisvolume\space
    \coppemainstring{ofname}\space\coppe@nvolumes
% The closing \par matters: on the capa the next thing is \vspace{\stretch{3}},
% which does not end a paragraph, so without it the city ran on into the volume
% line ("Volume 2 de 3 RIO DE JANEIRO").
    \par
  \fi}
```

`\advisor` Defines globally the title, name and academic degree of the advisors.

```
\newcount\@advisor\@advisor0
% The optional first argument is the member's institution, required by
% 3.1.2.1.3(e). Omitting it keeps every pre-4.1 call working unchanged.
\newcommand\advisor[5][]{%
  \global\@namedef{CoppeAdvisorInst:\expandafter\the\@advisor}{#1}
  \global\@namedef{CoppeAdvisorTitle:\expandafter\the\@advisor}{#2}
  \global\@namedef{CoppeAdvisorName:\expandafter\the\@advisor}{#3}
  \global\@namedef{CoppeAdvisorSurname:\expandafter\the\@advisor}{#4}
  \global\@namedef{CoppeAdvisorDegree:\expandafter\the\@advisor}{#5}
  \global\advance\@advisor by 1
  \ifnum\@advisor>1
```

```
    \renewcommand\local@advisorstring{Orientadores}
    \renewcommand\foreign@advisorstring{Advisors}
  \fi
}
```

`\examiner`

```
\newcount\@examiner\@examiner0
% Until v4.1 this stored only "#1 #2" and DISCARDED the third argument: the
% titulacao every document passed was silently thrown away and never printed,
% though 3.1.2.1.3(e) requires it. Each part is now kept separately, and the
% optional first argument carries the institution.
% CoppeExaminer: is retained with its old meaning for anything reading it.
\newcommand\examiner[4][]{%
  \global\@namedef{CoppeExaminerInst:\expandafter\the\@examiner}{#1}
  \global\@namedef{CoppeExaminerTitle:\expandafter\the\@examiner}{#2}
  \global\@namedef{CoppeExaminerName:\expandafter\the\@examiner}{#3}
  \global\@namedef{CoppeExaminerDegree:\expandafter\the\@examiner}{#4}
  \global\@namedef{CoppeExaminer:\expandafter\the\@examiner}{#2\ #3}
  \global\advance\@examiner by 1
}
```

`\author` It was redefined to allow the identification of the author's first names and surname.

```
\renewcommand\author[2]{%
  \global\def\@authname{#1}
  \global\def\@authsurn{#2}
}
```

`\date` Sets the *document*'s nominal month and year (used by the cover, the folha de rosto, and the catalog). Stores them in private slots (`\coppe@docmonth` / `\coppe@docyear`) rather than mutating TeX's `\month` / `\year` primitives, which would clobber `\today`.

```
\renewcommand\date[2]{%
  \def\coppe@docmonth{#1}%
  \def\coppe@docyear{#2}}
```

`\local@monthname`

```
\newcommand\local@monthname{\ifcase\coppe@docmonth\or
  Janeiro\or Fevereiro\or Mar{\c c}o\or Abril\or Maio\or Junho\or
  Julho\or Agosto\or Setembro\or Outubro\or Novembro\or Dezembro\fi}
```

`\foreign@monthname`

```
\newcommand\foreign@monthname{\ifcase\coppe@docmonth\or
  January\or February\or March\or April\or May\or June\or
  July\or August\or September\or October\or November\or December\fi}
```

`\keyword`

```
\newcounter{keywords}
\newcommand\keyword[1]{%
  \global\@namedef{CoppeKeyword:\expandafter\the\c@keywords}{#1}
  \global\addtocounter{keywords}{1}
}
% Keywords for the foreign abstract. Annex F of the manual shows the English
```

```
% abstract closing with English keywords, and \keyword alone cannot serve both
% pages. When no \foreignkeyword is given the foreign abstract falls back to
% the \keyword list, which is what a document written before v4.1 expects.
\newcounter{foreignkeywords}
\newcommand\foreignkeyword[1]{%
  \global\@namedef{CoppeForeignKeyword:\expandafter\the\c@foreignkeywords}{#1}
  \global\addtocounter{foreignkeywords}{1}
}
% Keywords for the optional third abstract (Portuguese, for the banca). Same
% shape and same fallback as \foreignkeyword: without it the Portuguese page
% would close with the main-language keywords under a Portuguese label, which
% is what a Spanish-, French- or Italian-main work would produce.
\newcounter{braziliankeywords}
\newcommand\braziliankeyword[1]{%
  \global\@namedef{CoppeBrazilianKeyword:\expandafter\the\c@braziliankeywords}{#1}
  \global\addtocounter{braziliankeywords}{1}
}
% 3.1.2.1.4: the keywords close the resumo, preceded by the term, a colon, and
% separated from one another by semicolons.
% #1 label, #2 csname prefix, #3 how many.
\newcommand\coppe@printkeywords[3]{%
  \ifnum#3>0
    \par\vspace*{7mm}%
    \noindent{\singlespacing\textbf{#1:}\enspace
    \count1=0
    \@whilenum\count1<#3\do{%
      \ifnum\count1>0 ;\space\fi
      \csname #2\the\count1\endcsname
      \advance\count1 by 1}%
    .\par}%
  \fi}
```

\areaconcentracao \linhapesquisa \tipoproducao \projetovinculado \nomeprojeto \agenciafomento

Data for the folha adicional (Coleta CAPES) and for the natureza block of the folha de rosto and folha de aprovacao. Every one of them defaults to empty; \makefolhaadicional prints an empty rule where a value is missing, so the programme can complete the sheet by hand if need be.

```
\newcommand\coppe@areaconc{}
\newcommand\areaconcentracao[1]{\gdef\coppe@areaconc{#1}}
\newcommand\coppe@dataaprov{}
\newcommand\dataaprovacao[1]{\gdef\coppe@dataaprov{#1}}
\newcommand\coppe@linhapesq{}
\newcommand\linhapesquisa[1]{\gdef\coppe@linhapesq{#1}}
\newcommand\coppe@tipoproducao{}
\newcommand\tipoproducao[1]{\gdef\coppe@tipoproducao{#1}}
\newcommand\coppe@projetovinc{}
\newcommand\projetovinculado[1]{\gdef\coppe@projetovinc{#1}}
\newcommand\coppe@nomeprojeto{}
\newcommand\nomeprojeto[1]{\gdef\coppe@nomeprojeto{#1}}
\newcounter{coppeagencias}
\newcommand\agenciafomento[2]{%
  \global\@namedef{CoppeAgencia:\expandafter\the\c@coppeagencias}{#1}%
  \global\@namedef{CoppeAgenciaSigla:\expandafter\the\c@coppeagencias}{#2}%
  \global\addtocounter{coppeagencias}{1}}
```

`\freeconfig` This command allows easy changing of core class parameters.

```
\newcommand\freeconfig[6]{%
 \providecommand\@degree{}
 \renewcommand\@degree{#1}
 \providecommand\@degreename{}
 \renewcommand\@degreename{#2}
 \providecommand\local@degname{}
 \renewcommand\local@degname{#3}
 \providecommand\foreign@degname{}
 \renewcommand\foreign@degname{#4}
 \providecommand\local@doctype{}
 \renewcommand\local@doctype{#5}
 \providecommand\foreign@doctype{}
 \renewcommand\foreign@doctype{#6}%
}
%   \end{macrocode}
% \end{macro}
%
% \begin{macro}{\frontmatter}
% The number of pages for both frontmatter and mainmatter printed
% in the cataloging details page is computed by means of simple
% \LaTeX\ labels.
%    \begin{macrocode}
\renewcommand\frontmatter{%
  \cleardoublepage
  \@mainmatterfalse
% The count already started at the folha de rosto (see \maketitle) and simply
% continues here: 2.7 requires every folha from the folha de rosto onwards to
% be counted sequentially, printing arabic numerals only from the first textual
% folha. Resetting the counter here -- as this class did until v4.1 -- made the
% Introducao come out as page 1, which the norm does not allow.
%
% The numeraisromanos option keeps the legacy behaviour (lowercase roman in the
% pre-textual part, arabic restarting at 1 in the textual part). That departs
% from 2.7, which has the pre-textual folhas counted but NOT numbered, so the
% option warns.
  \if@copperoman
    \ClassWarningNoLine{coppe}{Option ‘numeraisromanos’ prints roman numerals
      on the pre-textual folhas and restarts the arabic count at 1 in the
      textual part. Section 2.7 of the SiBI manual (9th rev. ed., 2026) has
      those folhas counted but not numbered, and the arabic numbering
      continuing the count. Drop the option to comply}%
    \pagenumbering{roman}%
  \fi
  \thispagestyle{empty}
  \makefrontpage
  \clearpage
  \if@copperoman
    \fancypagestyle{plain}{\fancyhf{}\fancyhead[RO,LE]{\thepage}%
      \renewcommand{\headrulewidth}{0pt}}%
    \pagestyle{coppe}%
  \else
    \fancypagestyle{plain}{\fancyhf{}\renewcommand{\headrulewidth}{0pt}}%
    \pagestyle{empty}%
```

```
  \fi
}
```

\mainmatter

```
\renewcommand\mainmatter{%
  \coppe@mainBegin
  \cleardoublepage
  \@mainmattertrue
  \fancypagestyle{plain}{\fancyhf{}\fancyhead[RO,LE]{\thepage}%
    \renewcommand{\headrulewidth}{0pt}}%
  \pagestyle{coppe}%
% No \pagenumbering{arabic} in the default path: it would reset the counter to
% 1 and make the Introducao folha 1, contrary to 2.7. The counter has been
% arabic and running since the folha de rosto; all that changes here is that
% the folio starts being PRINTED. Only the legacy numeraisromanos path, which
% switched to roman in the front matter, has to switch back -- and there the
% restart at 1 is intrinsic to that (non-conforming) scheme.
  \if@copperoman\pagenumbering{arabic}\fi}
```

\backmatter

```
\renewcommand\backmatter{%
  \if@openright
    \cleardoublepage
  \else
    \clearpage
  \fi}
%
```

\coppetexfinalpage Optional colophon-style closing page intended to face the rear cover. The macro forces a verso (even) page so that, in a two-sided print, the reader sees the colophon on the left and the inside of the rear cover on the right; pushes the text to the bottom of the page with `\vfill`; and types it in Portuguese unconditionally (the colophon is institutional metadata, not body content, so it is rendered in pt-BR regardless of the thesis main language). Three slot macros are user-overridable: `\coppefinalengine` (engine name; auto-detected via `iftex`), `\coppefinalfont` (text body font; defaults to Computer Modern, the LaTeX default), and `\coppefinalmanual` (the reference manual title; defaults to the current COPPE/UFRJ norm). Override them before `\coppetexfinalpage` with `\renewcommand` if the document uses a non-default font or follows an earlier edition of the norm.

```
\providecommand{\coppefinalengine}{%
  \ifPDFTeX\hologo{pdfLaTeX}\fi
  \ifLuaTeX\hologo{LuaLaTeX}\fi
  \ifXeTeX\hologo{XeLaTeX}\fi}
\providecommand{\coppefinalfont}{Computer Modern (padr\~ao \LaTeX{})}
\providecommand{\coppefinalmanual}{%
  \emph{Norma para a Elabora\c c\~ao Gr\'afica de Teses e Disserta\c c\~oes
  da COPPE/UFRJ}}
\newcommand{\coppetexfinalpage}{%
  \clearpage
  % Force the colophon onto a verso (even) page so it faces the rear cover.
  % If the current page is odd (recto), fill it with an empty verso first.
  \if@twoside
```

```
    \ifodd\c@page
      \hbox{}\thispagestyle{empty}\newpage
    \fi
  \fi
  \thispagestyle{empty}%
  \null\vfill
  \begin{otherlanguage}{brazilian}%
    \begin{center}
      {\footnotesize\setstretch{1.0}%
      \textsc{Colof\~ao}\par
      \vspace{0.7em}
      \begin{minipage}{0.75\textwidth}\centering
      Este documento foi composto em \LaTeX{} com \coppefinalengine{}
      em \today, na fonte \coppefinalfont. A bibliografia foi processada
      por Bib\LaTeX{} e Biber. Foi utilizada a classe \CoppeTeX,
      vers\~ao~\coppe@version, dispon\'\i vel no reposit\'orio GitHub
      \mbox{\url{https://github.com/COPPE-UFRJ/CoppeTeX}}, em
      conformidade com a \coppefinalmanual.
      \end{minipage}\par}
    \end{center}
  \end{otherlanguage}%
  \vspace*{2cm}%
  \clearpage}
```

`\makecover` The ABNT cover (capa, NBR 14724) suggested in the manual's Anexo A. It is the institution block — the university, the full COPPE name, and the graduate program taken from `\department` — followed by the author, the title, and the city and year, all centred and uppercased. The optional UFRJ and COPPE logos head the page. The capa is not counted: it precedes the (roman) front matter and is produced before the `titlepage` (folha de rosto).

```
\newcommand\makecover{%
  \begin{titlepage}%
  \centering
  \makebox[\linewidth]{%
    \includegraphics[height=2cm]{ufrj-logo}\hfill
    \includegraphics[height=2cm]{coppe-logo}}\par
  \vspace{1.5cm}
  {\singlespacing
 \MakeUppercase{\local@universityname}\par
   \nohyphens{\MakeUppercase{Instituto Alberto Luiz Coimbra de \\
     P{\' o}s-Gradua{\c c}{\~ a}o e Pesquisa de Engenharia}}\par
   \MakeUppercase{Programa de \local@deptname}\par}
  \vspace{\stretch{2}}
  \textbf{\MakeUppercase{\textbf{\@authname\ \@authsurn}}}\par
  \vspace{\stretch{2}}
  \nohyphens{\coppe@titleblock}\par
  \coppe@volumeline
  \vspace{\stretch{3}}
  \MakeUppercase{\local@cityname}\par
  \coppe@docyear
  \end{titlepage}%
}
```

\maketitle

```
\renewcommand\maketitle{%
% The capa is neither counted nor numbered (2.7), so it lives in its own
% throwaway numbering scheme; \pagenumbering{arabic} right after it restarts
% the counter, making the FOLHA DE ROSTO page 1 -- which is where 2.7 says the
% count begins. Nothing is printed here: \frontmatter keeps \pagestyle{empty}
% until the textual part starts.
  \pagenumbering{alph}
  \ltx@ifpackageloaded{hyperref}{\coppe@hypersetup}{}%
  \makecover
  \pagenumbering{arabic}
  \begin{titlepage}
  \begin{flushleft}
  \vspace*{1.5mm}
  \setlength\baselineskip{0pt}
  \setlength\parskip{1mm}
  \makebox[\linewidth]{%
    \includegraphics[height=2cm]{ufrj-logo}\hfill
    \includegraphics[height=2cm]{coppe-logo}}
  \end{flushleft}
  \vspace{1.05cm}
  \begin{center}
   \nohyphens{\@authname\ \@authsurn}\par
   \vspace*{3cm}
   \nohyphens{\coppe@titleblock}\par
   \coppe@volumeline
  \end{center}
  \vspace*{2.1cm}
  \begin{flushright}
% 2.4: on the folha de rosto the natureza/objetivo/instituicao block is set
% SINGLE spaced, and aligned from the middle of the mancha grafica to the right
% margin. 0.5\textwidth tracks the mancha, so the block follows automatically if
% the margins ever change -- the old fixed 8.45cm did not.
  \begin{minipage}{0.5\textwidth}
  \singlespacing\frontcover@maintext
  \end{minipage}\par
  \end{flushright}
% Only the natureza block is indented to the right (2.4). The linha de pesquisa
% and the orientadores return to the LEFT MARGIN, as the CAPES model of the
% folha de rosto and Annex B of the manual show. Until v4.1 they sat inside the
% right-hand block, which is what a recent review flagged.
  \vspace*{12mm}
  {\singlespacing
  \ifthenelse{\equal{\coppe@linhapesq}{}}{}{%
    \noindent Linha de pesquisa: \coppe@linhapesq\par
    \vspace*{5mm}}%
  \nohyphens{%
  \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
    \local@advisorstring: &
    \count1=0
    \toks@={}
    \@whilenum \count1<\@advisor \do{%
    \ifcase\count1 % same as \ifnum0=\count1
      \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
```

```
        \expandafter\endcsname\expandafter\space%
        \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
    \else
      \toks@=\expandafter\expandafter\expandafter{%
        \expandafter\the\expandafter\toks@%
        \expandafter&\expandafter\space%
        \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
        \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
      }%
    \fi
    \advance\count1 by 1}
    \the\toks@
  \end{tabularx}}%
  \ifnum\@coadvisor>0
    \par\vspace*{2mm}%
    \nohyphens{%
    \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
      \local@coadvisorstring: &
      \count1=0
      \toks@={}
      \@whilenum \count1<\@coadvisor \do{%
      \ifcase\count1
        \toks@=\expandafter{\csname CoppeCoadvisorName:\the\count1%
          \expandafter\endcsname\expandafter\space%
          \csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
      \else
        \toks@=\expandafter\expandafter\expandafter{%
          \expandafter\the\expandafter\toks@%
          \expandafter&\expandafter\space%
          \csname CoppeCoadvisorName:\the\count1\expandafter\endcsname%
          \expandafter\space\csname CoppeCoadvisorSurname:\the\count1\endcsname\\
        }%
      \fi
      \advance\count1 by 1}
      \the\toks@
    \end{tabularx}}%
  \fi
  \par}
  \vspace*{\fill}
% 3.1.2.1.1(g): the folha de rosto carries the ANO DE DEPOSITO alone -- the year
% the revised final version is handed to the Programa. The month was never asked
% for here (the capa, Annex A, also shows only the year), and printing it
% invited confusion with the defence date. \date still takes month and year:
% the month is what the abstract pages and the colophon use.
  \begin{center}
  \local@cityname\par
  \coppe@docyear
  \end{center}
  \end{titlepage}
  % Ficha catalográfica on the verso of the folha de rosto (NBR 14724, R57):
  % the title page is a recto, so this lands on the immediately following verso,
  % ahead of the approval sheet produced by \frontmatter.
% LaTeX's titlepage environment ends with \setcounter{page}{1} whenever the
% document is one-sided (see \endtitlepage in classes.dtx). That silently ate
```

```
% the folha de rosto from the count, putting the folha de aprovacao at 1 instead
% of 2. The count is therefore restated here: the folha de rosto WAS folha 1, so
% whatever follows it is folha 2.
  \setcounter{page}{2}%
% The page carrying the ficha catalografica is explicitly excluded from the
% count -- 2.7 of the 2026 manual ("a folha adicional que contem a ficha
% catalografica nao pode ser contada ou numerada"), and the 2025 edition said
% the same of the verso of the folha de rosto. Hence the counter is stepped
% back after it.
  \ifthenelse{\boolean{maledoc}}{}{%
    \makefolhaadicional\clearpage\addtocounter{page}{-1}}%
  \global\let\maketitle\relax%
  \global\let\and\relax}

% 3.1.2.1.3 lists the elements of the folha de aprovacao in order: (a) the
% AUTHOR, (b) the title in full, (c) the natureza/objetivo/instituicao/area de
% concentracao, (d) the date of approval, (e) the examining board. Until v4.1
% this page opened with the title in caps and the author under it, and set the
% natureza as a wall of uppercase (\frontpage@maintext). Annex D shows neither.
% One board member. \coppe@membroid is the identification line required by
% 3.1.2.1.3(e) -- name, titulacao and institution -- with the last two omitted
% when not given, so a document that never supplied them looks as it did before.
\newcommand\coppe@membroid[4]{%
  #1\ #2%
  \ifthenelse{\equal{#3}{}}{}{, #3}%
  \ifthenelse{\equal{#4}{}}{}{, #4}}
% The gap above each signature rule is \coppe@membrogap, set from the size of
% the board just before the list is assembled (see \makefrontpage). A fixed 9mm
% fitted five members and overflowed at seven -- the sheet silently continued
% onto a second page, which also printed a folio in the pre-textual part, where
% 2.7 forbids one. The shrink lives in the length itself -- \newlength makes a
% skip register, so "9mm minus 2mm" survives \setlength and reaches \vspace as
% real glue, and a title long enough to wrap an extra line is absorbed instead
% of costing a page. It has to go through \vspace and not a bare \vskip: the
% board list is emitted inside \nohyphens, i.e. in horizontal mode, where
% \vskip is an error and TeX then typesets the "plus ... minus ..." after it as
% ordinary text -- which is exactly what landed on the page the first time.
\newlength\coppe@membrogap
\setlength\coppe@membrogap{9mm minus 2mm}
\newcount\coppe@bancasize
\newcommand\coppe@membro[4]{%
  \if@coppeassin
    \vspace{\coppe@membrogap}%
    \begin{center}
      \rule{95mm}{0.4pt}\par
      \nopagebreak{\footnotesize\coppe@membroid{#1}{#2}{#3}{#4}}\par
    \end{center}%
  \else
    \noindent\hspace*{8mm}%
    \begin{minipage}{\dimexpr\textwidth-8mm\relax}
      \coppe@membroid{#1}{#2}{#3}{#4}
    \end{minipage}\par\vspace{2mm}%
  \fi}
\newcommand\makefrontpage{%
```

```
\begin{center}
  \nohyphens{\@authname\ \@authsurn}\par
  \vspace*{7mm}
  \sloppy\nohyphens{\coppe@selecttitle
    \@ifundefined{coppe@subtitle@\coppe@mainlang}{}{:
      \coppe@selectsubtitle}}\par
\end{center}\par
\vspace*{8mm}
% Same block, same indentation and same single spacing as the folha de rosto:
% 2.4 names the two pages together.
\begin{flushright}
\begin{minipage}{0.5\textwidth}
\singlespacing\frontcover@maintext
\end{minipage}
\end{flushright}
\vspace*{12mm}
% (d) date of approval. Annex D prints "Aprovada em:" followed by a blank to be
% filled in; \dataaprovacao supplies it when the date is already known.
\noindent\local@approvedonname:\enspace
\ifthenelse{\equal{\coppe@dataaprov}{}}{\hrulefill}{\coppe@dataaprov}\par
\vspace*{12mm}
% (e) the examining board, with the ORIENTADOR FIRST, as president of the board
% (3.1.2.1.3(e)). Until v4.1 the advisors were printed in a block of their own
% above a second list headed "Aprovada por:", which split the board in two and
% did not mark the advisor as its president. They are now one list.
%
% The list is assembled into \coppe@bancalist with \g@addto@macro instead of
% being built inside a tabularx: each entry now carries four fields (treatment,
% name, titulacao, institution) and the old \toks@/\expandafter accumulation
% could not carry them without becoming unreadable. \protected@edef keeps
% accents and other protected commands in the names intact.
% Size of the board decides the spacing above. The thresholds are measured, not
% guessed: with the 9mm gap a seven-member board pushed three members onto a
% second page, 7mm brought six back onto one sheet and 5mm did the same for
% seven and eight. Boards of five or fewer keep the original 9mm, so every
% document written before this change renders exactly as it did.
\coppe@bancasize=\@advisor
\advance\coppe@bancasize by \@coadvisor
\advance\coppe@bancasize by \@examiner
\ifnum\coppe@bancasize>6
  \setlength\coppe@membrogap{5mm minus 2mm}%
\else\ifnum\coppe@bancasize>5
  \setlength\coppe@membrogap{7mm minus 2mm}%
\fi\fi
\gdef\coppe@bancalist{}%
\count1=0
\@whilenum\count1<\@advisor\do{%
  \protected@edef\coppe@tmp{\noexpand\coppe@membro
    {\csname CoppeAdvisorTitle:\the\count1\endcsname}%
    {\csname CoppeAdvisorName:\the\count1\endcsname\space
     \csname CoppeAdvisorSurname:\the\count1\endcsname}%
    {\csname CoppeAdvisorDegree:\the\count1\endcsname}%
    {\csname CoppeAdvisorInst:\the\count1\endcsname}}%
  \expandafter\g@addto@macro\expandafter\coppe@bancalist
```

```
      \expandafter{\coppe@tmp}%
    \advance\count1 by 1}%
% Coorientadores come right after the orientadores and before the rest of the
% board, keeping the order 3.1.2.1.3(e) prescribes.
  \count1=0
  \@whilenum\count1<\@coadvisor\do{%
    \protected@edef\coppe@tmp{\noexpand\coppe@membro
      {\csname CoppeCoadvisorTitle:\the\count1\endcsname}%
      {\csname CoppeCoadvisorName:\the\count1\endcsname\space
       \csname CoppeCoadvisorSurname:\the\count1\endcsname}%
      {\csname CoppeCoadvisorDegree:\the\count1\endcsname}%
      {\csname CoppeCoadvisorInst:\the\count1\endcsname}}%
    \expandafter\g@addto@macro\expandafter\coppe@bancalist
      \expandafter{\coppe@tmp}%
    \advance\count1 by 1}%
  \count1=0
  \@whilenum\count1<\@examiner\do{%
    \protected@edef\coppe@tmp{\noexpand\coppe@membro
      {\csname CoppeExaminerTitle:\the\count1\endcsname}%
      {\csname CoppeExaminerName:\the\count1\endcsname}%
      {\csname CoppeExaminerDegree:\the\count1\endcsname}%
      {\csname CoppeExaminerInst:\the\count1\endcsname}}%
    \expandafter\g@addto@macro\expandafter\coppe@bancalist
      \expandafter{\coppe@tmp}%
    \advance\count1 by 1}%
  \noindent\local@approvedname:\par
  \nohyphens{\singlespacing\coppe@bancalist}\par
% No city/date block at the foot: 3.1.2.1.3 ends the folha de aprovacao at the
% examining board, and Annex D shows nothing below the signature lines. The date
% that belongs on this page is the date of APPROVAL, printed above.
% \frontpage@bottomtext stays defined for anyone calling it directly.
  \vspace*{\fill}}

\newcommand\coppe@hypersetup{%
\begingroup
  % changes to \toks@ and \count@ are kept local;
  % it’s not necessary for them, but it is usually the case
  % for \count1, because the first ten counters are written
  % to the DVI file, thus you got lucky because of PDF output
  \toks@={}% in this special case not necessary
  \count@=0 %
  \@whilenum\count@<\value{keywords}\do{%
    % * a keyword separator is not necessary,
    %    if there is just one keyword
    % * \csname CoppeKeyword:\the\count@\endcsname must be expanded
    %    at least once, to get rid of the loop depended \count@
    \ifcase\count@ % same as \ifnum0=\count@
      \toks@=\expandafter{\csname CoppeKeyword:\the\count@\endcsname}%
    \else
      \toks@=\expandafter\expandafter\expandafter{%
        \expandafter\the\expandafter\toks@
        \expandafter;\expandafter\space
        \csname CoppeKeyword:\the\count@\endcsname
      }%
```

```
    \fi
    \advance\count@ by 1 %
  }%
  \edef\x{\endgroup
    \noexpand\hypersetup{%
      pdfkeywords={\the\toks@}%
    }%
  }%
\x
\hypersetup{%
  pdfauthor={\@authname\ \@authsurn},
  pdftitle={\local@title},
  pdfsubject={\local@doctype\ de \@degreename\ em \local@deptname\ da COPPE/UFRJ},
  pdfcreator={LaTeX with CoppeTeX toolkit},
  breaklinks={true},
  raiselinks={true},
  pageanchor={true},
}}
```

`\makecatalog` When the document has illustrations, it is required to insert ": il.;" between the number of pages of the textual part and the page dimension. We have created a label to flag the existence of lists of figures. It is checked to be undefined using the plain TEX command `\@isundefined` (cf. the TEX FAQ).

```
\newcommand\makecatalog{%
  \vspace*{\fill}
  \begin{center}
    \setlength{\fboxsep}{5mm}
    \framebox[120mm][c]{\makebox[5mm][c]{}%
      \begin{minipage}[c]{105mm}
      \setlength{\parindent}{5mm}
      \noindent\sloppy\nohyphens\@authsurn,
      \nohyphens\@authname\par
      \nohyphens{%
        \coppe@selecttitle/\@authname\ \@authsurn. -- \local@cityname:
      UFRJ/COPPE, \coppe@docyear.}\par
      \pageref{front:pageno},
      \pageref{LastPage}
      p.\@ifundefined{r@cat:lofflag}{}{\pageref{cat:lofflag}} $29,7$cm.\par
      % There is an issue here. When the last entry must be split between lines,
      % the spacing between it and the next paragraph becomes smaller.
      % Should we manually introduce a fixed space? But how could we know that
      % a name was split? Is this happening yet?
      \nohyphens{%
      \begin{tabularx}{100mm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
        \local@advisorstring: &
        \count1=0
        \toks@={}
        \@whilenum \count1<\@advisor \do{%
        \ifcase\count1 % same as \ifnum0=\count1
          \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
          \expandafter\endcsname\expandafter\space%
          \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
        \else
          \toks@=\expandafter\expandafter\expandafter{%
```

```
            \expandafter\the\expandafter\toks@
            \expandafter&\expandafter\space
            \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
            \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
          }%
        \fi
        \advance\count1 by 1}
        \the\toks@
      \end{tabularx}}\par
      \nohyphens{\local@doctype\ ({\MakeLowercase\@degreename}) --
      UFRJ/COPPE/Programa de \local@deptname, \coppe@docyear.}\par
      Refer{\^ e}ncias Bibliogr{\' a}ficas: p. \pageref{bib:begin} -- \pageref{bib:end}.\par
      \count1=0
      \count2=1
      \nohyphens{\@whilenum \count1<\value{keywords} \do {%
        \number\count2. \csname CoppeKeyword:\the\count1 \endcsname.
        \advance\count1 by 1
        \advance\count2 by 1}
      I. \csname CoppeAdvisorSurname:0\endcsname,%
      \ \csname CoppeAdvisorName:0\endcsname%
      \ifthenelse{\@advisor>1}{\ \emph{et~al.}{}}.
      II. \local@universityname, COPPE, Programa de \local@deptname.
      III. T{\' i}tulo.}
    \end{minipage}}
  \end{center}
  \vspace*{\fill}}
```

`\fichacatalografica` `\makefolhaadicional` The folha adicional required from August 2026 (SiBI manual, 9th rev. ed., 3.1.2.1.2 and Annex H): the CAPES collection fields plus the ficha catalografica, sitting immediately after the folha de rosto and counted by nobody. `\fichacatalografica` takes the PDF produced by the SiBI generator at <http://fichacatalografica.sibi.ufrj.br/>, optionally with a page number.

The sheet is institutional, so it is always in Portuguese whatever the main language of the work – and deliberately without `\selectlanguage`, since the brazilian babel language need not be loaded in, say, a Spanish-main document.

```
\newcommand\coppe@fichafile{}
\newcommand\coppe@fichapage{1}
\newcommand\fichacatalografica[2][1]{%
  \gdef\coppe@fichapage{#1}\gdef\coppe@fichafile{#2}}
\newcommand\coppe@marcatipo[1]{%
  \ifthenelse{\equal{\coppe@tipoproducao}{#1}}{$\times$}{\phantom{$\times$}}}
\newcommand\coppe@marcavinc[1]{%
  \ifthenelse{\equal{\coppe@projetovinc}{#1}}{$\times$}{\phantom{$\times$}}}
\newcommand\coppe@campo[1]{%
  \ifthenelse{\equal{#1}{}}{\hrulefill}{#1\enspace\hrulefill}}
\newcommand\coppe@checatipo{%
  \ifthenelse{\equal{\coppe@tipoproducao}{}}{}{%
    \ifthenelse{\equal{\coppe@tipoproducao}{bibliografica}\OR
                \equal{\coppe@tipoproducao}{artistica}\OR
                \equal{\coppe@tipoproducao}{tecnologica}\OR
                \equal{\coppe@tipoproducao}{tecnica}}{}{%
      \ClassWarning{coppe}{Unknown \string\tipoproducao{} value
        `\coppe@tipoproducao'. Annex H of the SiBI manual allows only
```

```
      bibliografica, artistica, tecnologica or tecnica; no box will be
      ticked}}}}
\newcommand\makefolhaadicional{%
  \clearpage
  \thispagestyle{empty}%
  \coppe@checatipo
  {\singlespacing
  \begin{center}
    {\large\bfseries Informa\c c\~oes Coleta CAPES}
  \end{center}
  \vspace{7mm}
  \noindent Tipo de produ\c c\~ao intelectual:\par
  \vspace{1mm}
  \hspace*{10mm}(\coppe@marcatipo{bibliografica})\enspace Bibliogr\'afica\par
  \hspace*{10mm}(\coppe@marcatipo{artistica})\enspace Art\'{\i}stica\par
  \hspace*{10mm}(\coppe@marcatipo{tecnologica})\enspace Tecnol\'ogica\par
  \hspace*{10mm}(\coppe@marcatipo{tecnica})\enspace T\'ecnica\par
  \vspace{5mm}
  \noindent Projeto de pesquisa vinculado \`a produ\c c\~ao intelectual?\par
  \vspace{1mm}
  \hspace*{10mm}(\coppe@marcavinc{sim})\enspace Sim\par
  \hspace*{10mm}(\coppe@marcavinc{nao})\enspace N\~ao\par
  \vspace{5mm}
  \noindent Nome do projeto de pesquisa:\par
  \vspace{1mm}
  \noindent\coppe@campo{\coppe@nomeprojeto}\par
  \vspace{5mm}
  \noindent \'Area de concentra\c c\~ao da produ\c c\~ao intelectual:\par
  \vspace{1mm}
  \noindent\coppe@campo{\coppe@areaconc}\par
  \vspace{5mm}
  \noindent Ag\^encia(s) de fomento financiadora(s) da produ\c c\~ao
  intelectual:\par
  \vspace{1mm}
  \ifthenelse{\value{coppeagencias}=0}{%
    \noindent\hrulefill\par}{%
    \count1=0
    \@whilenum\count1<\value{coppeagencias}\do{%
      \noindent\hspace*{10mm}\csname CoppeAgencia:\the\count1\endcsname
      \ (\csname CoppeAgenciaSigla:\the\count1\endcsname)\par
      \advance\count1 by 1}}%
  \vspace{9mm}
  \begin{center}
    {\large\bfseries Ficha catalogr\'afica}\par
    \vspace{4mm}
    \ifthenelse{\equal{\coppe@fichafile}{}}{%
      \if@copperascunho
        \makecatalog
      \else
        \framebox[125mm][c]{\parbox[c][62mm][c]{115mm}{\centering\footnotesize
          Espa\c co reservado \`a ficha catalogr\'afica.\\[1ex]
          Gere-a em \texttt{fichacatalografica.sibi.ufrj.br} e inclua o
          arquivo com \texttt{\string\fichacatalografica\{arquivo.pdf\}},
          ou solicite-a \`a biblioteca do seu Programa.}}%
```

```
      \fi}{%
      \includegraphics[page=\coppe@fichapage,width=125mm]{\coppe@fichafile}}%
  \end{center}
  \par}}
```

`\dedication`

```
\newcommand\dedication[1]{
  \gdef\@dedic{#1}
    \cleardoublepage
    \vspace*{\fill}
    \begin{flushright}
      \begin{minipage}{60mm}
        \raggedleft \it \normalsize \@dedic
      \end{minipage}
    \end{flushright}}
```

`abstract` (*env.*) This is a specialization of the abstract in the article standard class.

```
\newenvironment{abstract}{%
  \cleardoublepage
  \thispagestyle{plain}
  \abstract@toptext\par
  \vspace*{8.6mm}
  \begin{center}
    \sloppy\nohyphens{\MakeUppercase\local@title}\par
    \vspace*{13.2mm}
    \@authname\ \@authsurn \par
    \vspace*{7mm}
    \local@monthname/\coppe@docyear
  \end{center}\par
  \vspace*{1.5cm}
  \noindent%
  \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
    \local@advisorstring: &
    \count1=0
    \toks@={}
    \@whilenum \count1<\@advisor \do{%
    \ifcase\count1 % same as \ifnum0=\count1
      \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
      \expandafter\endcsname\expandafter\space%
      \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
    \else
      \toks@=\expandafter\expandafter\expandafter{%
        \expandafter\the\expandafter\toks@
        \expandafter&\expandafter\space
        \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
        \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
      }%
    \fi
    \advance\count1 by 1}
    \the\toks@
  \end{tabularx}\par
  \vspace*{2mm}
  \noindent\local@deptstring: \local@deptname\par
  \vspace*{7mm}}{%
```

```
  \coppe@printkeywords{\coppemainstring{keywordsname}}%
    {CoppeKeyword:}{\value{keywords}}%
  \vspace*{\fill}}
```

foreignabstract (*env.*)

```
\newenvironment{foreignabstract}{%
  \cleardoublepage
  \thispagestyle{plain}
  \begin{otherlanguage}{\coppe@foreignlang}
  \foreignabstract@toptext\par
  \vspace*{8.6mm}
  \begin{center}
    \sloppy\nohyphens{\MakeUppercase\foreign@title}\par
    \vspace*{13.2mm}
    \@authname\ \@authsurn \par
    \vspace*{7mm}
    \foreign@monthname/\coppe@docyear
  \end{center}\par
  \vspace*{1.5cm}
  \noindent%
  \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
    \foreign@advisorstring: &
    \count1=0
    \toks@={}
    \@whilenum \count1<\@advisor \do{%
    \ifcase\count1 % same as \ifnum0=\count1
      \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
      \expandafter\endcsname\expandafter\space%
      \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
    \else
      \toks@=\expandafter\expandafter\expandafter{%
        \expandafter\the\expandafter\toks@
        \expandafter&\expandafter\space
        \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
        \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
      }%
    \fi
    \advance\count1 by 1}
    \the\toks@
  \end{tabularx}\par
  \vspace*{2mm}
  \noindent\foreign@deptstring: \foreign@deptname\par
  \vspace*{7mm}}{%
  \ifnum\value{foreignkeywords}>0
    \coppe@printkeywords{\coppeforeignstring{keywordsname}}%
      {CoppeForeignKeyword:}{\value{foreignkeywords}}%
  \else
    \coppe@printkeywords{\coppeforeignstring{keywordsname}}%
      {CoppeKeyword:}{\value{keywords}}%
  \fi
  \end{otherlanguage}
  \vspace*{\fill}
  \global\let\@author\@empty
  \global\let\@date\@empty
```

```
  \global\let\foreign@title\@empty
  \global\let\foreign@title\relax
  \global\let\local@title\@empty
  \global\let\local@title\relax
  \global\let\author\relax
  \global\let\author\relax
  \global\let\date\relax}
```

`brazilianabstract` (*env.*) Optional third abstract for non-pt main languages: a Brazilian Portuguese abstract printed in addition to `abstract` (main language) and `foreignabstract` (English by default). Wraps its body in `\begin{otherlanguage}{brazilian}` so the babel typography (hyphenation, quotation marks) is correct regardless of the current main language. Uses the same `\local@...` Portuguese strings the `abstract` env uses today. Meaningful only when main is neither `brazilian` nor `english`; harmless otherwise (the document just won't include it).

```
\newenvironment{brazilianabstract}{%
  \cleardoublepage
  \thispagestyle{plain}
  \begin{otherlanguage}{brazilian}
  \abstract@toptext\par
  \vspace*{8.6mm}
  \begin{center}
    \sloppy\nohyphens{\MakeUppercase\local@title}\par
    \vspace*{13.2mm}
    \@authname\ \@authsurn \par
    \vspace*{7mm}
    \local@monthname/\coppe@docyear
  \end{center}\par
  \vspace*{1.5cm}
  \noindent%
  \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
    \local@advisorstring: &
    \count1=0
    \toks@={}
    \@whilenum \count1<\@advisor \do{%
    \ifcase\count1
      \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
      \expandafter\endcsname\expandafter\space%
      \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
    \else
      \toks@=\expandafter\expandafter\expandafter{%
        \expandafter\the\expandafter\toks@
        \expandafter&\expandafter\space
        \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
        \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
      }%
    \fi
    \advance\count1 by 1}
    \the\toks@
  \end{tabularx}\par
  \vspace*{2mm}
  \noindent\local@deptstring: \local@deptname\par
  \vspace*{7mm}}{%
```

```
% \coppestring takes ONE argument and resolves against the current language;
% this runs before \end{otherlanguage}, so \languagename is already brazilian.
  \ifnum\value{braziliankeywords}>0
    \coppe@printkeywords{\coppestring{keywordsname}}%
      {CoppeBrazilianKeyword:}{\value{braziliankeywords}}%
  \else
    \coppe@printkeywords{\coppestring{keywordsname}}%
      {CoppeKeyword:}{\value{keywords}}%
  \fi
  \end{otherlanguage}
  \vspace*{\fill}}
```

\listoffigures

```
\renewcommand\listoffigures{%
    \coppe@hasLof
    \if@twocolumn
      \@restonecoltrue\onecolumn
    \else
      \@restonecolfalse
    \fi
    \chapter*{\listfigurename}%
      \addcontentsline{toc}{chapter}{\listfigurename}%
      \@mkboth{\MakeUppercase\listfigurename}%
              {\MakeUppercase\listfigurename}%
    \@starttoc{lof}%
    \if@restonecol\twocolumn\fi
    }
```

\listoftables

```
\renewcommand\listoftables{%
    \if@twocolumn
      \@restonecoltrue\onecolumn
    \else
      \@restonecolfalse
    \fi
    \chapter*{\listtablename}%
      \addcontentsline{toc}{chapter}{\listtablename}%
      \@mkboth{%
          \MakeUppercase\listtablename}%
         {\MakeUppercase\listtablename}%
    \@starttoc{lot}%
    \if@restonecol\twocolumn\fi
    }
```

\printlosymbols

```
\newcommand\printlosymbols{%
\renewcommand\glossaryname{\listsymbolname}%
\@input@{\jobname.los}}
```

\makelosymbols

```
\def\makelosymbols{%
  \newwrite\@losfile
  \immediate\openout\@losfile=\jobname.syx
  \newcommand\symbl[3][]{\@bsphack\begingroup
```

```
  \ifstrempty{##1}{\def\@tempsymbl{##2=}}{\def\@tempsymbl{##1=}}%
              \@sanitize%
              \@wrlos{\@tempsymbl}{##2}{##3}}\typeout%
  {Writing index of symbols file \jobname.syx}%
  \let\makelosymbols\@empty%
}%
\@onlypreamble\makelosymbols
```

```
\AtBeginDocument{%
\@ifpackageloaded{hyperref}{%
  \newcommand\@wrlos[3]{%
    \protected@write\@losfile{}%
      {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
    \endgroup%
    \@esphack}}{%
  \newcommand\@wrlos[3]{%
    \protected@write\@losfile{}%
      {\string\indexentry{#1[#2] #3}{\thepage}}%
    \endgroup%
    \@esphack}}}%
```

\printloabbreviations

```
\newcommand\printloabbreviations{%
\renewcommand\glossaryname{\listabbreviationname}%
\@input@{\jobname.lab}}
```

\makeloabbreviations

```
\def\makeloabbreviations{%
  \newwrite\@labfile
  \immediate\openout\@labfile=\jobname.abx
   \newcommand\abbrev[3][]{\@bsphack\begingroup
               \ifstrempty{##1}{\def\@tempsymbl{##2=}}{\def\@tempsymbl{##1=}}
               \@sanitize
               \@wrlab{\@tempsymbl}{##2}{##3}}\typeout
  {Writing index of abbreviations file \jobname.abx}%
  \let\makeloabbreviations\@empty
}
\@onlypreamble\makeloabbreviations
```

```
\AtBeginDocument{%
\@ifpackageloaded{hyperref}{%
  \newcommand\@wrlab[3]{%
    \protected@write\@labfile{}%
      {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
    \endgroup%
    \@esphack}}{%
  \newcommand\@wrlab[3]{%
    \protected@write\@labfile{}%
      {\string\indexentry{#1[#2] #3}{\thepage}}%
    \endgroup%
    \@esphack}}}%
```

`\newsigla` `\sigla` Acronyms (siglas), manual2024 2.8 / R43. `\newsigla{key}{SIGLA}{full form}` declares one; `\sigla{key}` prints “full form (SIGLA)” on first use and registers it in the list of abbreviations and acronyms (through `\abbrev`, so `\makeloabbreviations` must be active), then just "SIGLA" on later uses. `\sigla*{key}` forces the full form. The first-use flag is set globally so it survives groups (floats, footnotes).

```
\newcommand\newsigla[3]{%
  \@namedef{coppe@sigla@s@#1}{#2}%
  \@namedef{coppe@sigla@l@#1}{#3}}
\newcommand\coppe@sigla@reg[1]{%
  \@ifundefined{abbrev}{}%
    {\abbrev{\@nameuse{coppe@sigla@s@#1}}{\@nameuse{coppe@sigla@l@#1}}}}
\newcommand\coppe@sigla@full[1]{%
  \@nameuse{coppe@sigla@l@#1}\space(\@nameuse{coppe@sigla@s@#1})}
\newcommand\coppe@sigla@undef[1]{%
  \PackageWarning{coppe}{Sigla '#1' nao declarada com \string\newsigla}%
  \textbf{??#1??}}
\newcommand\coppe@sigla@first[1]{%
  \global\@namedef{coppe@sigla@done@#1}{}\coppe@sigla@reg{#1}\coppe@sigla@full{#1}}
\newcommand\coppe@sigla@norm[1]{%
  \@ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
    {\@ifundefined{coppe@sigla@done@#1}{\coppe@sigla@first{#1}}%
      {\@nameuse{coppe@sigla@s@#1}}}}
\newcommand\coppe@sigla@star[1]{%
  \@ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
    {\global\@namedef{coppe@sigla@done@#1}{}\coppe@sigla@full{#1}}}
\newcommand\sigla{\@ifstar\coppe@sigla@star\coppe@sigla@norm}
```

```
%%%% \AtBeginDocument{%
%%%% \@ifpackageloaded{hyperref}{%
%%%%   \def\@wrlab#1#2{%
%%%% \protected@write\@labfile{}%
%%%%       {\string\indexentry{[#1] #2|hyperpage}{\thepage}}%
%%%%     \endgroup
%%%%     \@esphack}}{%
%%%%   \def\@wrlab#1#2{%
%%%%     \protected@write\@labfile{}%
%%%%       {\string\indexentry{[#1] #2}{\arabic{page}}}%
%%%%     \endgroup
%%%%     \@esphack}}}
```

```
% Catalog-label writers. Two language-related traps must be avoided here:
%   1. The label NAMES contain ':' and '.', which babel makes active in
%      some languages (e.g. Brazilian's "shorthand" set). If the write
%      argument is tokenised while one of those is active, pdfTeX aborts
%      with "<inserted text> }\endwrite" because the shorthand expansion
%      injects unbalanced braces. \detokenize emits its argument as
%      catcode-12 characters regardless of the current activation, so the
%      label NAMES survive whatever language is current at writer time.
%   2. Spanish babel redefines \@roman (and therefore \roman{...}) to
%      produce small-caps roman numerals via \es@scroman, whose expansion
%      contains an \uppercase group that is unbalanced inside \write. The
%      same crash signature appears. The fix is to bypass LaTeX's
%      \roman/\arabic counter wrappers entirely and use the raw e-TeX
%      primitives \romannumeral and \number directly on the \c@page count
%      register --- both are fully expandable, language-agnostic, and
%      always produce plain ASCII digits or lowercase roman letters.
%      \Roman{page} is left in place because babel-spanish only redefines
```

```
%      \@Roman under the (non-default) ucroman option.
\AtBeginDocument{%
  \ltx@ifpackageloaded{hyperref}{%
    \def\coppe@bibEnd{%
      \immediate\write\@auxout{%
        \string\newlabel{\detokenize{bib:end}}%
          {{}{\number\c@page}{}{\detokenize{page.}\number\c@page}{}}}}%
    \def\coppe@bibBegin{%
      \immediate\write\@auxout{%
        \string\newlabel{\detokenize{bib:begin}}%
          {{}{\number\c@page}{}{\detokenize{page.}\number\c@page}{}}}}%
    \def\coppe@mainBegin{%
      \immediate\write\@auxout{%
        \string\newlabel{\detokenize{front:pageno}}%
          {{}{\Roman{page}}{}{\detokenize{page.}\romannumeral\c@page}{}}}}%
    \def\coppe@hasLof{%
      \immediate\write\@auxout{%
        \string\newlabel{\detokenize{cat:lofflag}}%
          {{}{\detokenize{:~il.;}}{}{\detokenize{page.}\romannumeral\c@page}{}}}}%
  }{%
    \def\coppe@bibEnd{%
      \immediate\write\@auxout{%
        \string\newlabel{\detokenize{bib:end}}{{}{\number\c@page}{}}}}%
    \def\coppe@bibBegin{%
      \immediate\write\@auxout{%
        \string\newlabel{\detokenize{bib:begin}}{{}{\number\c@page}{}}}}%
    \def\coppe@mainBegin{%
      \immediate\write\@auxout{%
        \string\newlabel{\detokenize{front:pageno}}{{}{\Roman{page}}{}}}}%
    \def\coppe@hasLof{%
      \immediate\write\@auxout{%
        \string\newlabel{\detokenize{cat:lofflag}}{{}{\detokenize{:~il.;}}{}}}}%
  }%
}
% The reference list is now produced by biblatex's \printbibliography (see the
% bibliography setup after babel, above). The former thebibliography
% redefinition and its \bibindent are no longer used and have been removed; the
% bib:begin/bib:end catalog labels are now written by \AtBeginBibliography /
% \AtEndBibliography via \coppe@bibBegin and \coppe@bibEnd, defined just above.
```

longquote (*env.*)

```
\newlength{\recuolongquote}
\setlength{\recuolongquote}{4cm}

\newenvironment*{longquote}[1][default]{%
 \begin{list}{}{%
   \setlength{\leftmargin}{\recuolongquote}%
   \setlength{\rightmargin}{0pt}%
   \setlength{\itemindent}{0pt}%
   \setlength{\listparindent}{0pt}%
   \setlength{\parsep}{0pt}%
   \setlength{\topsep}{\baselineskip}%
  }%
  \item\relax
```

```
  \footnotesize
  \singlespacing
  \ifthenelse{\not\equal{#1}{default}}%
  {\itshape\selectlanguage{#1}}%
  {}%
  \ignorespaces
 }{%
 \end{list}%
 \ignorespacesafterend
}

\newenvironment{theglossary}{%
  \if@twocolumn%
    \@restonecoltrue\onecolumn%
  \else%
    \@restonecolfalse%
  \fi%
  \@mkboth{\MakeUppercase\glossaryname}%
  {\MakeUppercase\glossaryname}%
  \chapter*{\glossaryname}%
  \addcontentsline{toc}{chapter}{\glossaryname}
  \list{}
  {\setlength{\listparindent}{0in}%
   \setlength{\labelwidth}{1.0in}%
   \setlength{\leftmargin}{1.5in}%
   \setlength{\labelsep}{0.5in}%
   \setlength{\itemindent}{0in}}%
   \sloppy}%
  {\if@restonecol\twocolumn\fi%
\endlist}
%
\renewenvironment{theindex}{%
  \if@twocolumn
    \@restonecolfalse
  \else
    \@restonecoltrue
  \fi
  \twocolumn[\@makeschapterhead{\indexname}]%
  \@mkboth{\MakeUppercase\indexname}%
  {\MakeUppercase\indexname}%
  \thispagestyle{plain}\parindent\z@
  \addcontentsline{toc}{chapter}{\indexname}
  \parskip\z@ \@plus .3\p@\relax
  \columnseprule \z@
  \columnsep 35\p@
  \let\item\@idxitem}
  {\if@restonecol\onecolumn\else\clearpage\fi}
\newcommand\listabbreviationname{\coppemainstring{listabbreviation}}
\newcommand\listsymbolname{\coppemainstring{listsymbol}}
\newcommand\glossaryname{\coppemainstring{glossary}}
%
\newcommand\local@advisorstring{Orientador}
\newcommand\foreign@advisorstring{Advisor}
% Fed by the string dispatcher instead of a hard-coded Portuguese literal.
```

```
% Expansion is lazy (it happens inside \makefrontpage), so \coppe@mainlang is
% already set by the time this is used.
\ifthenelse{\boolean{maledoc}}{%
  \newcommand\local@approvedname{\coppemainstring{approvedmale}}%
}{%
  \newcommand\local@approvedname{\coppemainstring{approvedfemale}}%
}
\ifthenelse{\boolean{maledoc}}{%
  \newcommand\local@approvedonname{\coppemainstring{approvedonmale}}%
}{%
  \newcommand\local@approvedonname{\coppemainstring{approvedonfemale}}%
}
\newcommand\local@universityname{Universidade Federal do Rio de Janeiro}
\newcommand\local@deptstring{Programa}
\newcommand\foreign@deptstring{Department}
\newcommand\local@cityname{Rio de Janeiro}
\newcommand\local@statename{RJ}
\newcommand\local@countryname{Brasil}
%
% 3.1.2.1.1(e) of the 2026 manual requires the natureza, the objetivo, the
% name of the institution AND the area de concentracao in this block. The area
% was missing; it is appended as its own sentence when \areaconcentracao was
% given, and simply omitted otherwise.
\newcommand\frontcover@maintext{
  \sloppy\nohyphens{\local@doctype\ de \@degreename\
  \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
  ao Programa de P{\' o}s-gradua{\c c}{\~ a}o em \local@deptname,
  COPPE, da \local@universityname, como parte dos requisitos
  necess{\' a}rios {\` a} obten{\c c}{\~ a}o do t{\' i}tulo de
  \local@degname\ em \local@deptname.}%
  \ifthenelse{\equal{\coppe@areaconc}{}}{}{%
    \par\vspace{1ex}%
    \sloppy\nohyphens{\'Area de concentra{\c c}{\~ a}o: \coppe@areaconc.}}%
}
%
\newcommand\frontpage@maintext{
  \noindent {\MakeUppercase\local@doctype}
  \ifthenelse{\boolean{maledoc}}{SUBMETIDO}{SUBMETIDA}
  \sloppy\nohyphens{AO CORPO DOCENTE DO INSTITUTO ALBERTO LUIZ COIMBRA
  DE P{\' O}S-GRADUA{\c C}{\~ A}O E PESQUISA DE ENGENHARIA DA
  UNIVERSIDADE FEDERAL DO RIO DE JANEIRO COMO PARTE DOS REQUISITOS
  NECESS{\' A}RIOS PARA A OBTEN{\c C}{\~ A}O DO GRAU DE
  {\MakeUppercase\local@degname} EM CI{\^E}NCIAS EM
  {\MakeUppercase\local@deptname.\par}}%
}
%
\newcommand\frontpage@bottomtext{%
  \begin{center}
  {\MakeUppercase{\local@cityname, \local@statename\ -- \local@countryname}}\par
  {\MakeUppercase\local@monthname\ DE \coppe@docyear}
  \end{center}%
}
%
\newcommand\abstract@toptext{%
```

```
  \noindent Resumo \ifthenelse{\boolean{maledoc}}{do}{da}
  \local@doctype\ \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
  \sloppy\nohyphens{{\` a} COPPE/UFRJ como parte dos requisitos
  necess{\' a}rios para a obten{\c c}{\~ a}o do grau de
  \local@degname\ em Ci{\^ e}ncias (\@degree)}
}
\newcommand\foreignabstract@toptext{%
  \noindent \sloppy\nohyphens{Abstract of \foreign@doctype\ presented to
  COPPE/UFRJ as a partial fulfillment of the requirements for the
  degree of \foreign@degname\ of Science (\@degree)}
}
%
```

Late-load packages: setspace, hyperref, biblatex

These three packages *must* be the last things the class loads, and in this exact order:

- `setspace` – supplies `\singlespacing`, `\onehalfspacing` and friends. It is loaded here so the line-spacing state it carries around is settled before `hyperref` patches all the cross-reference and TOC plumbing.
- `hyperref` – patches an enormous number of LaTeX internals (`\section`, `\label`, `\ref`, `\bibcite`, list-of-* commands, ...). Loading it as late as possible is the standard rule of thumb so it sees the final version of those macros, including the ones the `coppe` class redefines (e.g. `\@chapter`, `\listofquadros`).
- `biblatex` – must come after `hyperref` so its citation links and back-references hook into hyperref correctly; otherwise the `\cite` commands print as plain (non-clickable) text.

```
\RequirePackage{setspace}
% Apply the default line spacing now that setspace is available. The flag was
% set above by \DeclareOption{doublespacing}; absent the option, one-and-a-half.
\if@coppedoublespace\doublespacing\else\onehalfspacing\fi
% pdfx MUST precede hyperref -- it loads hyperref itself, with the options
% PDF/A needs, and cannot fix up an already-loaded one. Level a-2b rather than
% a-1b (the "ISO 19005-1" of Annex I): a-1b forbids transparency, which
% tcolorbox and several graphics packages produce, so a-1b would fail on
% documents that are otherwise perfectly fine. a-2b is ISO 19005-2, accepted by
% Pantheon and by the SiBI's own instructions, which also list PDF/A-2b.
\if@coppepdfa
  \RequirePackage[a-2b]{pdfx}
% pdfTeX stamps /PTEX.Fullbanner into the document information dictionary.
% It is not one of the keys ISO 19005-2 6.6.2.3.2 maps to an XMP property, and
% no extension schema declares it, so it is a DocInfo entry with no XMP
% counterpart -- a finding a PDF/A checker has every reason to raise, and one
% that costs nothing to remove. -1 sets every suppression bit.
  \pdfsuppressptexinfo=-1
% pdfx takes the XMP metadata from \jobname.xmpdata, which it reads when it is
% LOADED -- before the preamble has run, so before \title, \author and
% \keyword exist. The file is therefore WRITTEN by the class and picked up on
% the NEXT run. The build already runs pdflatex three times for cross-
```

```
% references, so the deposited PDF always carries current metadata; only a
% first-ever compilation of a brand-new file emits pdfx's "no \jobname.xmpdata"
% warning and lacks dc:title.
%
% The write happens at the END of the document, not at \begin{document}. This
% class has always documented -- and every example follows it -- that \title,
% \author, \department and \keyword go AFTER \begin{document}, next to
% \maketitle. A write at \begin{document} therefore ran before any of them
% existed and died with "Undefined control sequence" on \@authname, so the
% pdfa option was unusable in exactly the style the manual teaches. It only
% ever worked for a document that set its metadata in the preamble. At the end
% everything is set whichever style the author chose, and \immediate still puts
% the file on disk in time for the next run.
%
% The hook is enddocument/afteraux, and the stream is \@mainaux rather than one
% of our own. TeX has sixteen output streams and example.tex was already using
% all of them (toc, lof, lot, loa, lol, loq, los, idx, glo, aux, bcf, ...), so
% \newwrite here failed with "No room for a new \write" on precisely the
% document that most needs validating. enddocument/afteraux fires after LaTeX
% has closed the .aux, which frees that stream for the one \immediate\openout
% we need and costs the document nothing. Kernels older than 2020-10-01 have no
% such hook; there the class falls back to \AtEndDocument with a stream of its
% own, which is the pre-existing behaviour.
%
% Every field is guarded, and the guards are not cosmetic. \coppe@selecttitle
% is \@nameuse{coppe@title@<lang>}, and \csname on a name that does not exist
% yet MAKES it \relax: unexpandable, so \write puts the control sequence NAME
% into the file instead of the title. The .xmpdata then carried the literal
% \coppe@title@brazilian, pdfx read it back on the following run, and the
% document stopped compiling at all -- a single bad run bricked the file until
% someone deleted the .xmpdata by hand. A field that is not set is now simply
% omitted, which pdfx handles, instead of poisoning the next run.
%
% The writes must be \immediate. \protected@write defers to the shipout, and
% the stream is closed here and now, so deferred writes would land nowhere --
% which is exactly what happened first time round: a 0-byte .xmpdata.
% \let\protect\string inside the group gives the protection that
% \protected@write would have given: accents and other robust commands are
% written literally instead of being expanded. pdfx reads the .xmpdata as
% LaTeX, so \'a arrives there intact.
  \ifdefined\AddToHook
    \let\coppe@xmpout\@mainaux
    \def\coppe@xmphook{\AddToHook{enddocument/afteraux}}%
  \else
    \newwrite\coppe@xmpout
    \let\coppe@xmphook\AtEndDocument
  \fi
  \coppe@xmphook{%
    \gdef\coppe@xmpkeys{}%
    \count@=0
    \@whilenum\count@<\value{keywords}\do{%
      \ifnum\count@>0
        \g@addto@macro\coppe@xmpkeys{\string\sep\space}%
      \fi
```

```
      \protected@edef\coppe@tmp{\csname CoppeKeyword:\the\count@\endcsname}%
      \expandafter\g@addto@macro\expandafter\coppe@xmpkeys
        \expandafter{\coppe@tmp}%
      \advance\count@ by 1}%
    \immediate\openout\coppe@xmpout=\jobname.xmpdata
    \begingroup
      \let\protect\string
      \@ifundefined{coppe@title@\coppe@mainlang}{}{%
        \immediate\write\coppe@xmpout{\string\Title{\coppe@selecttitle}}}%
      \@ifundefined{@authname}{}{%
        \immediate\write\coppe@xmpout{%
          \string\Author{\@authname\space\@authsurn}}}%
      \immediate\write\coppe@xmpout{\string\Keywords{\coppe@xmpkeys}}%
      \ifthenelse{\equal{\coppe@deptcode}{}}{}{%
        \immediate\write\coppe@xmpout{\string\Subject{\local@doctype\space de
          \@degreename. Programa de \meta@deptname\space(\coppe@deptcode),
          COPPE/UFRJ}}}%
      \immediate\write\coppe@xmpout{\string\Publisher{\local@universityname}}%
      \@ifundefined{coppe@docyear}{}{%
        \immediate\write\coppe@xmpout{\string\Date{\coppe@docyear}}}%
    \endgroup
    \immediate\closeout\coppe@xmpout}%
\fi
\RequirePackage{hyperref}
% Bibliography engine: the full coppe BibLaTeX/biber ABNT style, replacing the
% former natbib+bibtex setup. The numbers class option selects the numeric
% citation style; otherwise the author-date default is used. natbib=true keeps
% \citet/\citep available for users migrating from natbib.
\if@coppenumeric
  \RequirePackage[backend=biber,datamodel=coppe,%
    bibstyle=coppe-numeric,citestyle=coppe-numeric,natbib=true]{biblatex}
\else
  \RequirePackage[backend=biber,datamodel=coppe,%
    bibstyle=coppe,citestyle=coppe,natbib=true]{biblatex}
\fi
% Reference-list heading "Referencias" (pt) / "References" (en) -- manual2024
% 3.1.4.1 requires the post-textual list to be titled "Referencias", not
% "Bibliografia" (babel's default \bibname). The title is rendered through
% \coppestring so it follows the current main language in BOTH the chapter
% heading and the .toc entry (biblatex's \bibstring proved unreliable across
% the .toc write, and babel's \bibname caption override was equally fragile).
\defbibheading{bibliography}{%
  \chapter*{\coppestring{references}}%
  \addcontentsline{toc}{chapter}{\coppestring{references}}}
% The catalog page range (bib:begin/bib:end) was written by the old
% thebibliography redefinition, which biblatex's \printbibliography does not
% use. biblatex offers \AtBeginBibliography but no end counterpart, so the
% begin label uses that hook and the end label is emitted by wrapping
% \printbibliography to write it just after the reference list.
\AtBeginBibliography{\coppe@bibBegin}
\let\coppe@orig@printbibliography\printbibliography
\renewcommand\printbibliography[1][]{%
  \coppe@orig@printbibliography[#1]%
  \coppe@bibEnd}
```

```
⟨/class⟩
⟨*glossary⟩
actual '='
quote '!'
level '>'
%%%% delim_0   ", p. "
delim_0 "\\dotfill "
lethead_flag  0
headings_flag 0
preamble
"\n\\begin{theglossary}\n  \\makeatletter"
postamble
"\n  \\end{theglossary}\n"
⟨/glossary⟩
⟨*class⟩
\newcommand{\annex}{
    \renewcommand{\appendixname}{\coppestring{annex}}%
    \appendix
    \@coppeannextrue
}
⟨/class⟩
```

BibLaTeX style files

Generated from this .dtx by coppe.ins (one docstrip module each).

```
⟨*dbx⟩
\ProvidesFile{coppe.dbx}[2026/05/28 v4.0 CoppeTeX ABNT datamodel]
% Custom entry types beyond biblatex's standard datamodel (games, TV, maps).
% map = cartographic document (NBR 6023 4.2.12); biblatex has no native @map.
% The remaining ABNT "exotic" types (legislation, jurisdiction, legal, patent,
% standard, music, audio, image, artwork, video, dataset, letter, periodical,
% performance) are already in biblatex's standard datamodel.
\DeclareDatamodelEntrytypes{game,videogame,boardgame,tvshow,map}
% Custom fields.
\DeclareDatamodelFields[type=field,datatype=literal]{course}      % "(... em <course>)"
\DeclareDatamodelFields[type=field,datatype=literal]{coppedegree} % degree word
\DeclareDatamodelFields[type=field,datatype=literal]{platform}    % videogame platform
\DeclareDatamodelFields[type=field,datatype=literal]{scale}       % map scale (escala)
\DeclareDatamodelFields[type=list,datatype=literal]{artist}       % game/illustration art
\DeclareDatamodelFields[type=list,datatype=literal]{director}     % movie/tvshow/video
\DeclareDatamodelFields[type=list,datatype=literal]{producer}
\DeclareDatamodelFields[type=field,datatype=literal]{caderno}     % newspaper section
\DeclareDatamodelFields[type=field,datatype=literal]{relator}     % jurisprudence rapporteur
\DeclareDatamodelEntryfields{course,coppedegree}
\DeclareDatamodelEntryfields[game,videogame,boardgame,software,movie,tvshow,video]%
  {artist,platform,director,producer,version}
\DeclareDatamodelEntryfields[map]{scale}
\DeclareDatamodelEntryfields[article]{caderno}
\DeclareDatamodelEntryfields[jurisdiction]{relator}
⟨/dbx⟩
⟨*bbx⟩
\ProvidesFile{coppe.bbx}[2026/05/28 v4.0 CoppeTeX ABNT bibliography style]
% The reference list is single-spaced (ABNT 2.4) via \singlespacing in
```

```
% \AtBeginBibliography. coppe.cls already loads setspace, but the style must work
% standalone too (e.g. \usepackage[bibstyle=coppe]{biblatex} in a plain article),
% so load setspace here as well (idempotent).
\RequirePackage{setspace}
% ABNT reference list keeps the date at the end (imprenta), same for author-date
% and numeric modes, so we derive from the standard style, not authoryear.
\RequireBibliographyStyle{standard}

% Note: standard style never dashes repeated authors, which already matches the
% post-2020 ABNT rule (repeat the entry in full), so no 'dashed' option is needed.
\ExecuteBibliographyOptions{%
  maxbibnames=99,      % ABNT 4.3.1.3: list all authors by default
  giveninits=false,
  uniquename=false,
  uniquelist=false,
  labeldateparts=true, % needed for author-date extradate (1999a/1999b)
  sorting=nyt,
}

% Language strings (Disponível em / Acesso em / In, etc.)
\DeclareLanguageMapping{brazilian}{brazilian-coppe}
\DeclareLanguageMapping{english}{english-coppe}

% --- Names: all reversed "FAMILY, Given", separated by ";" in the list ---
\DeclareNameAlias{author}{family-given}
\DeclareNameAlias{editor}{family-given}
\DeclareNameAlias{translator}{family-given}
\DeclareNameAlias{default}{family-given}
\DeclareDelimFormat[bib,biblist]{multinamedelim}{\addsemicolon\space}
\DeclareDelimFormat[bib,biblist]{finalnamedelim}{\addsemicolon\space}

% ABNT (NBR 6023 8.1.1.4): a collection entered under the name of its
% organizer prints the kind of responsibility in parentheses after the
% name, abbreviated, lowercase and in the singular even for several
% organizers -- e.g. "SOUSA, A. M. de; OLIVEIRA, L. da C. (org.)". The
% standard style prints ", Org." (comma, no parentheses) when the editor
% takes the author slot, so override editor+others to wrap the role string
% in "(...)". The host position ("In: NAME (org.)") already parenthesises
% the role via coppe:bookeditor below. The role word itself is the
% lowercase, singular "org." (and "ed.", "cur.", ...) set in the language
% packs (brazilian-coppe.lbx et al.).
\renewbibmacro*{editor+others}{%
  \ifboolexpr{ test \ifuseeditor and not test {\ifnameundef{editor}} }
    {\printnames{editor}%
     \setunit*{\addspace}%
     \printtext{\mkbibparens{\usebibmacro{editor+othersstrg}}}%
     \clearname{editor}}
    {}}
% Same parenthesised role for drivers that take the editor through the plain
% editor macro instead of editor+others (e.g. @manual).
\renewbibmacro*{editor}{%
  \ifboolexpr{ test \ifuseeditor and not test {\ifnameundef{editor}} }
    {\printnames{editor}%
     \setunit*{\addspace}%
```

```
      \printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
      \clearname{editor}}
     {}}

% Uppercase the surname (family + prefix) only inside the bibliography list;
% citations keep normal caps (post-2020 rule).
% ABNT date (NBR 6023 §4.3.5.5.2): "dia mês. ano" -- without the brazilian "de"
% between the parts (e.g. "4 fev. 1992", "abr. 1996"). #1=year, #2=month, #3=day
% field names; the month stays abbreviated via \mkbibmonth.
\protected\def\coppe@abntdate#1#2#3{%
  \iffieldundef{#3}{}{\stripzeros{\thefield{#3}}\nobreakspace}%
  \iffieldundef{#2}{}{\mkbibmonth{\thefield{#2}}\iffieldundef{#1}{}{\space}}%
  \iffieldundef{#1}{}{\stripzeros{\thefield{#1}}}}
\AtBeginBibliography{%
  \singlespacing                        % entries single-spaced internally (2.4)
  \setlength{\bibitemsep}{\baselineskip}% one blank line between entries (2.4)
  \renewcommand*{\mkbibnamefamily}[1]{\MakeUppercase{#1}}%
  \renewcommand*{\mkbibnameprefix}[1]{\MakeUppercase{#1}}%
  \let\mkbibdatelong\coppe@abntdate     % ABNT date format (drop the "de")
  \let\mkbibdateshort\coppe@abntdate
}

% --- Titles: bold for whole-document types; plain for parts; entry-by-title
%     (no author/editor) -> not bold, first word in CAPS (ABNT 4.2). ---
% \coppeUCtitle uppercases the first SIGNIFICANT word of the title (ABNT 4.2).
% If the title starts with an article (o/a/os/as/um/uma/uns/umas, the/an), the
% article AND the next word are uppercased; otherwise just the first word. The
% delimited macros split off words; \coppeUCkeep strips the trailing-space
% sentinel so no spurious space is left before the following punctuation.
\protected\def\coppeUCtitle#1{\coppeUCtitleA#1 \coppeUCstop}
\protected\def\coppeUCtitleA#1 #2\coppeUCstop{%
  \coppeUCifart{#1}
    {\MakeUppercase{#1}\ifblank{#2}{}{\space\coppeUCtitleB#2\coppeUCstop}}%
    {\MakeUppercase{#1}\ifblank{#2}{}{\space\coppeUCkeep#2\coppeUCstop}}}
\protected\def\coppeUCtitleB#1 #2\coppeUCstop{%
  \MakeUppercase{#1}\ifblank{#2}{}{\space\coppeUCkeep#2\coppeUCstop}}
\protected\def\coppeUCkeep#1 \coppeUCstop{#1}
% \coppeUCifart{word}{if-article}{if-not}: case-insensitive membership test.
\protected\def\coppeUCifart#1{%
  \begingroup
  \lowercase{\def\coppeUC@w{#1}}%
  \edef\coppeUC@w{,\coppeUC@w,}%
  \expandafter\in@\expandafter{\coppeUC@w}{,a,o,as,os,um,uma,uns,umas,the,an,}%
  \ifin@\aftergroup\@firstoftwo\else\aftergroup\@secondoftwo\fi
  \endgroup}
\DeclareFieldFormat{title}{%
  \ifboolexpr{ test {\ifnameundef{labelname}} }
    {#1}%
    {\mkbibbold{#1}}}
% First word of the title in CAPS for entry-by-title (ABNT 4.2). The titlecase
% format receives the RAW title text (the title format above only sees the
% field-printing commands), so \coppeUCtitle can split off and uppercase the
% first word here.
\DeclareFieldFormat{titlecase}{%
```

```
  \ifboolexpr{ test {\ifnameundef{labelname}} }
    {\coppeUCtitle{#1}}%
    {#1}}
\DeclareFieldFormat[article,incollection,inbook,inproceedings,suppperiodical,review]{title}{
\DeclareFieldFormat[thesis]{title}{\mkbibbold{#1}}
% biblatex's standard style quotes patent (and unpublished) titles; ABNT bolds
% the title of a whole document, so override those back to bold.
\DeclareFieldFormat[patent]{title}{\mkbibbold{#1}}
% Legal documents (NBR 6023 4.2.6): the law's identification/ementa is NOT
% highlighted; the entry leads with the jurisdiction (entity author, CAPS).
\DeclareFieldFormat[legislation,jurisdiction,legal]{title}{#1}
\DeclareFieldFormat{journaltitle}{\mkbibbold{#1}}
\DeclareFieldFormat{booktitle}{\mkbibbold{#1}}
\DeclareFieldFormat{maintitle}{\mkbibbold{#1}}

% --- Pages: ABNT uses "p." (also for ranges); theses use "f." for pagetotal ---
\DeclareFieldFormat{pagetotal}{#1\addnbspace p\adddot}
\DeclareFieldFormat[thesis]{pagetotal}{#1\addnbspace f\adddot}
\DeclareFieldFormat{pages}{p\adddot\addnbspace#1}

% --- Periodical volume / number prefixes ---
\DeclareFieldFormat[article]{volume}{v\adddot\addnbspace#1}
\DeclareFieldFormat[article]{number}{n\adddot\addnbspace#1}

% --- "In:" before host title (incollection/inproceedings/inbook) ---
\renewbibmacro*{in:}{\printtext{\bibstring{in}\addcolon\space}}

% --- Online: "Disponível em: <url>." + "Acesso em: <date>." via lbx strings ---
\DeclareFieldFormat{url}{\bibstring{availablefrom}\addcolon\space\url{#1}}

% --- ABNT article driver: no "In:"; Journal, Local, v., n., p., date, DOI/URL.
%     Newspapers (entrysubtype=newspaper, set by the @artigodejornal source map)
%     differ: the newspaper name (journaltitle) and city, then either
%       "<date>. <caderno>, p. <pages>"   (when a caderno/section is given), or
%       "p. <pages>, <date>"              (page before date, when none),
%     per NBR 6023 §4.2.3.6. ---
\DeclareBibliographyDriver{article}{%
  \usebibmacro{bibindex}%
  \usebibmacro{begentry}%
  \usebibmacro{author/translator+others}%
  \setunit{\labelnamepunct}\newblock
  \usebibmacro{title}%
  \newunit\newblock
  \printfield{journaltitle}%
  \setunit{\addcomma\space}\printlist{location}%
  \iffieldequalstr{entrysubtype}{newspaper}
    {\iffieldundef{caderno}
       {\setunit{\addcomma\space}\printfield{pages}%
        \setunit{\addcomma\space}\printdate}%
       {\setunit{\addcomma\space}\printdate%
        \setunit{\adddot\space}\printfield{caderno}%
        \setunit{\addcomma\space}\printfield{pages}}}%
    {\setunit{\addcomma\space}\printfield{volume}%
     \setunit{\addcomma\space}\printfield{number}%
```

```
     \setunit{\addcomma\space}\printfield{pages}%
     \setunit{\addcomma\space}\printdate}%
  \newunit\newblock
  \printfield{doi}%
  \newunit\newblock
  \usebibmacro{url+urldate}%
  \usebibmacro{finentry}}

% --- incollection: ABNT puts the host organizer before the booktitle, as
%     "AUTOR. Parte. In: ORGANIZADOR (Org.). Booktitle. Local: Editora, ano,
%     p. X-Y." The organizer role string is "Org./Orgs." (see the .lbx). ---
\newbibmacro*{coppe:bookeditor}{%
  \ifnameundef{editor}
    {}
    {\printnames{editor}%
     \setunit*{\addspace}\printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
     \clearname{editor}%
     \setunit{\adddot\space}}}
\newbibmacro*{coppe:partdriver}{%
  \usebibmacro{bibindex}\usebibmacro{begentry}%
  \usebibmacro{author/translator+others}%
  \setunit{\labelnamepunct}\newblock
  \usebibmacro{title}%
  \newunit\newblock
  \usebibmacro{in:}%
  \usebibmacro{coppe:bookeditor}%
  \printfield{booktitle}%
  \newunit\newblock
  \printfield{edition}%
  \newunit\newblock
  \printlist{location}%
  \setunit*{\addcolon\space}\printlist{publisher}%
  \setunit*{\addcomma\space}\printdate%
  \newunit\newblock
  \printfield{pages}%
  \setunit{\addcomma\space}\printfield{chapter}%
  \newunit\newblock
  \usebibmacro{url+urldate}%
  \usebibmacro{finentry}}
\DeclareBibliographyDriver{incollection}{\usebibmacro{coppe:partdriver}}
\DeclareBibliographyDriver{inbook}{\usebibmacro{coppe:partdriver}}

% --- type field: resolve a known bibliography string (babel-aware), else literal ---
\DeclareFieldFormat{type}{\ifbibstring{#1}{\bibstring{#1}}{#1}}

% --- Thesis: "<type> (em <course>)" using the custom course field (ABNT). ---
\DeclareFieldFormat{course}{\mkbibparens{\coppestring{in:}\addspace#1}}
\DeclareBibliographyDriver{thesis}{%
  \usebibmacro{bibindex}\usebibmacro{begentry}%
  \usebibmacro{author}%
  \setunit{\labelnamepunct}\newblock
  \usebibmacro{title}%
  \newunit\newblock
  \printfield{type}\setunit*{\addspace}\printfield{course}%
```

```
  \newunit\newblock
  \printlist{institution}%
  \setunit*{\addcomma\space}\printlist{location}%
  \setunit*{\addcomma\space}\printdate%
  \newunit\newblock
  \printfield{pagetotal}%
  \newunit\newblock
  \printfield{note}%
  \newunit\newblock
  \usebibmacro{url+urldate}%
  \usebibmacro{finentry}}
% --- Games / audiovisual: render the custom fields (artist, platform,
%     director, producer, version) the generic misc driver ignores. ---
\newbibmacro*{coppe:media}{%
  \iflistundef{artist}{}{\setunit{\addperiod\space}%
    \printtext{\coppestring{art}\addcolon\space}\printlist{artist}}%
  \iffieldundef{platform}{}{\setunit{\addperiod\space}%
    \printtext{\coppestring{platform}\addcolon\space}\printfield{platform}}%
  \iflistundef{director}{}{\setunit{\addperiod\space}%
    \printtext{\coppestring{directedby}\addcolon\space}\printlist{director}}%
  \iflistundef{producer}{}{\setunit{\addperiod\space}%
    \printtext{\coppestring{producedby}\addcolon\space}\printlist{producer}}}
\newbibmacro*{coppe:mediadriver}{%
  \usebibmacro{bibindex}\usebibmacro{begentry}%
  \usebibmacro{author/translator+others}%
  \setunit{\labelnamepunct}\newblock
  \usebibmacro{title}%
  \newunit\newblock
  \usebibmacro{coppe:media}%
  \newunit\newblock
  \printlist{publisher}%
  \setunit*{\addcomma\space}\printlist{location}%
  \setunit*{\addcomma\space}\printdate%
  \newunit\newblock
  \printfield{version}%
  \newunit\newblock
  \printfield{note}%
  \newunit\newblock
  \usebibmacro{url+urldate}%
  \usebibmacro{finentry}}
\DeclareBibliographyDriver{game}{\usebibmacro{coppe:mediadriver}}
\DeclareBibliographyDriver{videogame}{\usebibmacro{coppe:mediadriver}}
\DeclareBibliographyDriver{boardgame}{\usebibmacro{coppe:mediadriver}}
\DeclareBibliographyDriver{movie}{\usebibmacro{coppe:mediadriver}}
\DeclareBibliographyDriver{tvshow}{\usebibmacro{coppe:mediadriver}}
% Cartographic document (map, NBR 6023 4.2.12): like a book (author/title/
% imprenta) plus the mandatory scale ("Escala 1:N").
\DeclareFieldFormat{scale}{\bibstring{coppescale}\addnbspace#1}
\DeclareBibliographyDriver{map}{%
  \usebibmacro{bibindex}\usebibmacro{begentry}%
  \usebibmacro{author/translator+others}%
  \setunit{\labelnamepunct}\newblock
  \usebibmacro{title}%
  \newunit\newblock
```

```
  \printlist{location}%
  \setunit*{\addcolon\space}\printlist{publisher}%
  \setunit*{\addcomma\space}\printdate%
  \newunit\newblock
  \printfield{scale}%
  \newunit\newblock
  \usebibmacro{url+urldate}%
  \usebibmacro{finentry}}
% Patent (NBR 6023 4.2.5): inventor. \textbf{title}. Depositante: holder. type
% number. date. The holder (depositante) is printed in normal case, not the
% ALL-CAPS family treatment the bibliography gives the leading author.
\newbibmacro*{coppe:depositor}{%
  \ifnameundef{holder}
    {}
    {\printtext{\bibstring{depositor}\addcolon\space}%
     \begingroup
       \def\mkbibnamefamily##1{##1}%
       \def\mkbibnameprefix##1{##1}%
       \printnames{holder}%
     \endgroup}}
\DeclareBibliographyDriver{patent}{%
  \usebibmacro{bibindex}%
  \usebibmacro{begentry}%
  \usebibmacro{author/translator+others}%
  \setunit{\labelnamepunct}\newblock
  \usebibmacro{title}%
  \newunit\newblock
  \usebibmacro{coppe:depositor}%
  \newunit\newblock
  \printfield{type}%
  \setunit*{\addspace}\printfield{number}%
  \newunit\newblock
  \printdate%
  \newunit\newblock
  \usebibmacro{url+urldate}%
  \usebibmacro{finentry}}
% Exotic ABNT types that biblatex's standard style leaves without a driver:
% give each a baseline (book-like for published docs: norma/partitura; misc for
% the rest) so they print. Per-type ABNT refinements can follow later.
\DeclareBibliographyDriver{standard}{\usedriver{}{book}}
\DeclareBibliographyDriver{music}{\usedriver{}{book}}
% Legal documents (NBR 6023 4.2.6-4.2.8): jurisdiction/entity (CAPS author),
% the law id + ementa (plain, not bold), then location and date.
\newbibmacro*{coppe:legaldriver}{%
  \usebibmacro{bibindex}\usebibmacro{begentry}%
  \usebibmacro{author/translator+others}%
  \setunit{\labelnamepunct}\newblock
  \usebibmacro{title}%
  \newunit\newblock
  % Jurisprudência: "Relator: <nome>." after the decision title (NBR 6023).
  \iffieldundef{relator}{}{\printtext{Relator\addcolon\space}\printfield{relator}\newunit\ne
  \printlist{location}%
  \setunit{\addcomma\space}\printdate%
  \newunit\newblock
```

```
  \usebibmacro{url+urldate}%
  \usebibmacro{finentry}}
\DeclareBibliographyDriver{legislation}{\usebibmacro{coppe:legaldriver}}
\DeclareBibliographyDriver{jurisdiction}{\usebibmacro{coppe:legaldriver}}
\DeclareBibliographyDriver{legal}{\usebibmacro{coppe:legaldriver}}
\DeclareBibliographyDriver{audio}{\usedriver{}{misc}}
\DeclareBibliographyDriver{video}{\usedriver{}{misc}}
\DeclareBibliographyDriver{software}{\usedriver{}{misc}}
\DeclareBibliographyDriver{image}{\usedriver{}{misc}}
\DeclareBibliographyDriver{artwork}{\usedriver{}{misc}}
\DeclareBibliographyDriver{performance}{\usedriver{}{misc}}
\DeclareBibliographyDriver{dataset}{\usedriver{}{misc}}

% --- biber source map: Portuguese synonyms (entry types + fields) + thesis family ---
\DeclareSourcemap{%
  \maps[datatype=bibtex]{%
    % entry-type PT synonyms -> canonical English
    \map{\step[typesource=livro,typetarget=book]}
    \map{\step[typesource=artigo,typetarget=article]}
    \map{\step[typesource=artigodejornal,typetarget=article,final]\step[fieldset=entrysubtyp
    \map{\step[typesource=partedelivro,typetarget=incollection]}
    \map{\step[typesource=emlivro,typetarget=inbook]}
    \map{\step[typesource=verbete,typetarget=inreference]}
    \map{\step[typesource=anais,typetarget=proceedings]}
    \map{\step[typesource=trabalhodeevento,typetarget=inproceedings]}
    \map{\step[typesource=periodico,typetarget=periodical]}
    \map{\step[typesource=relatorio,typetarget=report]}
    \map{\step[typesource=relatoriotecnico,typetarget=report]}
    \map{\step[typesource=manual,typetarget=manual]}
    \map{\step[typesource=correspondencia,typetarget=letter]}
    \map{\step[typesource=patente,typetarget=patent]}
    \map{\step[typesource=diverso,typetarget=misc]}
    \map{\step[typesource=jogo,typetarget=game]}
    \map{\step[typesource=jogoeletronico,typetarget=videogame]}
    \map{\step[typesource=jogodetabuleiro,typetarget=boardgame]}
    \map{\step[typesource=filme,typetarget=movie]}
    \map{\step[typesource=serietv,typetarget=tvshow]}
    \map{\step[typesource=programadetv,typetarget=tvshow]}
    % Exotic ABNT types (NBR 6023 4.2.5-4.2.14); targets are biblatex-native
    % except map (declared in coppe.dbx).
    \map{\step[typesource=legislacao,typetarget=legislation]}
    \map{\step[typesource=atonormativo,typetarget=legislation]}
    \map{\step[typesource=jurisprudencia,typetarget=jurisdiction]}
    \map{\step[typesource=documentojuridico,typetarget=legal]}
    \map{\step[typesource=documentocivil,typetarget=legal]}
    \map{\step[typesource=norma,typetarget=standard]}
    \map{\step[typesource=partitura,typetarget=music]}
    \map{\step[typesource=iconografico,typetarget=image]}
    \map{\step[typesource=obradearte,typetarget=artwork]}
    \map{\step[typesource=tridimensional,typetarget=artwork]}
    \map{\step[typesource=cartografico,typetarget=map]}
    \map{\step[typesource=mapa,typetarget=map]}
    \map{\step[typesource=audiovisual,typetarget=video]}
    \map{\step[typesource=conjuntodedados,typetarget=dataset]}
```

```
    \map{\step[typesource=fasciculo,typetarget=suppperiodical]}
    \map{\step[typesource=trabalhoapresentado,typetarget=unpublished]}
    % thesis family: retype to thesis and preset the (babel-aware) type word
    \map{\step[typesource=tese,typetarget=thesis]}
    \map{\step[typesource=dissertacao,typetarget=thesis]}
    \map{\step[typesource=masterdissertation,typetarget=thesis,final]\step[fieldset=type,fie
    \map{\step[typesource=mscdissertation,typetarget=thesis,final]\step[fieldset=type,fieldv
    \map{\step[typesource=dissertacaomestrado,typetarget=thesis,final]\step[fieldset=type,fi
    \map{\step[typesource=dscthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=d
    \map{\step[typesource=tesedoutorado,typetarget=thesis,final]\step[fieldset=type,fieldval
    \map{\step[typesource=phdthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=p
    \map{\step[typesource=tesephd,typetarget=thesis,final]\step[fieldset=type,fieldvalue=phd
    \map{\step[typesource=tcc,typetarget=thesis,final]\step[fieldset=type,fieldvalue=tcc]}
    % field PT synonyms (unaccented) -> canonical English
    \map{\step[fieldsource=periodico,fieldtarget=journaltitle]}
    \map{\step[fieldsource=revista,fieldtarget=journaltitle]}
    \map{\step[fieldsource=jornal,fieldtarget=journaltitle]}
    \map{\step[fieldsource=titulo,fieldtarget=title]}
    \map{\step[fieldsource=subtitulo,fieldtarget=subtitle]}
    \map{\step[fieldsource=autor,fieldtarget=author]}
    \map{\step[fieldsource=editora,fieldtarget=publisher]}
    \map{\step[fieldsource=local,fieldtarget=location]}
    \map{\step[fieldsource=ano,fieldtarget=year]}
    \map{\step[fieldsource=data,fieldtarget=date]}
    \map{\step[fieldsource=edicao,fieldtarget=edition]}
    \map{\step[fieldsource=paginas,fieldtarget=pages]}
    \map{\step[fieldsource=numero,fieldtarget=number]}
    \map{\step[fieldsource=tradutor,fieldtarget=translator]}
    \map{\step[fieldsource=organizador,fieldtarget=editor]}
    \map{\step[fieldsource=instituicao,fieldtarget=institution]}
    \map{\step[fieldsource=tipo,fieldtarget=type]}
    \map{\step[fieldsource=nota,fieldtarget=note]}
    \map{\step[fieldsource=dataacesso,fieldtarget=urldate]}
    \map{\step[fieldsource=curso,fieldtarget=course]}
    \map{\step[fieldsource=grau,fieldtarget=coppedegree]}
    \map{\step[fieldsource=artista,fieldtarget=artist]}
    \map{\step[fieldsource=plataforma,fieldtarget=platform]}
    \map{\step[fieldsource=diretor,fieldtarget=director]}
    \map{\step[fieldsource=produtor,fieldtarget=producer]}
    \map{\step[fieldsource=versao,fieldtarget=version]}
    \map{\step[fieldsource=escala,fieldtarget=scale]}
  }%
}

% --- ABNT list layout: flush left, single spacing, blank line between entries ---
\setlength{\bibhang}{0pt}
\setlength{\bibitemsep}{\baselineskip}

⟨/bbx⟩
⟨*cbx⟩
\ProvidesFile{coppe.cbx}[2026/05/28 v4.0 CoppeTeX ABNT author-date citations]
% Author-date is the CoppeTeX 4.0 default. authoryear-comp gives compressed
% author-year cites with normal-caps names (post-2020 rule) out of the box.
\RequireCitationStyle{authoryear-comp}
```

```
% --- NBR 10520:2023 author separator ----------------------------------------
% Parenthetical citations separate authors with ";" -> (Silva; Souza, 2020).
% Narrative citations (\textcite, natbib \citet) keep comma + "e/and" ->
% Silva, Souza e Oliveira (2020). The reference list keeps its own ";" via
% coppe.bbx ([bib,biblist]); these contextless formats cover the cite side.
\DeclareDelimFormat{multinamedelim}{\addsemicolon\space}
\DeclareDelimFormat{finalnamedelim}{\addsemicolon\space}
\DeclareDelimFormat[textcite]{multinamedelim}{\addcomma\space}
\DeclareDelimFormat[textcite]{finalnamedelim}{\addspace\bibstring{and}\space}

% --- apud: citation of a citation (ABNT NBR 6023 §4.1.2.2 / NBR 10520) ---------
% ABNT lists only the work actually CONSULTED. So only the consulted work (#4, a
% .bib key) is cited and enters the reference list; the ORIGINAL work (#2 author,
% #3 year) was not consulted and is given as plain text -- it is never a cite key,
% so it is never added to the bibliography.
%   \citetapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
%     -> Orig Author (year) apud Consulted (yr[, page])        [narrative]
%   \citepapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
%     -> (Orig Author, year apud CONSULTED, yr[, page])        [parenthetical]
% #1 = optional postnote (page/loc) on the consulted work; "apud" is latin -> italic.
\DeclareRobustCommand{\citetapud}[4][]{%
  #2\space(#3)\space\mkbibemph{apud}\space\textcite[#1]{#4}}
\DeclareRobustCommand{\citepapud}[4][]{%
  \mkbibparens{#2\addcomma\space#3\space\mkbibemph{apud}\space
    \citeauthor{#4}\addcomma\space\citeyear{#4}%
    \ifblank{#1}{}{\addcomma\space\mkpageprefix[pagination]{#1}}}}
⟨/cbx⟩
⟨*lbxbr⟩
\ProvidesFile{brazilian-coppe.lbx}[2026/05/28 v4.0 CoppeTeX brazilian strings]
\InheritBibliographyExtras{brazilian}
\NewBibliographyString{availablefrom,mscdiss,dscthesis,tcc,depositor,coppescale}
\DeclareBibliographyStrings{%
  inherit       = {brazilian},
  availablefrom = {{Dispon\'{\i}vel em}{Dispon\'{\i}vel em}},
  urlseen       = {{Acesso em}{Acesso em}},
  in            = {{In}{In}},
  editor        = {{organizador}{org\adddot}},
  editors       = {{organizadores}{org\adddot}},
  depositor     = {{Depositante}{Depositante}},
  coppescale    = {{Escala}{Escala}},
  mscdiss       = {{Disserta\c{c}\~{a}o de Mestrado}{Disserta\c{c}\~{a}o de Mestrado}},
  dscthesis     = {{Tese de Doutorado}{Tese de Doutorado}},
  phdthesis     = {{Tese de Doutorado}{Tese de Doutorado}},
  tcc           = {{Trabalho de Conclus\~{a}o de Curso}{Trabalho de Conclus\~{a}o de Curso}}
}
⟨/lbxbr⟩
⟨*lbxen⟩
\ProvidesFile{english-coppe.lbx}[2026/05/28 v4.0 CoppeTeX english strings]
\InheritBibliographyExtras{english}
\NewBibliographyString{availablefrom,mscdiss,dscthesis,tcc,depositor,coppescale}
\DeclareBibliographyStrings{%
  inherit       = {english},
  availablefrom = {{Available from}{Available from}},
```

```
  urlseen       = {{Access on}{Access on}},
  in            = {{In}{In}},
  editor        = {{editor}{ed\adddot}},
  editors       = {{editors}{ed\adddot}},
  depositor     = {{Depositor}{Depositor}},
  coppescale    = {{Scale}{Scale}},
  mscdiss       = {{M\adddot Sc\adddot\ Dissertation}{M\adddot Sc\adddot\ Diss\adddot}},
  dscthesis     = {{D\adddot Sc\adddot\ Thesis}{D\adddot Sc\adddot\ Thesis}},
  phdthesis     = {{Ph\adddot D\adddot\ Thesis}{Ph\adddot D\adddot\ Thesis}},
  tcc           = {{Undergraduate Thesis}{Undergraduate Thesis}},
}
⟨/lbxen⟩
⟨*lbxes⟩
\ProvidesFile{spanish-coppe.lbx}[2026/05/28 v4.0 CoppeTeX spanish strings]
\InheritBibliographyExtras{spanish}
\NewBibliographyString{availablefrom,mscdiss,dscthesis,tcc,depositor,coppescale}
\DeclareBibliographyStrings{%
  inherit       = {spanish},
  availablefrom = {{Disponible en}{Disponible en}},
  urlseen       = {{Consultado el}{Consultado el}},
  in            = {{En}{En}},
  editor        = {{editor}{ed\adddot}},
  editors       = {{editores}{ed\adddot}},
  depositor     = {{Solicitante}{Solicitante}},
  coppescale    = {{Escala}{Escala}},
  mscdiss       = {{Tesis de Maestr\'{\i}a}{Tesis de Maestr\'{\i}a}},
  dscthesis     = {{Tesis Doctoral}{Tesis Doctoral}},
  phdthesis     = {{Tesis Doctoral}{Tesis Doctoral}},
  tcc           = {{Trabajo de Fin de Carrera}{Trabajo de Fin de Carrera}},
}
⟨/lbxes⟩
⟨*lbxfr⟩
\ProvidesFile{french-coppe.lbx}[2026/05/28 v4.0 CoppeTeX french strings]
\InheritBibliographyExtras{french}
\NewBibliographyString{availablefrom,mscdiss,dscthesis,tcc,depositor,coppescale}
\DeclareBibliographyStrings{%
  inherit       = {french},
  availablefrom = {{Disponible \`a}{Disponible \`a}},
  urlseen       = {{Consult\'e le}{Consult\'e le}},
  in            = {{Dans}{Dans}},
  editor        = {{\'editeur}{\'ed\adddot}},
  editors       = {{\'editeurs}{\'ed\adddot}},
  depositor     = {{D\'eposant}{D\'eposant}},
  coppescale    = {{\'Echelle}{\'Echelle}},
  mscdiss       = {{M\'emoire de Master}{M\'emoire de Master}},
  dscthesis     = {{Th\`ese de Doctorat}{Th\`ese de Doctorat}},
  phdthesis     = {{Th\`ese de Doctorat}{Th\`ese de Doctorat}},
  tcc           = {{M\'emoire de fin d'\'etudes}{M\'emoire de fin d'\'etudes}},
}
⟨/lbxfr⟩
⟨*lbxit⟩
\ProvidesFile{italian-coppe.lbx}[2026/05/28 v4.0 CoppeTeX italian strings]
\InheritBibliographyExtras{italian}
\NewBibliographyString{availablefrom,mscdiss,dscthesis,tcc,depositor,coppescale}
```

```
\DeclareBibliographyStrings{%
  inherit       = {italian},
  availablefrom = {{Disponibile da}{Disponibile da}},
  urlseen       = {{Consultato il}{Consultato il}},
  in            = {{In}{In}},
  editor        = {{curatore}{cur\adddot}},
  editors       = {{curatori}{cur\adddot}},
  depositor     = {{Depositante}{Depositante}},
  coppescale    = {{Scala}{Scala}},
  mscdiss       = {{Tesi di Laurea Magistrale}{Tesi di Laurea Magistrale}},
  dscthesis     = {{Tesi di Dottorato}{Tesi di Dottorato}},
  phdthesis     = {{Tesi di Dottorato}{Tesi di Dottorato}},
  tcc           = {{Tesi di Laurea Triennale}{Tesi di Laurea Triennale}},
}
⟨/lbxit⟩
⟨*langes⟩
%% coppe-lang-spanish.def -- CoppeTeX 4.x Spanish class-string pack.
%% Loaded by coppe.cls when \documentclass[spanish]{coppe} (or any other
%% spanish slot) is active. Populates the dispatcher table for the
%% spanish language and registers the biblatex language mapping.
%% Institutional names (universityname/cityname/statename/countryname)
%% are deliberately kept Portuguese: the COPPE/UFRJ cover and folha de
%% rosto are a Brazilian institutional template that does not translate.
\ProvidesFile{coppe-lang-spanish.def}[2026/05/28 v4.0 CoppeTeX spanish class strings]
\copperdefstring{spanish}{appendix}        {Ap\'endice}
\copperdefstring{spanish}{annex}           {Anexo}
\copperdefstring{spanish}{frame}           {Cuadro}
\copperdefstring{spanish}{listframe}       {Lista de Cuadros}
\copperdefstring{spanish}{source}          {Fuente}
\copperdefstring{spanish}{program}         {Programa}
\copperdefstring{spanish}{listprogram}     {Lista de Programas}
\copperdefstring{spanish}{references}      {Referencias}
\copperdefstring{spanish}{in:}             {en}
\copperdefstring{spanish}{art}             {Arte}
\copperdefstring{spanish}{platform}        {Plataforma}
\copperdefstring{spanish}{directedby}      {Direcci\'on}
\copperdefstring{spanish}{producedby}      {Producci\'on}
\copperdefstring{spanish}{listabbreviation}{Lista de Abreviaturas y Siglas}
\copperdefstring{spanish}{listsymbol}      {Lista de S\'imbolos}
\copperdefstring{spanish}{glossary}        {Glosario}
\copperdefstring{spanish}{advisor}         {Director}
\copperdefstring{spanish}{advisors}        {Directores}
\copperdefstring{spanish}{dept}            {Programa}
\copperdefstring{spanish}{approvedmale}    {Aprobado por}
\copperdefstring{spanish}{approvedfemale}  {Aprobada por}
\copperdefstring{spanish}{approvedonmale}  {Aprobado el}
\copperdefstring{spanish}{approvedonfemale}{Aprobada el}
\copperdefstring{spanish}{keywordsname} {Palabras clave}
\copperdefstring{spanish}{volumename} {Volumen}
\copperdefstring{spanish}{coadvisor}  {Codirector}
\copperdefstring{spanish}{coadvisors} {Codirectores}
\copperdefstring{spanish}{ofname}     {de}
\copperdefstring{spanish}{universityname}{Universidade Federal do Rio de Janeiro}
\copperdefstring{spanish}{cityname}    {Rio de Janeiro}
```

```
\copperdefstring{spanish}{statename}   {RJ}
\copperdefstring{spanish}{countryname} {Brasil}
\copperdefstring{spanish}{monthname1} {enero}
\copperdefstring{spanish}{monthname2} {febrero}
\copperdefstring{spanish}{monthname3} {marzo}
\copperdefstring{spanish}{monthname4} {abril}
\copperdefstring{spanish}{monthname5} {mayo}
\copperdefstring{spanish}{monthname6} {junio}
\copperdefstring{spanish}{monthname7} {julio}
\copperdefstring{spanish}{monthname8} {agosto}
\copperdefstring{spanish}{monthname9} {septiembre}
\copperdefstring{spanish}{monthname10}{octubre}
\copperdefstring{spanish}{monthname11}{noviembre}
\copperdefstring{spanish}{monthname12}{diciembre}
\copperdefstring{spanish}{algorithm2eopt}{spanish}
\copperdefstring{spanish}{babelname}     {spanish}
%% biblatex is loaded by coppe.cls AFTER this .def file is read; defer the
%% language-mapping registration until \begin{document} so biblatex's
%% \DeclareLanguageMapping is available.
\AtBeginDocument{\DeclareLanguageMapping{spanish}{spanish-coppe}}
⟨/langes⟩
⟨*langfr⟩
%% coppe-lang-french.def -- CoppeTeX 4.x French class-string pack.
%% See coppe-lang-spanish.def for the structure. Institutional names
%% remain Portuguese; only document-content strings translate.
\ProvidesFile{coppe-lang-french.def}[2026/05/28 v4.0 CoppeTeX french class strings]
\copperdefstring{french}{appendix}        {Annexe}
\copperdefstring{french}{annex}           {Annexe}
\copperdefstring{french}{frame}           {Encadr\'e}
\copperdefstring{french}{listframe}       {Liste des encadr\'es}
\copperdefstring{french}{source}          {Source}
\copperdefstring{french}{program}         {Programme}
\copperdefstring{french}{listprogram}     {Liste des programmes}
\copperdefstring{french}{references}      {R\'ef\'erences}
\copperdefstring{french}{in:}             {dans}
\copperdefstring{french}{art}             {Illustration}
\copperdefstring{french}{platform}        {Plateforme}
\copperdefstring{french}{directedby}      {R\'ealis\'e par}
\copperdefstring{french}{producedby}      {Produit par}
\copperdefstring{french}{listabbreviation}{Liste des abr\'eviations et acronymes}
\copperdefstring{french}{listsymbol}      {Liste des symboles}
\copperdefstring{french}{glossary}        {Glossaire}
\copperdefstring{french}{advisor}         {Directeur}
\copperdefstring{french}{advisors}        {Directeurs}
\copperdefstring{french}{dept}            {Programme}
\copperdefstring{french}{approvedmale}    {Approuv\'e par}
\copperdefstring{french}{approvedfemale}  {Approuv\'ee par}
\copperdefstring{french}{approvedonmale}  {Approuv\'e le}
\copperdefstring{french}{approvedonfemale}{Approuv\'ee le}
\copperdefstring{french}{keywordsname} {Mots-cl\'es}
\copperdefstring{french}{volumename} {Volume}
\copperdefstring{french}{coadvisor}  {Co-directeur}
\copperdefstring{french}{coadvisors} {Co-directeurs}
\copperdefstring{french}{ofname}     {de}
```

```
\copperdefstring{french}{universityname}{Universidade Federal do Rio de Janeiro}
\copperdefstring{french}{cityname}    {Rio de Janeiro}
\copperdefstring{french}{statename}   {RJ}
\copperdefstring{french}{countryname} {Brasil}
\copperdefstring{french}{monthname1} {janvier}
\copperdefstring{french}{monthname2} {f\'evrier}
\copperdefstring{french}{monthname3} {mars}
\copperdefstring{french}{monthname4} {avril}
\copperdefstring{french}{monthname5} {mai}
\copperdefstring{french}{monthname6} {juin}
\copperdefstring{french}{monthname7} {juillet}
\copperdefstring{french}{monthname8} {ao\^ut}
\copperdefstring{french}{monthname9} {septembre}
\copperdefstring{french}{monthname10}{octobre}
\copperdefstring{french}{monthname11}{novembre}
\copperdefstring{french}{monthname12}{d\'ecembre}
\copperdefstring{french}{algorithm2eopt}{french}
\copperdefstring{french}{babelname}     {french}
\AtBeginDocument{\DeclareLanguageMapping{french}{french-coppe}}
⟨/langfr⟩
⟨*langit⟩
%% coppe-lang-italian.def -- CoppeTeX 4.x Italian class-string pack.
%% See coppe-lang-spanish.def for the structure. The algorithm2e
%% language option for Italian is spelled "italiano" -- that's the
%% spelling algorithm2e ships, even though babel calls the language
%% "italian".
\ProvidesFile{coppe-lang-italian.def}[2026/05/28 v4.0 CoppeTeX italian class strings]
\copperdefstring{italian}{appendix}        {Appendice}
\copperdefstring{italian}{annex}           {Allegato}
\copperdefstring{italian}{frame}           {Riquadro}
\copperdefstring{italian}{listframe}       {Elenco dei riquadri}
\copperdefstring{italian}{source}          {Fonte}
\copperdefstring{italian}{program}         {Programma}
\copperdefstring{italian}{listprogram}     {Elenco dei programmi}
\copperdefstring{italian}{references}      {Riferimenti}
\copperdefstring{italian}{in:}             {in}
\copperdefstring{italian}{art}             {Illustrazione}
\copperdefstring{italian}{platform}        {Piattaforma}
\copperdefstring{italian}{directedby}      {Regia di}
\copperdefstring{italian}{producedby}      {Prodotto da}
\copperdefstring{italian}{listabbreviation}{Elenco di abbreviazioni e sigle}
\copperdefstring{italian}{listsymbol}      {Elenco dei simboli}
\copperdefstring{italian}{glossary}        {Glossario}
\copperdefstring{italian}{advisor}         {Relatore}
\copperdefstring{italian}{advisors}        {Relatori}
\copperdefstring{italian}{dept}            {Programma}
\copperdefstring{italian}{approvedmale}    {Approvato da}
\copperdefstring{italian}{approvedfemale}  {Approvata da}
\copperdefstring{italian}{approvedonmale}  {Approvato il}
\copperdefstring{italian}{approvedonfemale}{Approvata il}
\copperdefstring{italian}{keywordsname} {Parole chiave}
\copperdefstring{italian}{volumename} {Volume}
\copperdefstring{italian}{coadvisor}  {Correlatore}
\copperdefstring{italian}{coadvisors} {Correlatori}
```

```
\copperdefstring{italian}{ofname}     {di}
\copperdefstring{italian}{universityname}{Universidade Federal do Rio de Janeiro}
\copperdefstring{italian}{cityname}    {Rio de Janeiro}
\copperdefstring{italian}{statename}   {RJ}
\copperdefstring{italian}{countryname} {Brasil}
\copperdefstring{italian}{monthname1} {gennaio}
\copperdefstring{italian}{monthname2} {febbraio}
\copperdefstring{italian}{monthname3} {marzo}
\copperdefstring{italian}{monthname4} {aprile}
\copperdefstring{italian}{monthname5} {maggio}
\copperdefstring{italian}{monthname6} {giugno}
\copperdefstring{italian}{monthname7} {luglio}
\copperdefstring{italian}{monthname8} {agosto}
\copperdefstring{italian}{monthname9} {settembre}
\copperdefstring{italian}{monthname10}{ottobre}
\copperdefstring{italian}{monthname11}{novembre}
\copperdefstring{italian}{monthname12}{dicembre}
\copperdefstring{italian}{algorithm2eopt}{italiano}
\copperdefstring{italian}{babelname}     {italian}
\AtBeginDocument{\DeclareLanguageMapping{italian}{italian-coppe}}
⟨/langit⟩
⟨*numbbx⟩
\ProvidesFile{coppe-numeric.bbx}[2026/05/28 v4.0 CoppeTeX ABNT numeric bibliography]
% Numeric reference list for the ABNT numeric system (NBR 10520). It reuses the
% full coppe ABNT style (drivers, CAPS surnames, bold titles, "In:", p./f.,
% the PT->EN source map and the language mappings) and only adds the [n] list
% label and the numeric sorting on top.
\RequireBibliographyStyle{coppe}

% ABNT numeric system lists references in the order they are first cited
% (NBR 10520), i.e. unsorted. Override with sorting=nty for an alphabetical
% numbered list. 'labelnumber' generates the [n] field.
\ExecuteBibliographyOptions{labelnumber,sorting=none}

% Bracketed labels, both in the list and for any shorthand entries.
\DeclareFieldFormat{labelnumberwidth}{\mkbibbrackets{#1}}
\DeclareFieldFormat{shorthandwidth}{\mkbibbrackets{#1}}

% Reference list driven by the [n] label (copied from biblatex's numeric.bbx),
% keeping the ABNT blank line between entries via \bibitemsep set in coppe.bbx.
\defbibenvironment{bibliography}
  {\list
     {\printtext[labelnumberwidth]{%
        \printfield{labelprefix}%
        \printfield{labelnumber}}}
     {\setlength{\labelwidth}{\labelnumberwidth}%
      \setlength{\leftmargin}{\labelwidth}%
      \setlength{\labelsep}{\biblabelsep}%
      \addtolength{\leftmargin}{\labelsep}%
      \setlength{\itemsep}{\bibitemsep}%
      \setlength{\parsep}{\bibparsep}}%
      \renewcommand*{\makelabel}[1]{\hss##1}}
  {\endlist}
  {\item}
```

```
⟨/numbbx⟩
⟨*numcbx⟩
\ProvidesFile{coppe-numeric.cbx}[2026/05/28 v4.0 CoppeTeX ABNT numeric citations]
% Numeric citation system (ABNT NBR 10520): in-text references are bracketed
% numbers [1], compressed and sorted within a single cite group. This is the
% opt-in alternative to the author-date default (coppe.cbx); it pairs with
% bibstyle=coppe-numeric. With the biblatex natbib option, \citep -> [1] and
% \citet -> Author [1].
\RequireCitationStyle{numeric-comp}
⟨/numcbx⟩
```

10.2 The bibliographic databases

The two BibLaTeX/biber databases shipped with the class — `example.bib` (the sample database cited by the example monograph) and `coppe.bib` (the references of this manual) — are generated from this source too, so the whole distribution still has a single editable origin. They live inside an `\iffalse` block below: the documentation driver never typesets them, and only `coppe.ins` extracts them, through the `examplebib` and `coppebib` docstrip targets. Edit the entries here, never the generated `.bib` files.

10.3 The generated companion files

Everything the distribution ships is extracted from this file. The class, the BibLaTeX styles, the language packs, the databases and the `.ist` were already here; the demonstration documents, the `latexmkrc` and the regression tests joined them, so that a single `pdflatex coppe.ins` rebuilds the whole tree and there is no second place to edit. Each one sits in an `\iffalse` block: the documentation driver never typesets them, and only `coppe.ins` extracts them.

The tests live in `src/tests/` rather than a sibling `tests/`. That is not taste: TeX refuses `\openout` on a path containing `..`, so a `.ins` run in `src/` can write into a subdirectory and never into a parent. Putting the suite under `src/` is what lets it be generated at all.

What is NOT generated from here, and why: `tools/*.ps1` and `src/tests/run-tests.ps1` are the build harness, not artefacts of the class; `NORMA_COPPE_2026.tex` and `futuremanual2026.tex` are documents about the norm that happen to live in this repository, with their own life cycle.

10.3.1 `example_pt.tex -- pt-main demo`

example_pt.tex – CoppeTeX 4.0 multilingual demo, main language: brazilian. Short demonstration document (cover + folha + abstracts + 1 chapter) used by the CPGP submission package. The full pt-main example with bibliography and lists is the historical example.tex.

10.3.2 `example_en.tex -- en-main demo`

example_en.tex – CoppeTeX 4.0 multilingual demo, main language: english.

10.3.3 `example_es.tex -- es-main demo`

example_es.tex – CoppeTeX 4.0 multilingual demo, main language: spanish. Requires the language pack files coppe-lang-spanish.def and spanish-coppe.lbx to be installed alongside coppe.cls.

10.3.4 `example_fr.tex -- fr-main demo`

example_fr.tex – CoppeTeX 4.0 multilingual demo, main language: french. Requires the language pack files coppe-lang-french.def and french-coppe.lbx to be installed alongside coppe.cls.

10.3.5 `example_it.tex -- it-main demo`

example_it.tex – CoppeTeX 4.0 multilingual demo, main language: italian. Requires the language pack files coppe-lang-italian.def and italian-coppe.lbx to be installed alongside coppe.cls.

10.3.6 `example_pdfa.tex -- the full example as PDF/A`

example_pdfa.tex – o example.tex completo, compilado com a opção 'pdfa'.

Existe para dar ao veraPDF o documento mais exigente que a classe produz: o exemplo inteiro, com bibliografia, listas, figuras, tabelas, algoritmos e as caixas do tcolorbox – ou seja, transparência, que é justamente o que o a-1b proibiria e o a-2b admite. Validar só um documento curto não diria nada sobre o que vai de fato para o depósito.

\PassOptionsToClass tem de vir ANTES do \documentclass que o example.tex traz dentro de si; é o único jeito de acrescentar uma opção de classe a um arquivo gerado pelo docstrip sem duplicá-lo aqui e deixar as duas cópias divergirem com o tempo.

10.3.7 `covers_5languages.tex -- the five covers side by side`

covers_5languages.tex – One-page visual demonstration of the CoppeTeX 4.0 multilingual model: the five built-in main languages side by side, all sharing the same Portuguese institutional template, differing only in the title (and the body language, not visible on the cover).

This file uses the article class deliberately – it is a meta-document about the coppe class, not a thesis. It expects the five cover PNGs cover_pt.png, cover_en.png, cover_es.png, cover_fr.png, cover_it.png to live in the same directory (generated by pdftoppm from the example_<lang>.pdf demos).

10.3.8 `latexmkrc -- latexmk/Overleaf build configuration`

latexmkrc – CoppeTeX build configuration (works locally and on Overleaf).

Run with: latexmk example (or latexmk futuremanual2026, your thesis...) latexmk reads this file automatically when it is in the build directory. On Overleaf, place this file at the project root; Overleaf runs latexmk.

Engine: pdfLaTeX by default (fast). For LuaLaTeX (recommended, full Unicode) set $pdf_mode = 4 below, or use the "% !TeX program = lualatex" magic comment / the Overleaf "Menu > Compiler" setting.

10.3.9 `tests/test_brazilian_one_advisor.tex`

% % test_brazilian_one_advisor.tex – regression guard. % % Historical bug: when only ONE \advisor was given, the cover/folha-de-rosto % rendering used to fall over (the original \maketitle loop assumed at % least two advisors). This test pins the single-advisor path: it must % compile cleanly AND the cover label must read "Orientador" (singular), % not "Orientadores". % % Scope: just enough thesis to render \maketitle and the front pages. % No body chapters – they are covered by test_brazilian_two_advisors.tex. % % Run via ../tests/run-tests.ps1. %

10.3.10 `tests/test_brazilian_two_advisors.tex`

% % test_brazilian_two_advisors.tex – regression test for the coppe class. % Brazilian main language, two advisors (the common case; the cover label % should print "Orientadores"). % % Run via ../tests/run-tests.ps1, which sets TEXINPUTS so this file finds % coppe.cls and example.bib in ../src/. EXTEND THIS FILE whenever a new % bug is found in the two-advisor / common path: keep every existing case % working and add a minimal repro for the new one. % % Covered features (as of 2026/05/29, CoppeTeX v4.0, nlinguas branch): % * metadata: \title, \foreigntitle, \author, \advisor (x2), \examiner (x3), % \department, \date, \keyword (x3) % * \maketitle (cover + folha de rosto + ficha catalografica) % * \frontmatter: \dedication, agradecimentos chapter*, abstract, foreignabstract % * lists: \listoffigures, \listoftables, \listofquadros, \listofprogramas, % \printloabbreviations, \printlosymbols, \tableofcontents % * \mainmatter chapters covering: figures with \source/\fonte, booktabs % table, quadro float, subcaption subfigures, longtable, amsmath + % amssymb math, algorithm2e algorithm, listings Python block, longquote, % footnotes (singlespacing+filete via setspace), \sigla / \sigla*, % \symbl, \abbrev, \citet, \citep, \autoref cross-references % * back matter: \printbibliography, \appendix (x2), \annex % % Compile under pdfLaTeX. Under LuaLaTeX/XeLaTeX, the class skips amssymb % and the end-of-document ClassWarning will fire (test it by adding % \usepackage{unicode-math} + \setmathfont{Latin Modern Math} below). %

10.3.11 `tests/test_brazilian_three_advisors.tex`

% % test_brazilian_three_advisors.tex – upper-bound regression test. % % Three \advisor calls is the most the COPPE/UFRJ cover is expected to % typeset reasonably; the class must still render the cover and folha de % rosto cleanly, with the plural label "Orientadores". % % Scope: just enough thesis to render \maketitle and the front pages. % No body chapters – they are covered by test_brazilian_two_advisors.tex. % % Run via ../tests/run-tests.ps1. %

10.3.12 `tests/test_assinaturas.tex`

% % test_assinaturas.tex – regression test for the coppe class. % Exercises the pre-textual pages introduced/reshaped for the SiBI manual, % 9th revised edition (2026): % % * class option ‘assinaturas’ – folha de aprovacao with one signature % rule per board member instead of the compact list (Annex D) % * \examiner and \advisor with the OPTIONAL institution argument, and the % titulacao argument that pre-4.1 discarded silently % * folha adicional with the Coleta CAPES fields

(3.1.2.1.2, Annex H): % \tipoproducao, \projetovinculado, \nomeprojeto, \areaconcentracao, % \agenciafomento % * \linhapesquisa and \areaconcentracao on the folha de rosto % * \dataaprovacao on the folha de aprovacao % % Deliberately mixes calls WITH and WITHOUT the optional institution, so a % regression in the optional-argument handling shows up here. % % Run via ../tests/run-tests.ps1. Compile under pdfLaTeX. %

10.3.13 `tests/test_assinaturas_7.tex`

% % test_assinaturas_7.tex – regression test for the coppe class. % % The folha de aprovacao under the 'assinaturas' option gives every board % member a signature rule of its own (Annex D). The class had only ever been % exercised up to five members, and a doctoral board at the COPPE routinely % reaches seven, so this test pins the tall end of the range: two advisors % plus five examiners, one of them with a long name and a long institution, % which is the combination most likely to wrap a line and push the sheet over. % % What "passing" means here is narrower than "pdflatex exited 0": the folha % de aprovacao must still fit on ONE page. tools/check-aprovacao.py measures % that from the rendered PDF; run-tests.ps1 only sees the exit code, so a % silent overflow would show up as an Overfull \vbox in the log rather than % as a failure. Both are checked. % % Run via ../tests/run-tests.ps1. Compile under pdfLaTeX. %

10.3.14 `tests/test_pdfa.tex`

% % test_pdfa.tex – regression test for the coppe class. % Exercises the pretextual pages introduced/reshaped for the SiBI manual, % 9th revised edition (2026): % % * class option 'assinaturas' – folha de aprovacao with one signature % rule per board member instead of the compact list (Annex D) % * \examiner and \advisor with the OPTIONAL institution argument, and the % titulacao argument that pre-4.1 discarded silently % * folha adicional with the Coleta CAPES fields (3.1.2.1.2, Annex H): % \tipoproducao, \projetovinculado, \nomeprojeto, \areaconcentracao, % \agenciafomento % * \linhapesquisa and \areaconcentracao on the folha de rosto % * \dataaprovacao on the folha de aprovacao % % Deliberately mixes calls WITH and WITHOUT the optional institution, so a % regression in the optional-argument handling shows up here. % % Run via ../tests/run-tests.ps1. Compile under pdfLaTeX. %

Acknowledgments

Thanks to all CoppeTeX users who have reported their experience with this class. We also thank to professor Fernando Lizarralde and Heiko Oberdiek for their helpful comments. The authors would like to thank the National Council for Scientific and Technological Development (CNPq) of Brazil.

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Numbers written in italic refer to the page where the corresponding entry is described; numbers underlined refer to the code line of the definition; numbers in roman refer to the code lines where the entry is used.

Change History